

Symmetry-Compatible Principle for Optimizer Design: Embeddings, LM Heads, SwiGLU MLPs, and MoE Routers

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Abstract

A striking geometric disparity has long persisted in the practice of deep learning. While modern neural network architectures naturally exhibit rich symmetry and equivariance properties, popular optimization methods, such as ADAM and its variants, operate inherently coordinate-wise, rendering them unable to respect the equivariance structures of the parameter space. In this paper, we address this disparity by introducing a *symmetry-compatible principle* for optimizer design. Specifically, we argue that the gradient update rule should be equivariant under the symmetry group acting on the corresponding weight block, and should remove symmetry-redundant directions when the parameterization has quotient structure. Following this principle, we first provide a unified perspective on bi-orthogonally equivariant updates for general matrix layers, as employed by stochastic spectral descent, MUON, SCION, and polar gradient methods. More importantly, by moving from orthogonal groups to permutation and shared-shift symmetries, we derive new classes of symmetry-compatible optimizers tailored to parameter blocks whose symmetries differ from those of ordinary matrix layers: for embedding and LM head matrices, left-permutation and right-orthogonal equivariance leads to one-sided spectral, row-norm, and hybrid row-norm/spectral updates, with projected variants for LM heads; for SwiGLU MLP projections, intermediate-neuron permutation symmetry motivates row-aware and column-aware variants; and for MoE routers, expert-permutation symmetry together with shared-logit-shift invariance gives rise to projected centered row-norm and left-spectral updates. These constructions yield an end-to-end layerwise optimizer stack in which each major matrix-valued parameter class is assigned an update whose equivariance matches its symmetry group. We corroborate this optimizer design principle through extensive pre-training experiments on dense and sparse MoE language models, including Qwen3-0.6B-style, Gemma 3 1B-style, OLMoE-1B-7B-style, and downsized gpt-oss architectures. Across these experiments, symmetry-compatible update rules consistently improve final validation loss, reduce expert load imbalance in sparse MoE models, and in several cases control final vocabulary-logit growth, improve router stability, and overall training stability over the corresponding ADAMW updates.

 https://github.com/timloutk/equivariant_optimizers

1 Introduction

The most widely used optimizers in deep learning, such as ADAM [83], ADAFACTOR [139], RMSPROP [149], ADAGRAD [44, 116], and their variants, all belong to the broad family of *coordinate-wise adaptive gradient methods*. These methods treat model parameters as a single long concatenated vector and update each coordinate independently. Despite their empirical success, this design

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implicitly assumes that every entry of a weight matrix is an independent coordinate in a high-dimensional vector space. This assumption is rarely questioned, yet it strongly shapes the training dynamics of modern neural networks. In particular, such a geometry-blind treatment ignores the rich matrix structure of neural network parameters and fails to distinguish between the geometries of different layer types, such as embeddings, LM heads, dense linear layers, attention projections, SwiGLU MLP projections, and MoE routers.

At the same time, our theoretical understanding of optimizer behavior remains limited across the two major families most relevant to modern large-scale training: *coordinate-wise adaptive gradient optimizers* and *spectral optimizers*. In language model pre-training in particular, comparisons between these optimizer families are still largely empirical, relying on large-scale benchmarking exercises [155, 136] and speedrunning [77], with relatively little analysis of their different geometric behavior and training dynamics. Hyperparameter transfer rules [163] and scaling-law prescriptions [79, 69], for example, are often applied across optimizers, even though their original development was tied primarily to coordinate-wise adaptive methods, particularly ADAMW [111]. Another notable benchmarking effort is ALGOPERF: TRAINING ALGORITHMS [30, 80], which evaluates training speedups obtained solely from changes to the training algorithm and aims to provide a more comprehensive comparison of optimizers. However, ALGOPERF does not include a language modeling workload, and its workloads are far smaller than the language models considered in modern pre-training. Such benchmarking practices implicitly assume that different optimizer families are directly comparable and share similar training phenomena, which need not be the case.

The central thesis of this paper is that optimizer design for modern neural networks should be layerwise and symmetry-compatible. Rather than applying a single coordinate-wise optimizer to all parameters, we propose a layerwise symmetry-compatible principle: each major matrix-valued parameter class should be updated by an optimizer whose equivariance matches the symmetry of that parameter class. This leads to a broad family of *equivariant optimizers*, whose update laws are matched to the symmetry groups of the parameter blocks on which they act.

Figure 1 summarizes this shift. The coordinate-wise view treats matrix-valued parameters as vectorized collections of independent coordinates, leading to updates that can discard spectral structure and break natural equivariances. In contrast, the symmetry-aware matrix view starts from the layerwise geometry of each parameter class and derives optimizer updates whose equivariance matches that geometry.

Contributions. Our work makes the following contributions.

1. **A symmetry-compatible principle for matrix-gradient optimizer design.** We argue that optimizer updates should respect the symmetry structure of the parameter block on which they act. Popular coordinate-wise adaptive optimizers such as ADAM, ADAMW, and RMSPROP are often *geometrically mismatched* for matrix-valued parameters, since their updates generally fail to respect the natural equivariance and quotient structures of modern neural network layers. Fully-connected layers, attention projections, embedding and LM head matrices, SwiGLU MLP projections, and MoE router matrices all possess nontrivial row, column, permutation, spectral, or shared-shift geometries. Our central message is that neural network weight matrices live in layer-dependent geometries that coordinate-wise adaptive methods do not explicitly capture.
2. **A unifying equivariance view of spectral optimizers.** We show that bi-orthogonal equivariance naturally leads to the class of *spectral optimizers*. This class includes or provides a unifying interpretation of stochastic spectral descent (SSD) [21], MUON [78], SCION [124], and polar gradient methods (POLARGRAD) [91]. These methods compute, exactly or approximately,

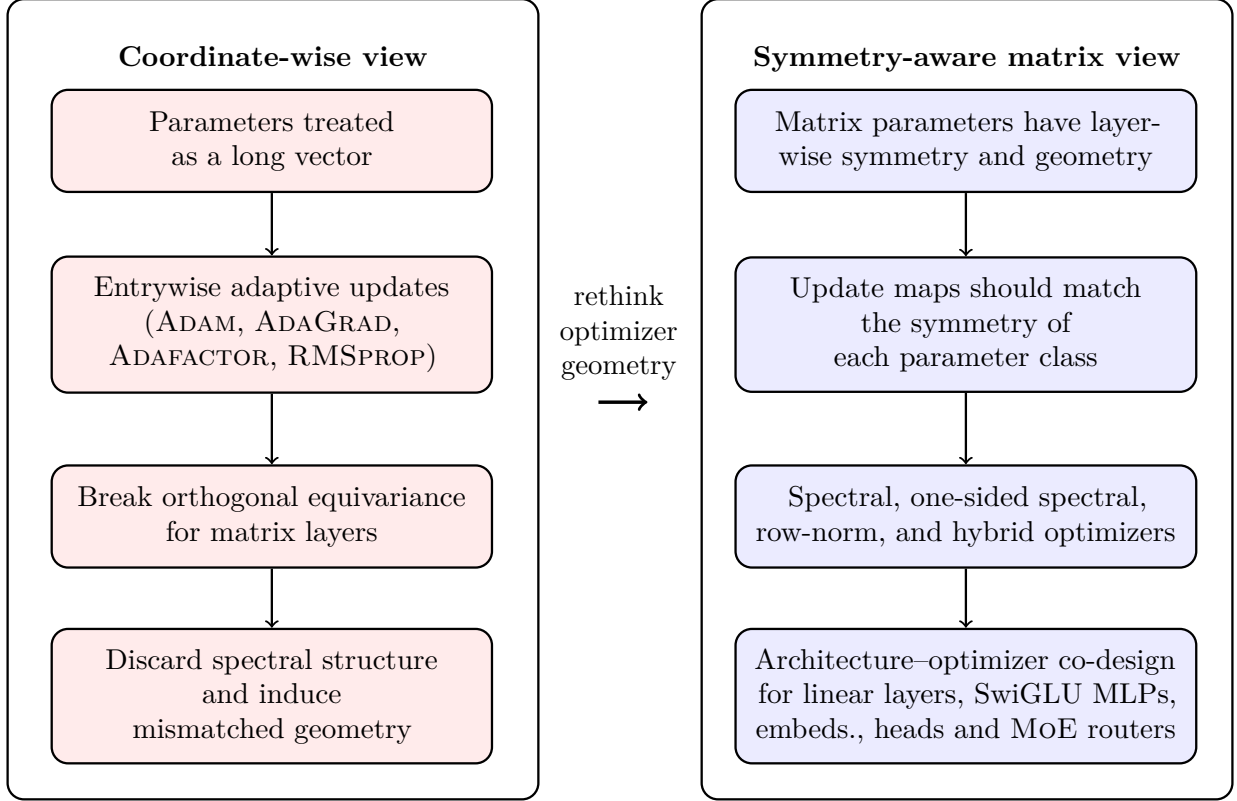


Figure 1: Two perspectives on deep learning optimization. Left: coordinate-wise adaptive methods treat matrix parameters as vectors and ignore matrix geometry. Right: the symmetry- and equivariance-based viewpoint developed in this paper leads to a family of equivariant, layer-specific optimizer classes and architecture-optimizer co-design.

the orthogonal polar factor of an update direction D , such as a gradient G or momentum M :

$$D = U\Sigma V^\top \quad \Rightarrow \quad U_p := \text{polar}(D) = UV^\top.$$

Such updates are bi-orthogonally equivariant, preserve the singular-vector geometry of the update direction, and arise naturally from matrix geometry. This perspective also gives a symmetry-based interpretation of the spectral-norm steepest descent principle underlying MUON [13, 14, 78]: because the spectral norm is unitarily invariant, the corresponding polar update is naturally bi-orthogonally equivariant.

- Layerwise equivariant optimizers for architecture-optimizer co-design.** Beyond full spectral optimizers for ordinary matrix layers, we derive symmetry-compatible optimizer classes for layers whose symmetries differ from those of standard linear maps. For embeddings and LM heads, left-permutation/right-orthogonal equivariance leads to one-sided spectral, row-norm, and hybrid row-norm/spectral updates; for LM heads, softmax shared-logit-shift invariance further requires projected updates on the horizontal quotient space. For SwiGLU MLP projections, intermediate-neuron permutation symmetry motivates row-aware updates for gate and up projections and column-aware updates for down projections. For MOE routers, expert-permutation symmetry together with shared-logit-shift invariance leads to projected centered row-norm, left-spectral, and hybrid updates. The corresponding practical momentum variants are denoted RIGHTPOLARGRADM, LEFTPOLARGRADM, ROWNORMM, and HYBRIDPOLARGRADM.

These constructions instantiate an architecture–optimizer co-design principle based on layerwise equivariance and quotient geometry.

4. **End-to-end pre-training evidence.** We evaluate the proposed layerwise optimizer assignments in dense and sparse MoE language model pre-training experiments (Section 4). To the best of our knowledge, these experiments provide the first end-to-end pre-training optimizer stack in which all major matrix-valued parameter classes in language models are assigned updates according to their layerwise symmetry. Replacing ADAMW on large vocabulary-indexed matrices with row-norm or hybrid symmetry-compatible updates consistently improves final validation loss. The gains are modest but visible for the smaller Qwen3-0.6B-style dense model, become more pronounced for the larger Gemma 3 1B-style model, and persist in sparse MoE experiments based on OLMoE-1B-7B and downsized gpt-oss (Figure 2). In dense models, hybrid row-norm/spectral updates for SwiGLU MLP projections further improve validation loss, and projected symmetry-compatible LM head updates reduce final vocabulary-logit growth in Gemma 3 1B-style pre-training. In the MoE setting, symmetry-compatible router updates improve over coordinate-wise router updates, reduce expert load imbalance, improve router stability as measured by router z-loss, and can reduce training loss spikes.

As a representative example, Figure 2 shows the effect of symmetry-compatible assignments in a sparse MoE pre-training experiment.

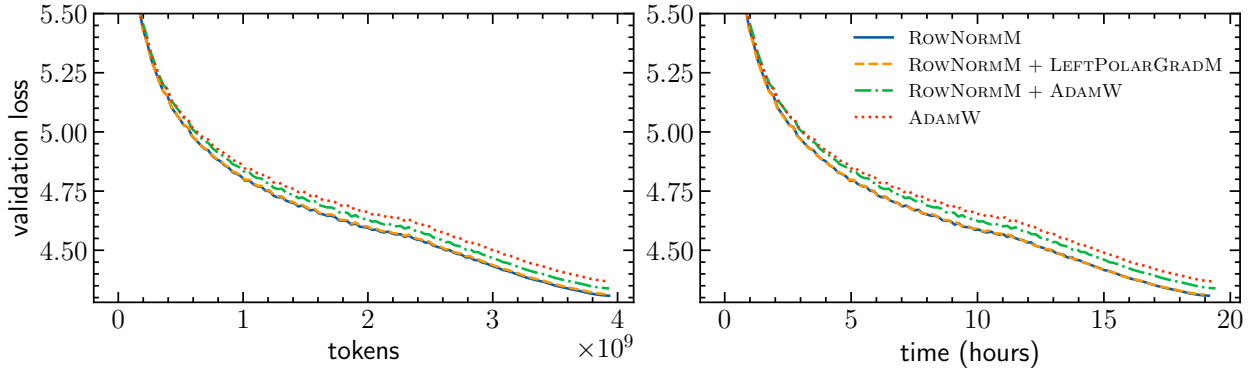


Figure 2: Validation losses for downsized gpt-oss pre-training. The configurations differ in the optimizers for the embedding, LM head, and router matrices; see Section 4.4 for details. Configurations (i) and (ii) use symmetry-compatible optimizers derived from the layerwise equivariance principle, while configuration (iii) replaces the router update by ADAMW and configuration (iv) uses ADAMW for the embedding, LM head, and router matrices.

Scope and limitations. Our goal is not to claim that equivariant optimizers dominate coordinate-wise adaptive methods in all regimes. Rather, we develop a layerwise equivariance principle for matrix-valued parameters and show that it leads to practical optimizer assignments that are competitive and often beneficial in representative pre-training settings. The empirical results should be viewed as evidence for the usefulness of the principle, not as an exhaustive large-scale optimizer benchmark.

Organization. We first introduce notation and closely related work in Section 2. In Section 3, we develop the layerwise symmetry-compatible principle, beginning from a linear-operator view of

matrix parameters and the resulting coordinate-free equivariance requirements. We then derive symmetry-compatible optimizer classes for embeddings, LM heads, SwiGLU MLP projections, and MoE routers, including one-sided spectral, row-norm, projected, and hybrid variants. In Section 3.8, we characterize the spectral optimizer class arising from bi-orthogonal equivariance for ordinary matrix layers. We present practical momentum implementations in Section 3.7 and dense and MoE language model pre-training experiments in Section 4. We conclude with a discussion of broader implications and future directions in Section 5.

2 Preliminaries and Related Work

In this section, we introduce necessary notation and related work for self-containedness. For an extended overview of related work, we refer the readers to Section A.

Notation. For any real-valued square matrix $S \in \mathbb{R}^{d \times d}$, $\text{diag}(S) \in \mathbb{R}^d$ denotes the vector of its diagonal entries, $\text{Diag}(S) \in \mathbb{R}^{d \times d}$ the diagonal matrix with diagonal entries equal to those of S , and $\text{tr}(S)$ is its trace. For any $x \in \mathbb{R}^d$, $\text{Diag}(x) \in \mathbb{R}^{d \times d}$ is the diagonal matrix with diagonal entries equal to the entries of x . For any $m \times n$ real-valued matrices $A := (a_{i,j})_{1 \leq i \leq m, 1 \leq j \leq n}$ and $B := (b_{i,j})_{1 \leq i \leq m, 1 \leq j \leq n}$, we denote the Frobenius inner product of A and B by $\langle\langle A, B \rangle\rangle_F := \text{tr}(A^\top B) = \sum_{i,j} a_{i,j} b_{i,j}$. For a matrix $A \in \mathbb{R}^{m \times n}$, we denote its Frobenius norm by $\|A\|_F := \sqrt{\langle\langle A, A \rangle\rangle_F}$, its spectral norm by $\|A\|_S := \sup_{x \in \mathbb{R}^n, x \neq 0} \{\|Ax\|_2 / \|x\|_2\}$, its nuclear norm by $\|A\|_{\text{nuc}} := \sum_{i=1}^{m \wedge n} \sigma_i(A)$, where $\sigma(A) = (\sigma_1(A), \dots, \sigma_{m \wedge n}(A))^\top$ is the vector of nonincreasing ordered singular values of A , and its max norm by $\|A\|_{\max} := \max_{1 \leq i \leq m, 1 \leq j \leq n} |a_{i,j}|$. The Schatten p -norm of A is denoted by $\|A\|_p := \|\sigma(A)\|_p$. The Hadamard product of $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{m \times n}$ is denoted by $A \odot B := (a_{i,j} b_{i,j})_{1 \leq i \leq m, 1 \leq j \leq n}$. For the matrix $A \in \mathbb{R}^{m \times n}$, we denote by $\text{vec}(A) \in \mathbb{R}^{mn}$ its vectorization by rows. Conversely, for $x \in \mathbb{R}^{mn}$, we write $\text{reshape}(x, m, n) \in \mathbb{R}^{m \times n}$ for the inverse operation, so that $\text{reshape}(\text{vec}(A), m, n) = A$ for all $A \in \mathbb{R}^{m \times n}$. Let $\mathbb{S}^d := \{A \in \mathbb{R}^{d \times d} : A = A^\top\}$ denote the space of real symmetric matrices in $\mathbb{R}^{d \times d}$, $\mathbb{S}_+^d := \{A \in \mathbb{S}^d : A \succcurlyeq 0\}$ the set of symmetric positive semidefinite matrices, and $\mathbb{S}_{++}^d := \{A \in \mathbb{S}^d : A \succ 0\}$ the set of symmetric positive definite matrices, where $\succcurlyeq$ and $\succ$ denote Löwner orders. Let $\mathbb{O}^d := \{A \in \mathbb{R}^{d \times d} : A^\top A = AA^\top = I_d\}$ denote the set of $d \times d$ orthogonal matrices, where $I_d \in \mathbb{R}^{d \times d}$ is the $d \times d$ identity matrix. Let $\mathbb{P}^d := \{P \in \{0, 1\}^{d \times d} : P \mathbf{1}_d = \mathbf{1}_d, P^\top \mathbf{1}_d = \mathbf{1}_d\}$ denote the set of $d \times d$ permutation matrices, where $\mathbf{1}_d$ is the all-ones vector in $\mathbb{R}^d$. Let $\mathcal{E}$ be a Euclidean space endowed with an inner product $\langle \cdot, \cdot \rangle$ and the induced norm $\|\cdot\|$. The domain of a function $f : \mathcal{E} \rightarrow \overline{\mathbb{R}} := \mathbb{R} \cup \{\pm\infty\}$ is $\text{dom } f := \{x \in \mathcal{E} : f(x) < \infty\}$. A function $f : \mathcal{E} \rightarrow \overline{\mathbb{R}}$ is said to be *proper* if it has a nonempty domain. The (convex) indicator function $\iota_{\mathcal{C}}(x)$ of a nonempty closed convex set $\mathcal{C}$ at x equals 0 if $x \in \mathcal{C}$ and $+\infty$ otherwise. The Euclidean projection of x onto a nonempty closed convex set $\mathcal{C}$ is denoted by $\text{proj}_{\mathcal{C}}(x)$. $\mathbb{N}$ denotes the set of nonnegative integers and $\mathbb{N}^* := \mathbb{N} \setminus \{0\}$ denotes the set of positive integers. For a function $f : \mathcal{E} \rightarrow \overline{\mathbb{R}}$, we use $\text{argmin } f$ to denote the unique minimizer of f .

2.1 Matrix-Gradient Optimizers

The recent release of MUON [78], together with its strong empirical performance in the modded-nanogpt speedrun [77], has renewed interest in matrix-gradient optimizers for deep learning. This has led to a rapidly growing line of work on geometry-aware and matrix-structured optimization methods [124, 91, 29, 152, 26, 87, 50, 73, 142, 173, 161, 75, 175, 58, 128, 162, 43]. Conceptually, MUON is closely related to stochastic spectral descent (SSD) [21–23], since both methods can be derived from steepest descent with respect to the spectral norm. We emphasize that this spectral-norm steepest

descent perspective is already closely aligned with the equivariance view developed here: because the spectral norm is unitarily invariant, its steepest descent direction is the orthogonal polar factor, and the resulting MUON update is implicitly bi-orthogonally equivariant. Our contribution is to make this equivariance principle explicit, to place MUON and related methods inside a broader class of spectral optimizers, and to extend the same symmetry-based design logic to layers whose symmetries are not fully bi-orthogonal, such as embeddings, LM heads, SwiGLU MLP projections, and MoE routers.

On the theoretical side, local one-step analyses of simplified MUON-type updates have been developed in [147, 33, 59], while several recent works study convergence rates and optimization guarantees under different assumptions [99, 91, 24, 86, 140, 81, 114]. Our work is aligned with this broad effort, but differs in emphasis: rather than viewing matrix-gradient optimizers primarily as normalization or preconditioning heuristics, we derive them from symmetry and equivariance principles for matrix-valued neural network parameters.

A separate but related line of work develops matrix-gradient optimizers from second-order or preconditioning perspectives. These include Kronecker-factored or layerwise preconditioners such as K-FAC [115, 46], SHAMPOO [64, 7, 141, 45], BFGS and L-BFGS-type methods [56], SOAP [153], KL-Shampoo and KL-SOAP [106], and learned or adaptive preconditioners such as preconditioned SGD (PSGD) [100, 125]. These methods typically approximate curvature or preconditioning structure, whereas spectral and polar updates can also be understood as enforcing equivariance properties of the update map itself. This distinction is important for our framework, since the appropriate optimizer geometry depends on the symmetry group of the layer, not only on curvature approximation.

Other related directions include imposing constraints directly on the weights, such as Stiefel-manifold interpretations and manifold-constrained optimizers [12, 20, 159, 164, 63], as well as variance reduction and low-rank gradient projection methods such as MARS-M and GaLore [108, 168, 179, 145]. These methods address complementary aspects of matrix-gradient optimization, including weight constraints, variance control, and computational efficiency. We refer readers to the recent review [123] for a broader overview of geometry-aware optimization methods in deep learning.

2.2 Matrix Optimization Problems, Löwner Operators, and Spectral Operators

Matrix optimization problems have long been studied as a distinct class of optimization problems because matrices carry algebraic and geometric structures, such as eigenvalues, singular values, ranks, invariant subspaces, and unitary symmetries, that are obscured by vectorization [39, 40]. The foundations for convex and unitarily invariant matrix functions, eigenvalue optimization, and spectral optimization were developed in convex matrix analysis and variational analysis [94, 95, 97, 96].

Our framework is also closely related to spectral functions and spectral operators [70, 16, 68, 148, 27]. For rectangular matrices, such operators act on singular values while preserving singular vectors, $G = U \text{Diag}(\sigma(G))V^\top \mapsto \mathcal{T}(G) = U \text{Diag}(\psi(\sigma(G)))V^\top$. This is the same operator-theoretic structure underlying spectral matrix-gradient optimizers such as stochastic spectral descent, MUON, SCION, and polar gradient methods.

2.3 Symmetry and Equivariance in Deep Learning

There is a long line of work recognizing symmetry and equivariance as organizing principles in neural networks, both for understanding optimization, generalization, and representation learning [120, 65, 135, 102, 1, 176, 177, 127, 178], and for designing equivariant architectures [104, 10, 84]. Our work is complementary: rather than imposing equivariance on the architecture or studying equivariance of existing training dynamics, we impose equivariance on the optimizer update map

acting on parameter tensors. Thus, our viewpoint extends the equivariance principle from architecture design to optimizer design, where the relevant symmetry is the internal geometry of the parameter space rather than only the symmetry of the input or output domain.

3 Equivariant Optimizers from Layerwise Symmetry

Modern deep learning architectures contain matrix-valued parameters with different symmetry structures. The common principle is that a parameter matrix does not always represent an arbitrary array of coordinates, but often represents a linear map between two structured spaces. If the coordinates of these spaces are changed, the parameter and its gradient transform accordingly, and a geometry-compatible optimizer should transform in the same way.

We first state this principle in a general form. Let $W \in \mathbb{R}^{m \times n}$ represent a linear map from an input space to an output space. Suppose the output and input coordinates are transformed by invertible matrices $P \in \text{GL}(m)$ and $Q \in \text{GL}(n)$. Then the same linear map is represented by $\widetilde{W} = PWQ^{-1}$. If $\widetilde{f}(\widetilde{W}) := f(P^{-1}\widetilde{W}Q)$, then standard matrix calculus gives

$$\nabla_{\widetilde{W}} \widetilde{f}(\widetilde{W}) = P^{-\top} \nabla_W f(W) Q^{\top}.$$

Thus, under a general change of coordinates, the gradient transforms contravariantly with respect to the output coordinates and covariantly with respect to the input coordinates.

In this work, we study the equivariance of the update map $\mathcal{U}: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ in matrix-optimizer iterations

$$(\forall k \in \mathbb{N}) \quad W_{k+1} = W_k - \gamma_k \mathcal{U}(D_k),$$

where D_k is an update direction, such as a gradient or momentum. The relevant requirement is not necessarily that the layerwise loss function be invariant under arbitrary transformations, but that the optimizer update transform consistently with the representation of its input direction. Thus, once a layer symmetry specifies a transformation law $D_k \mapsto g \cdot D_k$, we require

$$\mathcal{U}(g \cdot D_k) = g \cdot \mathcal{U}(D_k).$$

When D_k transforms equivariantly, the update $\mathcal{U}(D_k)$ therefore transforms equivariantly as well.

In this paper, however, we do not require equivariance under all invertible changes of coordinates. The relevant symmetry group depends on the layer. For ordinary linear and attention matrices, the natural coordinate changes are orthonormal changes of basis, so $P \in \mathbb{O}^m$ and $Q \in \mathbb{O}^n$. In this case $P^{-\top} = P$ and $Q^{-1} = Q^{\top}$, and both the parameter and gradient transform as $W \mapsto PWQ^{\top}$ and $G \mapsto PGQ^{\top}$. This leads to the bi-orthogonal equivariance condition

$$\mathcal{U}(PGQ^{\top}) = P \mathcal{U}(G) Q^{\top}.$$

For embedding and LM head matrices $W \in \mathbb{R}^{v \times d}$, the row axis indexes vocabulary items, so the admissible left action is not a general orthogonal rotation but a permutation $P \in \mathbb{P}^v$, while the hidden feature axis still admits right orthogonal transformations. For MoE routers, the row axis indexes experts and additionally has a shared-logit-shift invariance. For SwiGLU MLP projections, the relevant symmetry is permutation of intermediate neurons, which acts on the rows of the gate and up projections and on the columns of the down projection.

This gives a layerwise equivariance principle: the optimizer update map should commute with the symmetry group of the parameter block on which it acts. Full bi-orthogonal equivariance leads to spectral optimizers for ordinary matrix layers; left-permutation/right-orthogonal equivariance

leads to row-aware and right-spectral optimizers for embeddings and LM heads; intermediate-neuron permutation symmetry leads to row- and column-aware updates for SwiGLU MLP projections; and expert-permutation plus shared-shift symmetry leads to centered row-aware or left-spectral updates for MoE routers.

3.1 A General Symmetry-Induced Optimizer Geometry

Let $W \in \mathbb{R}^{m \times n}$ be a layer parameter and let $f: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ be the corresponding layerwise loss. Suppose a group $\mathcal{G}$ acts on the parameter space by transformations $W \mapsto g \cdot W$. In the matrix settings considered below, this action is typically of the form $g \cdot W = PWQ^{-1}$, or, after restricting to orthogonal or permutation symmetries, $g \cdot W = PWQ^\top$. We say that the parameterization admits the symmetry group $\mathcal{G}$ if $f(g \cdot W) = f(W)$ for all $g \in \mathcal{G}$. The corresponding optimizer update map should satisfy

$$(\forall g \in \mathcal{G}) \quad \mathcal{U}(g \cdot G) = g \cdot \mathcal{U}(G),$$

where G is an update direction, such as a gradient or momentum, expressed in the corresponding transformed coordinates. This condition ensures that the optimizer does not depend on arbitrary choices of representation that are invisible to the model.

3.2 Bi-Orthogonal Equivariance for Ordinary Matrix Layers

The general reparameterization $W \mapsto PWQ^{-1}$ specializes to $W \mapsto PWQ^\top$ when the admissible coordinate changes are orthogonal. This is the natural case for ordinary linear layers and attention projection matrices, where both input and output coordinates represent continuous feature bases. Therefore an update map for such layers should satisfy

$$(\forall P \in \mathbb{O}^m, \forall Q \in \mathbb{O}^n) \quad \mathcal{U}(PGQ^\top) = P\mathcal{U}(G)Q^\top.$$

This is exactly *bi-orthogonal equivariance*.

Definition 3.1 (Bi-orthogonal equivariance). Let $\mathcal{U}: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ be a matrix-valued map. We say that $\mathcal{U}$ is *bi-orthogonally equivariant* if, for all $G \in \mathbb{R}^{m \times n}$ and all $P \in \mathbb{O}^m, Q \in \mathbb{O}^n$,

$$\mathcal{U}(PGQ^\top) = P\mathcal{U}(G)Q^\top.$$

Thus, bi-orthogonal equivariance is exactly the requirement that if W and its gradient are transformed as $W \mapsto PWQ^\top$ and $G \mapsto PGQ^\top$, then the optimizer update transforms in the same way. More generally, if an update rule has the form $\Delta W = \mathcal{U}(W, G)$, one may require

$$\mathcal{U}(PWQ^\top, PGQ^\top) = P\mathcal{U}(W, G)Q^\top.$$

In this paper, we focus on update maps of the form $\mathcal{U}(G)$ or $\mathcal{U}(D)$, where D is a gradient-derived direction such as momentum. Terms that depend explicitly on W , such as decoupled weight decay or scalar step-size scaling, can typically be handled separately and preserve the same equivariance, so we suppress the dependence on W for simplicity.

For ordinary matrix layers, bi-orthogonal equivariance motivates polar and spectral update directions. In particular, the orthogonal polar factor [9, 66] satisfies

$$\text{polar}(PGQ^\top) = P \text{polar}(G)Q^\top, \tag{1}$$

and hence MUON-style and polar-gradient updates are symmetry-compatible for ordinary matrix layers. We defer the full characterization of bi-orthogonally equivariant maps as spectral operators

to Section 3.8. Standard momentum constructions such as EMA, Polyak, and Nesterov momentum preserve the same equivariance because their buffers are linear combinations of past gradients. We state this fact for Nesterov momentum in the following proposition.

Proposition 3.1 (Nesterov momentum is bi-orthogonally equivariant). *Let $(W_k)_{k \in \mathbb{N}} \subset \mathbb{R}^{m \times n}$ be a parameter sequence, and define $G_k := \nabla_W f(W_k)$ for $k \in \mathbb{N}$. Fix orthogonal matrices $P \in \mathbb{O}^m$ and $Q \in \mathbb{O}^n$, and define the transformed parameter sequence $\tilde{W}_k := PW_k Q^\top$ for $k \in \mathbb{N}$, with transformed gradients $\tilde{G}_k := \nabla_W f(\tilde{W}_k)$ for $k \in \mathbb{N}$. Let the momentum buffer be $M_k = \beta M_{k-1} + G_k$ with $M_{-1} = 0$, and define the update direction by $N_k := G_k + \beta M_k$. For the transformed sequence, let $\tilde{M}_k = \beta \tilde{M}_{k-1} + \tilde{G}_k$ with $\tilde{M}_{-1} = 0$, and $\tilde{N}_k := \tilde{G}_k + \beta \tilde{M}_k$. Then we have $\tilde{M}_k = PM_k Q^\top$, $\tilde{N}_k = PN_k Q^\top$, and $\text{polar}(\tilde{N}_k) = P \text{polar}(N_k) Q^\top$.*

All proofs are given in Section D.

Remark 3.1 (Momentum-first or polar-first POLARGRAD?). Lau et al. [91] studied two variants of POLARGRAD—*momentum-first*, which uses the orthogonal polar factor of the EMA momentum of the gradient as the update direction, and *polar-first*, which uses the EMA momentum of the orthogonal polar factor of the gradient as the update direction. They resemble previous orthogonalized gradient optimizers in a similar spirit. Tuddenham et al. [150] adopt a polar-first update, while Jordan et al. [78] adopt a momentum-first update, which outperforms the polar-first one in practice. Indeed, Proposition 3.1 provides an intuitive explanation for why momentum-first POLARGRAD is preferred to its polar-first counterpart. Only momentum-first updates are directly covered by the bi-orthogonal equivariance argument above: the gradient and its EMA momentum transform equivariantly, and the polar map is then applied to this equivariant momentum direction. In contrast, $\text{polar}(\beta M_{k-1} + (1 - \beta)G_k) \neq \beta \text{polar}(M_{k-1}) + (1 - \beta)\text{polar}(G_k)$ in general because $\text{polar}(\cdot)$ is nonlinear. Thus, the polar-first update is generally not obtained by applying a spectral map to an equivariant momentum direction. In the momentum-first update, the polar step instead extracts a “best orthogonal direction” from the smoothed matrix signal via EMA momentum, which tends to be less noisy and better behaved.

3.3 Optimizers for Embeddings and LM Heads via Left-Permutation Right-Orthogonal Equivariance

For vocabulary-indexed matrices such as input embeddings and untied LM heads, rows index vocabulary items and columns correspond to hidden features. This motivates left-permutation and right-orthogonal equivariant update maps. For an update direction $D \in \mathbb{R}^{v \times d}$, representative examples include row-norm updates, right-spectral updates, and hybrid row-norm/right-spectral updates. In the embedding case, these take the form

$$\mathcal{U}_{\text{row}}(D) = \text{Diag}(\eta(\|D_{1:}\|_2), \dots, \eta(\|D_{v:}\|_2))D,$$

$$\mathcal{U}_{\text{R}}(D) = D(D^\top D + \varepsilon I)^{-1/2},$$

and

$$\mathcal{U}_{\text{hyb}}(D) = \mathcal{U}_{\text{R}}(\mathcal{U}_{\text{row}}(D)) \quad \text{or} \quad \mathcal{U}_{\text{row}}(\mathcal{U}_{\text{R}}(D)).$$

The row-norm update acts locally on vocabulary rows, the right-spectral update acts globally through the hidden-feature Gram matrix, and the hybrid update combines the two.

For untied LM heads, there is an additional quotient geometry. Since softmax probabilities are invariant under adding the same scalar to all vocabulary logits, adding a shared row vector $\mathbf{1}_v a^\top$ to the LM head changes $W_{\text{out}} h$ only by a shared logit shift and therefore leaves the output distribution

unchanged. Thus, the shared-row direction is vertical, and the intrinsic LM head update should be horizontal. Let

$$\Pi_v^\perp := I_v - \frac{1}{v} \mathbf{1}_v \mathbf{1}_v^\top.$$

For LM head updates, we therefore apply the corresponding row-norm, right-spectral, or hybrid maps to the centered direction $D_c = \Pi_v^\perp D$, and project the final update back to the horizontal subspace when rowwise nonlinear operations are used. For example, the horizontal LM head row-norm update is

$$\mathcal{U}_{\text{row}}^{\text{LM}}(D) = \Pi_v^\perp \text{Diag}(\eta(\|D_{c,1}\|_2), \dots, \eta(\|D_{c,v}\|_2)) D_c, \quad D_c = \Pi_v^\perp D.$$

The final projection is not redundant: although $\mathbf{1}_v^\top D_c = 0$, the rowwise-rescaled matrix

$$\text{Diag}(\eta(\|D_{c,1}\|_2), \dots, \eta(\|D_{c,v}\|_2)) D_c$$

need not be row-centered, since the scaling factors vary across rows. Similarly, for hybrid row-norm/right-spectral LM head updates, we reproject after rowwise nonlinear stages and, in practice, also project the final update to ensure that the actual parameter update remains horizontal.

We now explain why these updates are natural from the symmetry of embeddings and LM heads. In empirical uses of MUON [78], it is often recommended that ADAMW be used for embedding and LM head matrices. For embeddings, this recommendation is motivated in part by modular norm theory [90]; for LM heads, it appears to be driven more by empirical considerations. Relatedly, SCION [124] derives embedding updates from induced operator norms in its linear minimization oracle framework. These approaches depend on a particular choice of norm. Here we instead derive optimizer classes directly from the symmetry of the parameterization.

Let $v \in \mathbb{N}^*$ denote the vocabulary size and $d \in \mathbb{N}^*$ the embedding dimension, typically with $v \gg d$. Consider an input embedding matrix $E \in \mathbb{R}^{v \times d}$ and an untied LM head matrix $W_{\text{out}} \in \mathbb{R}^{v \times d}$. In both cases, rows index vocabulary items, while columns correspond to hidden features. Thus, the row axis admits permutation symmetry, whereas the hidden feature axis admits right orthogonal symmetry. The natural equivariance condition for an update map is therefore

$$\mathcal{U}_{\text{LPRO}}(PDR^\top) = P \mathcal{U}_{\text{LPRO}}(D) R^\top, \quad (2)$$

for all $D \in \mathbb{R}^{v \times d}$, permutation matrices $P \in \mathbb{P}^v$, and orthogonal matrices $R \in \mathbb{O}^d$. We call such maps *left-permutation right-orthogonal* (LPRO) equivariant. For untied LM heads, the same principle is applied on the horizontal quotient space defined by $\Pi_v^\perp$.

Definition 3.2 (Left-permutation and right-orthogonal equivariant maps). A map $\mathcal{U}_{\text{LPRO}}: \mathbb{R}^{v \times d} \rightarrow \mathbb{R}^{v \times d}$ is said to be *left-permutation and right-orthogonal equivariant* if (2) holds for all $D \in \mathbb{R}^{v \times d}$, $P \in \mathbb{P}^v$, and $R \in \mathbb{O}^d$. We denote the set of such maps by $\mathcal{U}_{\text{LPRO}}^{v \times d}$.

3.3.1 Right-Spectral Optimizers

A first natural subclass of LPRO-equivariant maps is given by right-spectral updates,

$$\mathcal{U}_{\text{R}}(D) = D \Phi(D^\top D), \quad (3)$$

where $\Phi: \mathbb{S}_+^d \rightarrow \mathbb{R}^{d \times d}$ is an orthogonally equivariant spectral operator. Equivalently, if $D^\top D = V \text{Diag}(\lambda(D^\top D)) V^\top$, then

$$\Phi(D^\top D) = V \text{Diag}(\psi(\lambda(D^\top D))) V^\top$$

for some absolutely symmetric map $\psi: \mathbb{R}_+^d \rightarrow \mathbb{R}^d$.

Theorem 3.2 (Right-spectral updates are LPRO-equivariant). *Let $\mathcal{U}_R$ be of the form (3), where $\Phi(RXR^\top) = R\Phi(X)R^\top$ for all $X \in \mathbb{S}_+^d$ and $R \in \mathbb{O}^d$. Then*

$$\mathcal{U}_R(PDR^\top) = P\mathcal{U}_R(D)R^\top$$

for all $D \in \mathbb{R}^{v \times d}$, $P \in \mathbb{P}^v$, and $R \in \mathbb{O}^d$. We denote the set of right-spectral maps by $\mathcal{U}_R^{v \times d}$.

The choice $\Phi(X) = (X + \varepsilon I)^{-1/2}$ yields the damped right-polar update

$$\mathcal{U}_R(D) = D(D^\top D + \varepsilon I)^{-1/2}.$$

When $\varepsilon = 0$ and D has full column rank, this is the orthogonal polar factor of D . Thus, right-spectral updates are the one-sided analogue of spectral or polar-gradient updates, but they require only the smaller right Gram matrix $D^\top D \in \mathbb{R}^{d \times d}$.

For LM heads, right-spectral updates preserve horizontality once the input direction is centered. Indeed, if $\mathbf{1}_v^\top D_c = 0$, then

$$\mathbf{1}_v^\top D_c \Phi(D_c^\top D_c) = 0.$$

Thus, in exact arithmetic, the final projection is not necessary for purely right-spectral maps. We nevertheless use the quotient-space form

$$\mathcal{U}_R^{\text{LM}}(D) = \Pi_v^\perp D_c (D_c^\top D_c + \varepsilon I)^{-1/2}, \quad D_c = \Pi_v^\perp D,$$

as a numerically robust horizontal update, especially when the inverse square root or polar factor is computed by an inexact oracle.

This computational distinction is important for embeddings and LM heads, where $v \gg d$. Although coordinate-wise adaptive methods such as ADAM are often used for these matrices in practice, they are not LPRO-equivariant. The reason for their empirical use may be computational rather than geometric: accurately approximating polar factors of tall-skinny, ill-conditioned vocabulary matrices can be challenging with simple Newton–Schulz iterations. More robust polar decomposition algorithms, such as QDWH [118] and ZOLO-PD [119], can compute such updates more accurately [91].

Right-spectral normalization is also natural statistically. In a mini-batch of size b , gradients of embedding and LM head matrices have rank at most $O(b)$, since they factor through token occurrences and hidden features. Thus, even when the vocabulary dimension is very large, the gradient often lies in a low-dimensional feature subspace. Right-spectral updates act on this intrinsic singular geometry through $D^\top D$, rather than on individual coordinates.

3.3.2 Row-Norm and Hybrid LPRO-Equivariant Optimizers

Right-spectral maps do not exhaust all LPRO-equivariant updates. Restricting the left symmetry from the full orthogonal group to the permutation group allows update maps that depend on individual row norms. In particular, row-norm maps of the form

$$\mathcal{U}_{\text{row}}(D) = \text{Diag}(\eta(\|D_{1:}\|_2), \dots, \eta(\|D_{v:}\|_2))D \tag{4}$$

are LPRO-equivariant for any scalar function $\eta: \mathbb{R}_+ \rightarrow \mathbb{R}$, because left multiplication by a permutation matrix permutes the row norms and right orthogonal transformations preserve them. We denote the set of such maps by $\mathcal{U}_{\text{row}}^{v \times d}$.

For LM heads, the horizontal version is

$$\mathcal{U}_{\text{row}}^{\text{LM}}(D) = \Pi_v^\perp \text{Diag}(\eta(\|D_{c,1:}\|_2), \dots, \eta(\|D_{c,v:}\|_2))D_c, \quad D_c = \Pi_v^\perp D. \tag{5}$$

This map is permutation equivariant and right-orthogonal equivariant on the horizontal quotient space, and it satisfies

$$\mathbf{1}_v^\top \mathcal{U}_{\text{row}}^{\text{LM}}(D) = 0.$$

The final projection in (5) is essential for row-norm updates, because row-dependent rescaling does not preserve the zero-row-mean constraint.

Thus, right-spectral maps form a global Gram-based subclass, while row-norm maps form a local row-adaptive subclass:

$$\mathcal{U}_{\text{R}}^{v \times d} \subset \mathcal{U}_{\text{LPRO}}^{v \times d}, \quad \mathcal{U}_{\text{row}}^{v \times d} \subset \mathcal{U}_{\text{LPRO}}^{v \times d}.$$

Hybrid LPRO maps are obtained by composing these two types of maps. For LM heads, hybrid maps are obtained by the same compositions, but with horizontal projections inserted after rowwise nonlinear stages and on the final update. For example, a row-then-right-spectral horizontal hybrid update may be written as

$$D_0 = \Pi_v^\perp D, \quad D_1 = \Pi_v^\perp \text{Diag}(\eta(\|D_{0,1}\|_2), \dots, \eta(\|D_{0,v}\|_2)) D_0,$$

$$\mathcal{U}_{\text{hyb}}^{\text{LM}}(D) = \Pi_v^\perp D_1 (D_1^\top D_1 + \varepsilon I)^{-1/2}.$$

Proposition 3.3 (Closure under composition). *If $\mathcal{U}_1, \mathcal{U}_2: \mathbb{R}^{v \times d} \rightarrow \mathbb{R}^{v \times d}$ are both left-permutation and right-orthogonal equivariant, then $\mathcal{U}_2 \circ \mathcal{U}_1$ is also left-permutation and right-orthogonal equivariant.*

Definition 3.3 (Hybrid LPRO-equivariant maps). A map $\mathcal{U}$ is called a *hybrid LPRO-equivariant map* if it is a finite composition of right-spectral maps and row-norm maps. We denote the set of such maps by $\mathcal{U}_{\text{hyb}}^{v \times d}$.

Accordingly, LPRO-compatible optimizers for embeddings are obtained by applying maps in $\mathcal{U}_{\text{R}}^{v \times d}$, $\mathcal{U}_{\text{row}}^{v \times d}$, or $\mathcal{U}_{\text{hyb}}^{v \times d}$ to update directions that transform in the same way as the gradient, such as momentum buffers formed from past gradients. For untied LM heads, the corresponding optimizer should additionally be horizontal with respect to the shared-logit-shift quotient, which is enforced by the projections $\Pi_v^\perp$ above.

Remark 3.2. Unlike the bi-orthogonal case, LPRO equivariance does not characterize a single spectral form. Right-spectral updates are a canonical global subclass, but row-norm and hybrid maps are also symmetry-compatible. This distinction is important: embeddings and LM heads have a discrete vocabulary symmetry on the left, not a full continuous orthogonal symmetry, so row-aware operations are allowed by the layer geometry. For LM heads, softmax shift invariance imposes an additional horizontal constraint; the actual update should be projected back to the quotient space whenever rowwise nonlinear operations are used.

Examples. Several recent optimizers can be interpreted through this framework. SCALE [53] applies column normalization to the EMA momentum for LM heads; under our row-vocabulary convention, this corresponds to a row-norm-based update. Other row- or column-norm-based optimizers include RMNP [37], REG [109] and NORA [169]. Finally, NORMUON [101], MUON+ [174], and MUONEQ [25], which apply row-wise and/or column-wise normalization to the orthogonal polar factor of the EMA momentum, can be viewed as hybrid spectral/row-norm optimizers.

3.4 Optimizers for SwiGLU MLP Projections

We next consider SwiGLU MLP projection matrices [32, 137]. Unlike ordinary linear and attention projection matrices, SwiGLU projections do not possess full bi-orthogonal symmetry. Instead, their natural symmetry is the permutation symmetry of intermediate neurons. This suggests that the gate and up projections should use left-permutation/right-orthogonal equivariant updates, while the down projection should use the corresponding transposed geometry. Concretely, for $W_{\text{gate}}, W_{\text{up}} \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$ and $W_{\text{down}} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}$, we apply the LPRO-compatible optimizer classes of Section 3.3 to W_{gate} , W_{up} , and $W_{\text{down}}^\top$.

This viewpoint is closely related to Aurora [38], an optimizer designed for the up and gate projections in SwiGLU MLPs. Aurora alternates between row-normalization and polar steps on the momentum, which is similar in spirit to the hybrid row-norm/right-spectral optimizers developed above. Our contribution here is to derive this type of update from the intermediate-neuron permutation symmetry of the SwiGLU block.

Proposition 3.4 (Intermediate-neuron permutation symmetry of SwiGLU MLPs). *Let us consider the SwiGLU MLP block defined by*

$$\text{SwiGLU}(x; W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}) := W_{\text{down}}(\sigma(W_{\text{gate}}x) \odot (W_{\text{up}}x)),$$

where $W_{\text{gate}}, W_{\text{up}} \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$, $W_{\text{down}} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}$, σ is applied coordinatewise, and $\odot$ denotes the Hadamard product. Let $P \in \mathbb{P}^{d_{\text{ff}}}$ be a permutation matrix, and define $\widetilde{W}_{\text{gate}} := PW_{\text{gate}}$, $\widetilde{W}_{\text{up}} := PW_{\text{up}}$, and $\widetilde{W}_{\text{down}} := W_{\text{down}}P^\top$. Then, for every input $x \in \mathbb{R}^{d_{\text{model}}}$,

$$\text{SwiGLU}(x; \widetilde{W}_{\text{gate}}, \widetilde{W}_{\text{up}}, \widetilde{W}_{\text{down}}) = \text{SwiGLU}(x; W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}).$$

Remark 3.3 (Failure of full left-orthogonal symmetry). The permutation matrix P in Proposition 3.4 cannot generally be replaced by an arbitrary orthogonal matrix $Q \in \mathbb{O}^{d_{\text{ff}}}$. For an elementwise nonlinearity σ , one typically has $\sigma(Qz) \neq Q\sigma(z)$, and similarly $(Qa) \odot (Qb) \neq Q(a \odot b)$ for a general orthogonal matrix Q . Thus, the SwiGLU intermediate dimension has a discrete permutation symmetry, not a full left-orthogonal symmetry.

Proposition 3.4 implies that optimizers for W_{gate} and W_{up} should commute with permutations of their rows and orthogonal changes of basis in the model dimension. Thus, the same LPRO-compatible row-norm, right-spectral, and hybrid row-norm/right-spectral updates developed for embeddings and LM heads apply directly to these matrices. For the down projection, the intermediate-neuron axis appears as the column dimension, so the same principle applies to $W_{\text{down}}^\top$. Equivalently, one may view the down-projection update as a right-permutation/left-orthogonal analogue of the LPRO updates.

More generally, this intermediate-neuron permutation symmetry is not specific to SwiGLU. Any feed-forward block whose hidden nonlinearity is applied coordinatewise admits a permutation symmetry of the hidden units: simultaneously permuting the rows of the input projection and the corresponding columns of the output projection leaves the represented function unchanged. For gated MLPs such as GLU, GeGLU, ReGLU, and SwiGLU, the same symmetry holds provided the gate and value projections are permuted together, along with the corresponding columns of the down projection.

3.5 Optimizers for MoE Routers

We now consider the router matrix in a mixture-of-experts (MoE) model. Unlike ordinary linear layers, embeddings, and LM heads, the router has an additional symmetry: experts are interchangeable, and the softmax is invariant under adding a shared scalar offset to all expert logits.

Let $W \in \mathbb{R}^{e \times d}$ denote the router matrix, where $e \in \mathbb{N}^*$ is the number of experts and $d \in \mathbb{N}^*$ is the hidden dimension. The routing distribution is $p(x; W) = \text{softmax}(Wx)$ for $x \in \mathbb{R}^d$. Since permuting the rows of W only relabels the experts, and since $\text{softmax}(z + c\mathbf{1}_e) = \text{softmax}(z)$ for all $z \in \mathbb{R}^e$ and $c \in \mathbb{R}$, the router parameters admit the symmetry

$$(\forall P \in \mathbb{P}^e, \forall a \in \mathbb{R}^d) \quad W \mapsto PW + \mathbf{1}_e a^\top.$$

Indeed,

$$(\forall x \in \mathbb{R}^d) \quad (PW + \mathbf{1}_e a^\top)x = P(Wx) + (a^\top x)\mathbf{1}_e,$$

so the router logits are unchanged up to expert relabeling and a shared logit shift.

This symmetry suggests that router updates should be defined on the centered expert geometry. Let

$$\Pi_\perp := I_e - \frac{1}{e}\mathbf{1}_e\mathbf{1}_e^\top$$

be the orthogonal projector onto $\mathbf{1}_e^\perp$, and define $D_c := \Pi_\perp D$.

The centered direction D_c removes the shared-row component of D and captures the intrinsic variation across experts. This motivates router update maps built from the centered direction, for example through the centered left Gram matrix

$$D_c D_c^\top = \Pi_\perp D D^\top \Pi_\perp.$$

Definition 3.4 (Router-compatible update maps). A map $\mathcal{U}: \mathbb{R}^{e \times d} \rightarrow \mathbb{R}^{e \times d}$ is called *router-compatible* if it is expert-permutation equivariant, shared-row-shift invariant, and horizontal, namely,

$$\mathcal{U}(PD + \mathbf{1}_e a^\top) = P\mathcal{U}(D), \quad \mathbf{1}_e^\top \mathcal{U}(D) = 0,$$

for all $D \in \mathbb{R}^{e \times d}$, all permutation matrices $P \in \mathbb{P}^e$, and all $a \in \mathbb{R}^d$.

The horizontality condition reflects the quotient geometry induced by softmax shift invariance. An update component in the shared-row direction $\mathbf{1}_e a^\top$ changes all expert logits by the same scalar and therefore does not affect the router probabilities. A geometry-compatible router update should therefore lie in the horizontal subspace

$$\{U \in \mathbb{R}^{e \times d} : \mathbf{1}_e^\top U = 0\}.$$

We consider two basic router-compatible update families. The first is a left-spectral update in the centered expert geometry:

$$\mathcal{U}_L^{\text{router}}(D) = \Psi(D_c D_c^\top) D_c, \quad D_c = \Pi_\perp D,$$

where $\Psi: \mathbb{S}_+^e \rightarrow \mathbb{R}^{e \times e}$ is permutation equivariant and preserves the centered expert subspace. For example, the damped centered left-polar update corresponds to

$$\Psi(X) = (X + \varepsilon \Pi_\perp)^{-1/2}$$

on $\mathbf{1}_e^\perp$, or equivalently to applying the inverse square root on the nonzero centered expert subspace. Since left multiplication by a matrix acting on the centered expert space preserves $\mathbf{1}_e^\perp$, this update is horizontal:

$$\mathbf{1}_e^\top \mathcal{U}_L^{\text{router}}(D) = 0.$$

The second family is a centered row-norm update. Importantly, for row-norm maps it is not sufficient to center the input direction once. Although $D_c = \Pi_\perp D$ satisfies $\mathbf{1}_e^\top D_c = 0$, the rowwise-rescaled matrix

$$\text{Diag}(\eta(\|D_{c,1:}\|_2), \dots, \eta(\|D_{c,e:}\|_2))D_c$$

need not remain centered, because the scaling factors vary across expert rows. Therefore the router-compatible row-norm update is

$$\mathcal{U}_{\text{row}}^{\text{router}}(D) = \Pi_\perp \text{Diag}(\eta(\|D_{c,1:}\|_2), \dots, \eta(\|D_{c,e:}\|_2))D_c, \quad D_c = \Pi_\perp D,$$

where $\eta: \mathbb{R}_+ \rightarrow \mathbb{R}$ is applied pointwise to the centered expert-row norms. The final projection is essential: it ensures that the actual parameter update, not only the input to the row-norm map, lies in the horizontal router subspace.

Proposition 3.5 (Router-compatible update families). *Assume that Ψ is permutation equivariant and maps the centered expert subspace to itself. Then the centered left-spectral update*

$$\mathcal{U}_{\text{L}}^{\text{router}}(D) = \Psi(D_c D_c^\top)D_c, \quad D_c = \Pi_\perp D,$$

is router-compatible. Moreover, for any scalar function $\eta: \mathbb{R}_+ \rightarrow \mathbb{R}$, the projected centered row-norm update

$$\mathcal{U}_{\text{row}}^{\text{router}}(D) = \Pi_\perp \text{Diag}(\eta(\|D_{c,1:}\|_2), \dots, \eta(\|D_{c,e:}\|_2))D_c, \quad D_c = \Pi_\perp D,$$

is router-compatible. In particular, for all $D \in \mathbb{R}^{e \times d}$, $P \in \mathbb{P}^e$, and $a \in \mathbb{R}^d$,

$$\mathcal{U}_{\text{L}}^{\text{router}}(PD + \mathbf{1}_e a^\top) = P \mathcal{U}_{\text{L}}^{\text{router}}(D), \quad \mathcal{U}_{\text{row}}^{\text{router}}(PD + \mathbf{1}_e a^\top) = P \mathcal{U}_{\text{row}}^{\text{router}}(D),$$

and both updates satisfy

$$\mathbf{1}_e^\top \mathcal{U}_{\text{L}}^{\text{router}}(D) = 0, \quad \mathbf{1}_e^\top \mathcal{U}_{\text{row}}^{\text{router}}(D) = 0.$$

The converse is false in general: router compatibility does not force an update map to be left-spectral. Projected centered row-norm updates are already router-compatible, but they depend on individual centered expert-row norms rather than only on the centered Gram matrix $D_c D_c^\top$. Thus, left-spectral and row-norm updates should be viewed as two natural subclasses of router-compatible maps.

Hybrid router maps are obtained by composing router-compatible maps and projecting the final update back to the horizontal subspace. If $\mathcal{U}_1$ and $\mathcal{U}_2$ are router-compatible, then the horizontally projected composition

$$\Pi_\perp (\mathcal{U}_2 \circ \mathcal{U}_1)$$

is also router-compatible. Therefore, finite compositions of centered left-spectral and projected centered row-norm updates remain router-compatible. A representative row-after-left-spectral hybrid router update is

$$Z = \Psi(D_c D_c^\top)D_c, \quad D_c = \Pi_\perp D,$$

$$\mathcal{U}_{\text{hyb}}^{\text{router}}(D) = \Pi_\perp \text{Diag}(\eta(\|Z_{1:}\|_2), \dots, \eta(\|Z_{e:}\|_2))Z.$$

Such hybrid router optimizers combine the global expert-mixing geometry of left-spectral updates with the local expert-wise normalization of row-norm updates, while preserving expert-permutation equivariance, shared-row-shift invariance, and horizontality of the actual update.

Layer	Symmetry group	Optimizer classes
Linear / MLP / attention weights	$\mathbb{O}^{d_{\text{out}}} \times \mathbb{O}^{d_{\text{in}}}$	full spectral
Embedding	$\mathbb{P}^v \times \mathbb{O}^d$	LPRO: row-norm / right-spectral / hybrid
LM head	$\mathbb{P}^v \times \mathbb{O}^d$	LPRO: row-norm / right-spectral / hybrid
	$\mathbb{R}^{v \times d} / (\mathbf{1}_v \mathbb{R}^{1 \times d})$	projected updates
SwiGLU MLP ($W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}^\top$)	$\mathbb{P}^{d_{\text{ff}}} \times \mathbb{O}^d$	LPRO: row-norm / right-spectral / hybrid
MoE router	$\mathbb{P}^e \times (\mathbf{1}_e \mathbb{R}^{1 \times d})$	centered row-norm / left-spectral / hybrid
	$\mathbb{R}^{e \times d} / (\mathbf{1}_e \mathbb{R}^{1 \times d})$	projected updates

Table 1: **Optimizer classes across LLM layers induced by symmetry.** Each matrix parameter has a natural symmetry group, which determines the corresponding symmetry-compatible optimizer class. For LM heads and MoE routers, softmax shared-logit-shift invariance further induces quotient geometries; projected updates remove the shared-row direction so that the actual update acts on the non-redundant degrees of freedom.

Definition 3.5 (Router-compatible optimizers). A matrix optimizer for an MoE router is called *router-compatible* if its update rule has the form

$$(\forall k \in \mathbb{N}) \quad W_{k+1} = W_k - \gamma_k \mathcal{U}(D_k),$$

where $\mathcal{U}$ is router-compatible and $D_k \in \mathbb{R}^{e \times d}$ is an update direction that transforms under expert permutations and shared row shifts in the same way as the gradient.

Remark 3.4 (Why the final projection matters). The final projection in the row-norm and hybrid router updates is not a numerical detail. Centering the direction $D_c = \Pi_\perp D$ removes the shared-row component before computing row norms, but row-dependent rescaling generally reintroduces a shared-row component. The final projection $\Pi_\perp$ removes this component and ensures that the update is horizontal. This is the router analogue of the horizontal update condition for untied LM heads under softmax shared-logit-shift invariance.

Remark 3.5. Left- and right-spectral optimizers bear some formal resemblance to one-sided SHAMPOO [158, 98] and ASGO [6]. A key difference is that our framework applies the spectral transformation directly to the current update direction, or to a symmetry-compatible momentum direction, rather than maintaining moving averages of Gram matrices. More importantly, our design principle identifies which layer types such one-sided updates are appropriate for, according to the symmetry group of the corresponding parameter block.

Remark 3.6 (Connection to non-Euclidean norm-based steepest descent). This symmetry-based construction differs from non-Euclidean steepest descent approaches based on choosing a matrix norm [13, 152, 162]. Row-wise or column-wise normalizations generally preserve only one-sided or permutation symmetries, rather than full bi-orthogonal symmetry. Thus, while such geometries may be useful in layer-specific settings, they should be distinguished from fully spectral constructions for ordinary matrix layers.

3.6 Symmetry-to-Optimizer Principle and Architecture–Optimizer Co-Design

The preceding developments suggest a unifying principle for optimizer design in modern deep learning architectures: the optimizer geometry should be determined by the symmetry structure of the underlying parameterization. This does not require the full layerwise loss to be globally invariant under every group action considered. Rather, it is enough that the gradient, or more

generally the chosen update direction, transforms equivariantly under the relevant representation of the parameter block. The optimizer update should then transform in the same way, and should lie in the corresponding quotient or horizontal subspace whenever the parameterization contains symmetry-redundant directions.

Accordingly, whenever the update direction associated with a layer parameter W transforms under a symmetry group $\mathcal{G}$, a natural optimizer should use an update map $\mathcal{U}$ that is equivariant under the same induced action. For matrix-valued parameters, this principle leads to several canonical optimizer geometries: full spectral updates under bi-orthogonal symmetry; right-spectral updates under right-orthogonal symmetry; left-spectral updates under left-orthogonal symmetry; row-norm and hybrid updates under left-permutation/right-orthogonal symmetry; and projected centered left-spectral or row-norm updates under the expert-permutation and shared-row-shift quotient geometry of MoE routers. For LM heads, the same softmax shared-logit-shift invariance induces a horizontal quotient constraint on the vocabulary rows. Thus, the symmetry structure of the layer determines not only the appropriate optimizer class, but also whether the final update should be projected to remove redundant directions.

This principle also gives a practical recipe for architecture–optimizer co-design. Given a new architecture block, one should:

1. identify the symmetry group of the parameterization;
2. determine whether the symmetry acts on the left, on the right, on both sides, or only after quotienting out symmetry-redundant directions;
3. choose the matching symmetry-compatible optimizer class;
4. project the update onto the horizontal subspace when the parameterization has redundant shift directions; and
5. use the smallest Gram matrix or invariant statistic compatible with that symmetry.

In Section 3.7, we instantiate this recipe by extending momentum POLARGRAD [91] to one-sided and hybrid variants, including LEFTPOLARGRADM, RIGHTPOLARGRADM, ROWNORMM, and HYBRIDPOLARGRADM.

Remark 3.7 (Projected and proximal extensions). The symmetry-compatible optimizer classes above extend naturally to projected and proximal updates. If $\mathcal{T}(D_k)$ is $\mathcal{G}$ -equivariant and the constraint set $\mathcal{C}$ is $\mathcal{G}$ -invariant, then the projected update

$$W_{k+1} = \text{proj}_{\mathcal{C}}(W_k - \gamma_k \mathcal{T}(D_k))$$

is symmetry-compatible, since the Euclidean projection onto a $\mathcal{G}$ -invariant set is $\mathcal{G}$ -equivariant whenever it is uniquely defined. Similarly, if h is a $\mathcal{G}$ -invariant regularizer, then

$$W_{k+1} = \text{prox}_{\gamma_k h}(W_k - \gamma_k \mathcal{T}(D_k))$$

preserves the same symmetry whenever the proximal map is uniquely defined. In the bi-orthogonal setting, this includes unitarily invariant constraints and regularizers, such as spectral-norm, nuclear-norm, rank, and Schatten- p constraints or penalties. We leave a systematic study of such variants, including PROXPOLARGRAD, to future work.

3.7 Practical Optimizers for Embeddings, LM Heads, SwiGLU MLP Projections, and MoE Routers

The preceding subsections introduced three main classes of symmetry-compatible optimizers: one-sided spectral optimizers, row-norm-based optimizers, and hybrid variants obtained by composing row-wise normalization with one-sided spectral updates. We now describe their practical momentum implementations. The main computational issue is the efficient and stable approximation of matrix inverse square roots, or equivalently orthogonal polar factors, which we compute using GPU-friendly numerical linear algebra routines. In addition, for untied LM heads and MoE routers, the practical updates must respect the horizontal quotient geometry induced by softmax shared-logit-shift invariance. Thus, when row-wise nonlinear operations are used for these blocks, we project the final update back to the corresponding centered subspace.

3.7.1 One-Sided Spectral Optimizers

For embedding, LM head, and SwiGLU MLP projection matrices, the relevant matrices are often tall-skinny. In this regime, right-spectral updates are attractive because they only require the inverse square root of the smaller right Gram matrix

$$C_k := G_k^\top G_k \quad \text{or, with momentum,} \quad C_k := M_k^\top M_k.$$

For an untied LM head, we use the centered momentum direction

$$\Pi_v^\perp := I_v - \frac{1}{v} \mathbf{1}_v \mathbf{1}_v^\top, \quad M_{k,c} := \Pi_v^\perp M_k,$$

so that the update removes the shared vocabulary-logit-shift direction. The corresponding right Gram matrix is

$$C_k := M_{k,c}^\top M_{k,c}.$$

For MoE routers, the corresponding left-spectral update is applied in the centered expert geometry. Writing

$$\Pi_\perp := I_e - \frac{1}{e} \mathbf{1}_e \mathbf{1}_e^\top, \quad M_{k,c} := \Pi_\perp M_k,$$

the relevant Gram matrix is

$$C_k := M_{k,c} M_{k,c}^\top.$$

To compute the inverse square roots in these updates, we use Newton–Schulz iterations with the polynomial coefficients of POLAR EXPRESS [5]. This is motivated by the connection between polynomial iterations for polar decomposition and inverse-square-root computation [67]. For numerical stability, the Gram inverse-square-root implementation is performed in `float32`. Other fast inverse-square-root routines, such as PRISM [165], could be used in the same role. For RIGHTPOLARGRADM, we also provide a Gram Newton–Schulz implementation [172], which directly approximates

$$M_k (M_k^\top M_k)^{-1/2}$$

and supports more inner iterations in lower-precision formats such as `bfloat16` or `float16`. The resulting one-sided momentum polar-gradient algorithms are summarized in Algorithm 1.

At the level of exact polar decomposition, LEFTPOLARGRADM and RIGHTPOLARGRADM compute the same polar direction whenever both sides are well-defined. Their distinction is therefore computational and architectural: they differ in which Gram matrix is formed, which inverse square root is computed, and which layer symmetry they are intended to respect. For LM heads and MoE routers, the one-sided spectral updates are applied to centered momentum directions when the corresponding softmax shared-logit-shift quotient geometry is present.

Algorithm 1 LEFTPOLARGRADM and RIGHTPOLARGRADM

Input: $W_0 \in \mathbb{R}^{m \times n}$, $M_{-1} = 0$, learning rates $\{\gamma_k\}_{k \geq 0}$, momentum $\beta \in [0, 1]$, scaling exponent $\alpha \in [0, 1]$, damping $\varepsilon > 0$, weight decay $\lambda \geq 0$

```
1: for  $k = 0, \dots, K - 1$  do
2:    $G_k = \nabla_W f(W_k)$ 
3:    $M_k = \beta M_{k-1} + (1 - \beta)G_k$ 
4:   if LEFTPOLARGRADM then
5:      $C_k = M_k M_k^\top$ 
6:      $L_k = (C_k + \varepsilon I)^{-1/2}$  via POLAR EXPRESS or Newton–Schulz iteration
7:      $\nu_k = \max\{\text{tr}(C_k L_k), \varepsilon\}$ 
8:      $W_{k+1} = (1 - \gamma_k \lambda)W_k - \gamma_k \nu_k^\alpha L_k M_k$ 
9:   else if RIGHTPOLARGRADM then
10:     $C_k = M_k^\top M_k$ 
11:     $R_k = (C_k + \varepsilon I)^{-1/2}$  via POLAR EXPRESS or Newton–Schulz iteration
12:     $\nu_k = \max\{\text{tr}(C_k R_k), \varepsilon\}$ 
13:     $W_{k+1} = (1 - \gamma_k \lambda)W_k - \gamma_k \nu_k^\alpha M_k R_k$ 
14:   end if
15: end for
Return:  $W_K$ 
```

3.7.2 Row-Norm-Based and Hybrid Variants

For embedding, LM head, and SwiGLU MLP projection matrices, row-norm-based updates provide a cheaper LPRO-compatible alternative. Given a momentum direction M_k , define

$$D_\eta(M_k) := \text{Diag}(\eta(\|M_{k,1}\|_2), \dots, \eta(\|M_{k,m}\|_2)).$$

For embeddings and SwiGLU gate/up projections, a row-norm update takes the form

$$\mathcal{T}_{\text{row}}(M_k) = D_\eta(M_k)M_k.$$

Here η may be chosen as a bounded row-scaling rule or as a smoothed normalization rule such as $\eta(t) = 1/(t + \varepsilon)$.

For untied LM heads, the update must also remove the shared vocabulary-logit-shift direction. Let

$$\Pi_v^\perp := I_v - \frac{1}{v} \mathbf{1}_v \mathbf{1}_v^\top, \quad M_{k,c} := \Pi_v^\perp M_k.$$

The horizontal LM head row-norm update is

$$\mathcal{T}_{\text{row}}^{\text{LM}}(M_k) = \Pi_v^\perp D_\eta(M_{k,c})M_{k,c}.$$

The final projection is essential for row-norm updates: even though $\mathbf{1}_v^\top M_{k,c} = 0$, row-dependent rescaling by $D_\eta(M_{k,c})$ need not preserve the zero-row-mean constraint.

Hybrid variants combine row-wise scaling with a one-sided spectral step. For embeddings and SwiGLU gate/up projections, there are two natural orders:

$$\text{right-spectral/row-norm:} \quad Z_k = M_k(M_k^\top M_k + \varepsilon I)^{-1/2}, \quad \mathcal{T}_{\text{hyb}}(M_k) = D_\eta(Z_k)Z_k,$$

and

$$\text{row-norm/right-spectral:} \quad Z_k = D_\eta(M_k)M_k, \quad \mathcal{T}_{\text{hyb}}(M_k) = Z_k(Z_k^\top Z_k + \varepsilon I)^{-1/2}.$$

Both remain left-permutation and right-orthogonal equivariant. For LM heads, the same constructions are applied to the centered momentum $M_{k,c} = \Pi_v^\perp M_k$, with horizontal projections inserted after rowwise nonlinear stages and on the final update. For example, the row-norm/right-spectral order becomes

$$Z_k = \Pi_v^\perp D_\eta(M_{k,c})M_{k,c}, \quad \mathcal{T}_{\text{hyb}}^{\text{LM}}(M_k) = \Pi_v^\perp Z_k (Z_k^\top Z_k + \varepsilon I)^{-1/2}.$$

For MoE routers, the same principle applies in the centered expert geometry. Writing

$$\Pi_\perp := I_e - \frac{1}{e} \mathbf{1}_e \mathbf{1}_e^\top, \quad M_{k,c} = \Pi_\perp M_k,$$

the router row-norm update is

$$\mathcal{T}_{\text{row}}^{\text{router}}(M_k) = \Pi_\perp D_\eta(M_{k,c})M_{k,c}.$$

Hybrid router updates combine a centered left-spectral step with projected row-wise normalization across experts. We refer to these practical hybrid variants collectively as HYBRIDPOLARGRADM. The algorithms for ROWNORMM and HYBRIDPOLARGRADM are summarized in Algorithm 2.

For down projections in SwiGLU MLPs, we apply the same procedure to $W_{\text{down}}^\top$ and transpose the resulting update back. For routers, the hybrid update uses the left-spectral branch; for embeddings, LM heads, and SwiGLU gate/up projections, it uses the right-spectral branches.

The practical distinction among these variants is both geometric and computational. RIGHTPOLARGRADM preserves the right-orthogonal geometry through a Gram inverse square root, while ROWNORMM is a purely row-adaptive LPRO-compatible alternative. HYBRIDPOLARGRADM interpolates between the two by composing row normalization and one-sided spectral normalization. For LM heads and MoE routers, softmax shared-logit-shift invariance imposes an additional quotient geometry; therefore, row-norm and hybrid updates are projected back to the horizontal subspace so that the actual update removes the shared-row direction. For MoE routers, the same alternatives appear in the centered expert geometry: one may use a projected centered row-norm update, a centered left-spectral update, or a projected hybrid left-spectral/row-norm update.

Remark 3.8 (Projection and proximal extensions). The practical optimizer families described above can be combined with projection or proximal steps. For example, given a regularizer h or a feasible set $\mathcal{C}$, one may consider

$$W_{k+1} = \text{prox}_{\gamma_k h}(W_k - \gamma_k \mathcal{T}(D_k)) \quad \text{or} \quad W_{k+1} = \text{proj}_{\mathcal{C}}(W_k - \gamma_k \mathcal{T}(D_k)),$$

where $\mathcal{T}(D_k)$ is any symmetry-compatible update direction from the preceding constructions. If h or $\mathcal{C}$ is invariant under the same layer symmetry, then the resulting proximal or projected update preserves the same equivariance. Thus, the geometry-aware optimizer classes introduced here are compatible with standard regularization and constrained-optimization extensions.

3.8 Spectral Optimizers as the Bi-Orthogonally Equivariant Class

We have used bi-orthogonal equivariance as the symmetry principle for ordinary matrix layers. We now record the corresponding structural characterization: direction-wise update maps satisfying this equivariance are precisely spectral operators. This explains why SSD [23], MUON [78], SCION [124], POLARGRAD [91], and related spectral updates form the canonical optimizer class for ordinary matrix layers. The preceding sections show how different symmetry groups lead instead to different optimizer classes for embeddings, LM heads, SwiGLU MLP projections, and MoE routers.

Algorithm 2 ROWNORMM and HYBRIDPOLARGRADM with Optional Horizontal Projection

Input: W_0 , $M_{-1} = 0$, learning rates $\{\gamma_k\}_{k \geq 0}$, momentum $\beta \in [0, 1)$, scaling exponent $\alpha \in [0, 1]$, row-scaling rule η , damping $\varepsilon > 0$, weight decay $\lambda \geq 0$

- 1: **for** $k = 0, \dots, K - 1$ **do**
- 2: $G_k = \nabla_W f(W_k)$
- 3: $M_k = \beta M_{k-1} + (1 - \beta)G_k$
- 4: Set the working direction $\overline{M}_k$ and final projector Π by layer type:

$$(\overline{M}_k, \Pi) = \begin{cases} (M_k, I), & \text{embedding or SwiGLU gate/up,} \\ (\Pi_v^\perp M_k, \Pi_v^\perp), & \text{LM head,} \\ (\Pi_e^\perp M_k, \Pi_e^\perp), & \text{MoE router,} \end{cases}$$

where

$$\Pi_v^\perp = I_v - \frac{1}{v} \mathbf{1}_v \mathbf{1}_v^\top, \quad \Pi_e^\perp = I_e - \frac{1}{e} \mathbf{1}_e \mathbf{1}_e^\top.$$

- 5: **if** ROWNORMM **then**
 - 6: $D_k = \text{Diag}(\eta(\|\overline{M}_{k,1}\|_2), \dots, \eta(\|\overline{M}_{k,m}\|_2))$
 - 7: $U_k = \Pi D_k \overline{M}_k$
 - 8: **else if** HYBRIDPOLARGRADM, right-spectral/row-norm order **then**
 - 9: $C_k = \overline{M}_k^\top \overline{M}_k$
 - 10: $R_k = (C_k + \varepsilon I)^{-1/2}$ via POLAR EXPRESS or Newton–Schulz iteration
 - 11: $\nu_k = \max\{\text{tr}(C_k R_k), \varepsilon\}$
 - 12: $Z_k = \Pi \nu_k^\alpha \overline{M}_k R_k$
 - 13: $D_k = \text{Diag}(\eta(\|Z_{k,1}\|_2), \dots, \eta(\|Z_{k,m}\|_2))$
 - 14: $U_k = \Pi D_k Z_k$
 - 15: **else if** HYBRIDPOLARGRADM, row-norm/right-spectral order **then**
 - 16: $D_k = \text{Diag}(\eta(\|\overline{M}_{k,1}\|_2), \dots, \eta(\|\overline{M}_{k,m}\|_2))$
 - 17: $Z_k = \Pi D_k \overline{M}_k$
 - 18: $C_k = Z_k^\top Z_k$
 - 19: $R_k = (C_k + \varepsilon I)^{-1/2}$ via POLAR EXPRESS or Newton–Schulz iteration
 - 20: $\nu_k = \max\{\text{tr}(C_k R_k), \varepsilon\}$
 - 21: $U_k = \Pi \nu_k^\alpha Z_k R_k$
 - 22: **else if** MOE router and HYBRIDPOLARGRADM, left-spectral/row-norm order **then**
 - 23: $C_k = \overline{M}_k \overline{M}_k^\top$
 - 24: $L_k = (C_k + \varepsilon I)^{-1/2}$ via POLAR EXPRESS or Newton–Schulz iteration
 - 25: $\nu_k = \max\{\text{tr}(C_k L_k), \varepsilon\}$
 - 26: $Z_k = \Pi \nu_k^\alpha L_k \overline{M}_k$
 - 27: $D_k = \text{Diag}(\eta(\|Z_{k,1}\|_2), \dots, \eta(\|Z_{k,e}\|_2))$
 - 28: $U_k = \Pi D_k Z_k$
 - 29: **end if**
 - 30: $W_{k+1} = (1 - \gamma_k \lambda) W_k - \gamma_k U_k$
 - 31: **end for**
- Return:** W_K
-

We first recall the relevant matrix-analytic notions. A proper function $\varphi: \mathbb{R}^{m \times n} \rightarrow \overline{\mathbb{R}}$ is called *spectral* if there exists a proper function $\psi: \mathbb{R}^r \rightarrow \overline{\mathbb{R}}$, where $r = \min\{m, n\}$, such that $\varphi = \psi \circ \sigma$.

Its matrix-valued analogue is a *spectral operator*: a map $\mathcal{U}: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ is spectral if there exists an absolutely symmetric map $\psi: \mathbb{R}^r \rightarrow \mathbb{R}^r$ such that, for every singular value decomposition $D = U \text{Diag}(\sigma(D))V^\top$,

$$\mathcal{U}(D) = U \text{Diag}(\psi(\sigma(D)))V^\top.$$

Thus, spectral operators preserve the singular-vector geometry of their input and act only on singular values.

We next recall the standard gradient formula for spectral functions; see, e.g., Lewis [94].

Theorem 3.6 (Gradient formula for spectral functions). *Let $f: \mathbb{R}^r \rightarrow \bar{\mathbb{R}}$ be convex and absolutely symmetric. Then the corresponding spectral function $f \circ \sigma$ is differentiable at $W \in \mathbb{R}^{m \times n}$ if and only if f is differentiable at $\sigma(W)$. In this case, if $W = U \text{Diag}(\sigma(W))V^\top$ is a singular value decomposition of W , then*

$$\nabla(f \circ \sigma)(W) = U \text{Diag}(\nabla f(\sigma(W)))V^\top.$$

This formula shows that spectral scalar functions act through singular values while preserving singular directions. The same structure appears when one requires an update map to commute with arbitrary left and right orthogonal changes of coordinates. We now state the corresponding characterization.

Theorem 3.7 (Characterization of bi-orthogonally equivariant update maps). *A continuous matrix-valued map $\mathcal{U}: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ satisfies*

$$\mathcal{U}(PDQ^\top) = P \mathcal{U}(D)Q^\top$$

for all $D \in \mathbb{R}^{m \times n}$, $P \in \mathbb{O}^m$, and $Q \in \mathbb{O}^n$ if and only if it is a spectral operator. Equivalently, $\mathcal{U}$ is bi-orthogonally equivariant if and only if, for every singular value decomposition $D = U \text{Diag}(\sigma(D))V^\top$, there exists an absolutely symmetric map $\psi: \mathbb{R}^r \rightarrow \mathbb{R}^r$ such that

$$\mathcal{U}(D) = U \text{Diag}(\psi(\sigma(D)))V^\top, \quad r = \min\{m, n\}.$$

We denote the set of bi-orthogonally equivariant matrix maps from $\mathbb{R}^{m \times n}$ to $\mathbb{R}^{m \times n}$ by $\mathcal{U}_0^{m \times n}$. Theorem 3.7 shows that bi-orthogonal equivariance is not merely a desirable property: it is a complete structural characterization of direction-wise matrix update maps. Any such update must act through the singular values of the update direction and preserve its singular vectors.

Remark 3.9. Theorem 3.7 concerns equivariance of the optimizer update map with respect to transformations of the matrix gradient or update direction. It does not assume that the layerwise loss itself satisfies $f(PWQ^\top) = f(W)$ for every individual layer parameter W . The claim is instead that, whenever a matrix gradient or momentum direction is represented in a rotated basis, a symmetry-compatible optimizer should transform its update in the same way.

Accordingly, we call a matrix optimizer *spectral* if its update rule is bi-orthogonally equivariant. Concretely, its iterates satisfy

$$W_{k+1} = W_k - \gamma_k \mathcal{U}(D_k),$$

where $\gamma_k > 0$, $\mathcal{U} \in \mathcal{U}_0^{m \times n}$, and D_k is an update direction that itself transforms bi-orthogonally, such as the gradient or a symmetry-compatible momentum direction. By Theorem 3.7, every such direction-wise spectral optimizer is determined by a singular-value transformation ψ .

This characterization applies to maps acting on a single update direction. It does not exclude broader stateful matrix optimizers such as SHAMPOO, whose auxiliary states may themselves evolve equivariantly. Spectral optimizers should therefore be understood as the bi-orthogonally equivariant class of memoryless, or direction-wise, matrix update maps; stateful equivariant optimizers form a larger class.

3.8.1 Examples of Spectral and Equivariant Matrix Optimizers

The simplest spectral optimizer is vanilla gradient descent, for which the spectral operator is the identity map. Its Polyak, Nesterov, and EMA-momentum variants remain spectral because bi-orthogonally equivariant momentum constructions composed with spectral operators preserve spectrality. Other examples include stochastic spectral descent [21, 23], MUON [78], SCION [124], and POLARGRAD [91]. Power-type spectral maps $\psi(t) = t^p$, including $p \in \{1/2, 1/4\}$, are also studied in [128].

It is useful to distinguish spectral operators acting on the current update direction from history-dependent optimizers whose states evolve equivariantly. For example, SHAMPOO [64, 7] maintains left and right preconditioners based on moving averages of $G_k G_k^\top$ and $G_k^\top G_k$, and applies

$$L_k^{-1/4} G_k R_k^{-1/4}.$$

This update is bi-orthogonally equivariant when the state variables (L_k, R_k) transform accordingly, but it is not generally a spectral operator of the current gradient alone, since the preconditioners need not share the singular-vector basis of G_k . In special aligned cases, it reduces to a singular-value transformation and hence becomes spectral. One-sided SHAMPOO [158, 98] and ASGO [6] admit a similar interpretation.

SOAP [153] is also geometry-aware but generally not spectral in the strict direction-wise sense. It rotates gradients into learned eigenspaces and applies coordinate-wise adaptive scaling in those evolving coordinates. Thus, the update depends on history-dependent bases and coordinate-wise statistics, not only on the singular values of the current gradient. It becomes spectral only in special cases where the learned eigenspaces align with the singular-vector basis and the adaptive scaling acts symmetrically across singular directions.

Normalization preserves spectrality when the normalizing scalar is unitarily invariant. If $\mathcal{U}$ is spectral and $\alpha: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}_+$ is a positive spectral scalar function, then $D \mapsto \mathcal{U}(D)/\alpha(D)$ is again spectral. In particular, normalization by a unitarily invariant matrix norm preserves spectrality. By contrast, row-wise or column-wise normalization generally breaks full bi-orthogonal equivariance because it depends on a preferred coordinate system. Such normalizations may still be appropriate for layers with one-sided or permutation symmetries, but they are not spectral operators for ordinary matrix layers.

More broadly, steepest descent for matrix-valued parameters is compatible with bi-orthogonal symmetry when the underlying norm is unitarily invariant. The unit ball of a unitarily invariant norm is invariant under $D \mapsto PDQ^\top$, so the associated steepest descent direction transforms equivariantly under left and right orthogonal changes of coordinates. Non-unitarily invariant norms, such as general induced $\ell_p \rightarrow \ell_q$ operator norms with $(p, q) \neq (2, 2)$, generally fail this property and impose a coordinate-dependent geometry on the update.

4 Numerical Experiments

Our experiments test the layerwise equivariance principle in full language model pre-training. Rather than changing a single optimizer in isolation, we instantiate a symmetry-compatible optimizer stack in which each major matrix-valued parameter class is assigned an update whose equivariance matches its layerwise symmetry: attention matrices use spectral or head-wise spectral updates; embeddings and LM heads use row-norm or hybrid updates; dense and expert SwiGLU MLP projections use right-spectral, row-aware, column-aware, or hybrid updates along the appropriate intermediate-neuron axis; MoE routers use centered row-aware or left-spectral updates; and scalar or vector parameters use standard coordinate-wise optimizers.

We validate the proposed equivariant optimizer classes on four open-weight dense and MoE language model architectures spanning different vocabulary sizes, hidden sizes, embedding and LM head matrix dimensions, and numbers of trainable parameters. Our goal is to test the practical implications of the symmetry- and geometry-based design principles developed above across architectures with distinct matrix-parameter geometries. Because our focus is optimizer behavior rather than scaling-law-optimal pre-training, we do not train the models for the large token budgets typically prescribed by scaling laws. We pre-train all models on a 10B-token subset of FineWeb-Edu [113] with context length 1024. These settings allow us to examine optimizer behavior in controlled yet nontrivial pre-training regimes.

Unless otherwise specified, we use MUON [78] with POLAR EXPRESS coefficients [5] for hidden and attention matrices, and ADAMW [111] for scalar and vector parameters. For consistency with the intended layerwise geometry, fused attention weights are handled by applying POLAR EXPRESS to the momentum of each attention head or fused component separately, in a similar spirit to MUON SPLIT [55]. Likewise, for OLMoE-1B-7B and downsized gpt-oss, we treat fused expert SwiGLU projection tensors according to their intermediate-neuron geometry. The fused expert gate/up tensors are reshaped or interpreted so that gate/up channels associated with intermediate neurons receive row-aware or hybrid updates along the correct axis, while expert down projections are updated along the corresponding intermediate-neuron axis. This avoids applying a geometry-aware optimizer to an artifact of tensor storage rather than to the functional symmetry of the layer. We use untied input embeddings and output LM head weights, which allows us to assign different optimizers to these two large vocabulary-indexed matrices. Full experimental details are given in Section G. Code is available at https://github.com/timlautk/equivariant_optimizers.

4.1 Qwen3-0.6B-Style Pre-Training

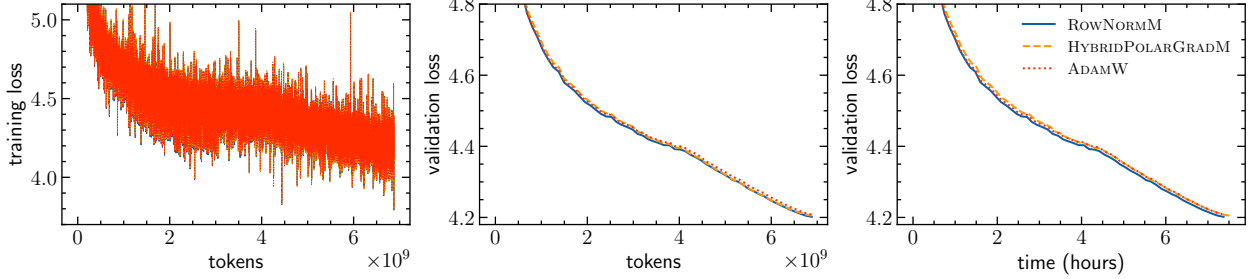
We first pre-train a Qwen3-0.6B-style dense language model [129] incorporating several recent architectural innovations, including Grouped Query Attention (GQA) [4], the SwiGLU activation function [32, 137], Rotary Positional Embeddings (RoPE) [146], pre-norm normalization [160] with RMSNORM [171, 76], and QK-Norm [36], without QKV bias terms. The model uses a vocabulary size of 151,936 and hidden dimension 1024. Since untying the embeddings increases the number of trainable parameters, we reduce the number of hidden layers from 28 to 20, resulting in a total of 625,784,832 trainable parameters.

We compare three optimizer assignments for the vocabulary-indexed matrices, namely the input embedding and LM head matrices: (i) ROWNORMM, (ii) HYBRIDPOLARGRADM with row-norm/right-spectral, (iii) ADAMW. For SwiGLU MLP projections, we compare (a) right-spectral updates, equivalently MUON-style updates, with (b) hybrid row-norm/right-spectral updates applied along the appropriate intermediate-neuron axis: row-aware for gate and up projections, and column-aware for down projections.

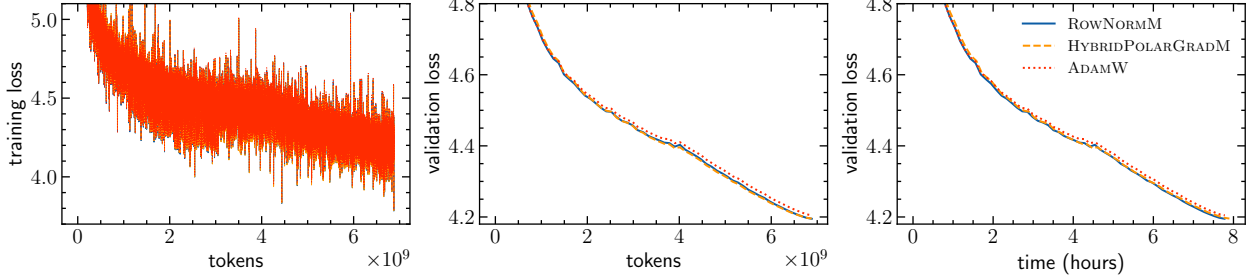
Although different LPRO-compatible optimizers could in principle be assigned to the embedding and LM head matrices, we use the same optimizer for both layers in each configuration for simplicity.

The final validation losses for configurations (i)–(iii) are 4.2017, 4.2050, and 4.2084 in Figure 3a, and 4.1950, 4.1955, and 4.2046 in Figure 3b, respectively. As shown in Figure 3, in both settings, configuration (iii) makes comparable initial progress to configuration (i) and has lower validation loss at the earlier stage of training, but is subsequently overtaken by both ROWNORMM and HYBRIDPOLARGRADM. The final gap between HYBRIDPOLARGRADM and ADAMW is smaller than that between ROWNORMM and ADAMW, but the validation loss still favors the symmetry-compatible update.

Across Figure 3a and Figure 3b, using HYBRIDPOLARGRADM for the SwiGLU MLP projec-



(a) SwiGLU MLP projection matrices use MUON, equivalently RIGHTPOLARGRADM with $\alpha = 0$.



(b) SwiGLU MLP projection matrices use HYBRIDPOLARGRADM with a row-norm/right-spectral composition.

Figure 3: Training and validation losses for Qwen3-0.6B-style pre-training. In each subfigure, the three configurations differ only in the optimizer assigned to the input embedding and LM head matrices: ROWNORMM, HYBRIDPOLARGRADM, or ADAMW.

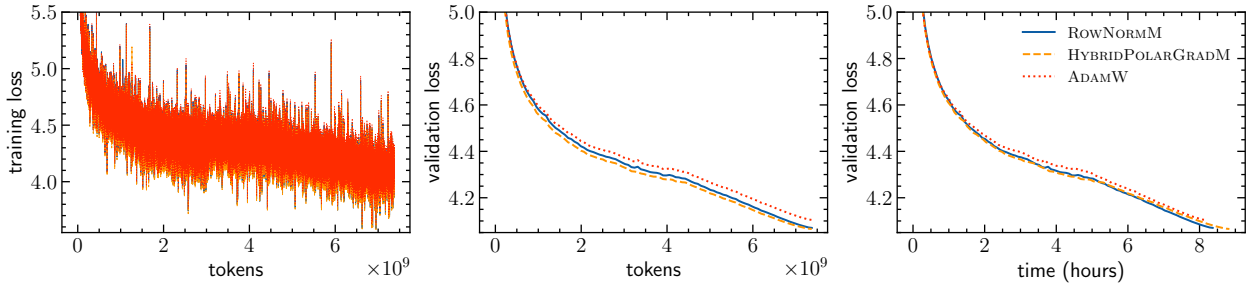
tion matrices improves all three embedding/LM head configurations relative to using MUON for the SwiGLU MLP projections. The final validation losses decrease from 4.2017 to 4.1950 for ROWNORMM, from 4.2050 to 4.1955 for HYBRIDPOLARGRADM, and from 4.2084 to 4.2046 for ADAMW. The improvement is largest when HYBRIDPOLARGRADM is also used for the input embedding and LM head matrices, suggesting a possible complementary effect between symmetry-compatible updates on vocabulary-indexed matrices and row-aware/right-spectral updates on SwiGLU MLP projections. Nevertheless, ROWNORMM remains the best-performing assignment for the input embedding and LM head matrices in both settings. Thus, the comparison between (a) and (b) suggests that applying row-norm/right-spectral hybrid updates to SwiGLU MLP projections can further improve validation loss, while the relative ranking among the three vocabulary-indexed optimizer assignments remains stable. This improvement is plausibly due to the tall-skinny geometry of the SwiGLU MLP projections. Since $d_{\text{model}} = 1024$ and $d_{\text{ff}} = 3072$, the gate and up projections have many more rows than columns, with rows corresponding to intermediate neurons. Row-scale imbalance can therefore be important. While MUON captures the right-spectral geometry, HYBRIDPOLARGRADM additionally normalizes intermediate-neuron rows before the spectral step, which may yield a better update geometry in this regime.

In terms of wall-clock training time, HYBRIDPOLARGRADM incurs additional overhead due to the inner Gram Newton–Schulz iterations used to approximate the right-spectral component. Consequently, configurations (ii) and (iii) require a similar amount of training time to reach comparable validation loss values, despite HYBRIDPOLARGRADM achieving a slightly lower final loss. Configurations (i) and (ii) both follow symmetry-compatible geometries for the embedding and LM head matrices, whereas configuration (iii) applies coordinate-wise ADAMW updates to these matrix-valued parameters, thereby introducing a geometry mismatch. Overall, these results are

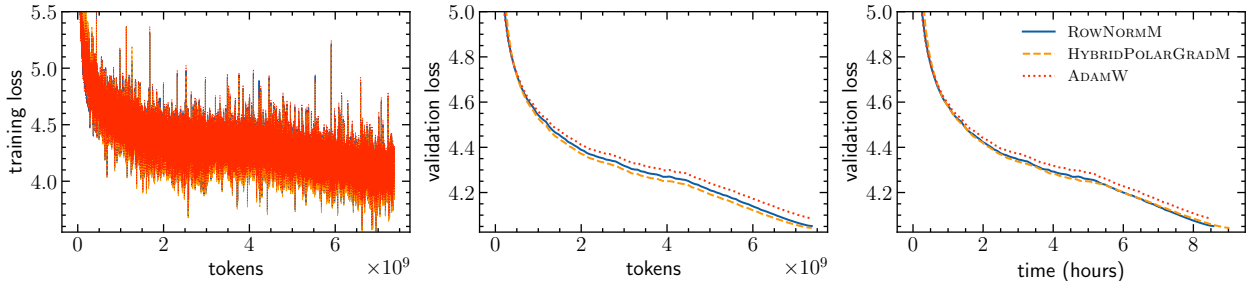
consistent with our symmetry-aware matrix view of optimizer design: even when applied only to the vocabulary-indexed matrices, geometry-compatible updates can improve the optimization trajectory and final validation loss.

4.2 Gemma 3 1B-Style Pre-Training

We next pre-train a Gemma 3 1B-style dense language model [51]. Compared with the Qwen3-0.6B-style experiment, this model has both a larger vocabulary size of 262,144, and a larger hidden dimension of 1152. Consequently, the input embedding and LM head matrices are substantially larger, and their matrix gradients may be more anisotropic or ill-conditioned. This setting therefore provides a more stringent test of geometry-aware optimizers for vocabulary-indexed matrix parameters. For right-spectral updates, such as those used inside RIGHTPOLARGRADM and HYBRIDPOLARGRADM, the larger hidden dimension also makes the Gram matrix inverse-square-root computation more demanding, often requiring more accurate or additional Newton–Schulz iterations. To keep the total model size close to one billion trainable parameters after using untied embeddings, we reduce the number of hidden layers from 26 to 18, resulting in 1,087,138,944 trainable parameters. We use the same three optimizer assignments for the input embedding and LM head matrices as in the Qwen3-0.6B-style experiment, and compare two assignments for the SwiGLU MLP projection matrices: (a) MUON and (b) HYBRIDPOLARGRADM.



(a) SwiGLU MLP projection matrices use MUON, equivalently RIGHTPOLARGRADM with $\alpha = 0$.



(b) SwiGLU MLP projection matrices use HYBRIDPOLARGRADM with a row-norm/right-spectral composition.

Figure 4: Training and validation losses for Gemma 3 1B-style pre-training. In each subfigure, the three configurations differ only in the optimizer assigned to the input embedding and LM head matrices: ROWNORMM, HYBRIDPOLARGRADM, or ADAMW.

The final validation losses for configurations (i)–(iii) are 4.0702, 4.0655, and 4.1046 in Figure 4a, and 4.0516, 4.0435, and 4.0862 in Figure 4b, respectively. In both settings, HYBRIDPOLARGRADM achieves the lowest final validation loss, while ROWNORMM also substantially outperforms ADAMW. This behavior is consistent with the hypothesis that geometry-compatible updates become increas-

ingly important as vocabulary-indexed matrices grow larger and their gradients become more anisotropic or ill-conditioned.

Comparing Figure 4a and Figure 4b, using HYBRIDPOLARGRADM for the SwiGLU MLP projection matrices improves all three embedding/LM head optimizer assignments. The final validation loss decreases from 4.0702 to 4.0516 for ROWNORMM, from 4.0655 to 4.0435 for HYBRIDPOLARGRADM, and from 4.1046 to 4.0862 for ADAMW. This improvement is plausibly related to the tall-skinny geometry of the SwiGLU MLP projections: in Gemma 3 1B-style models, $d_{\text{model}} = 1152 \ll d_{\text{ff}} = 6912$, so the gate and up projections have many more rows than columns, with rows corresponding to intermediate neurons. While MUON captures the right-spectral geometry, HYBRIDPOLARGRADM additionally normalizes intermediate-neuron rows before the spectral step, which can better account for row-scale imbalance in this regime.

At the same time, HYBRIDPOLARGRADM incurs higher computational overhead than ROWNORMM because it requires approximating matrix inverse square roots through inner Gram Newton–Schulz iterations. In contrast, ROWNORMM performs only row-wise normalization and therefore avoids spectral computations altogether. Thus, the Gemma 3 1B-style experiment highlights a practical tradeoff: the hybrid row-norm/right-spectral update achieves the best validation loss, while the row-norm-only update provides a computationally cheaper symmetry-compatible alternative that still improves markedly over the coordinate-wise ADAMW baseline. We also provide additional experimental results for a base learning rate sweep and two extra random seeds in Section G.2. Finally, Section G.2.3 shows that projected symmetry-compatible LM head updates reduce final vocabulary-logit growth in Gemma 3 1B-style pre-training, providing an optimizer-side mechanism that can reduce the reliance on final logit soft-capping.

4.3 OLMoE-1B-7B-Style Pre-Training

In addition to dense language models, we also pre-train a sparse Mixture-of-Experts (MoE) model, a widely used architecture in recent open-weight language models [74, 35, 122, 130, 55, 144, 60, 34]. We use AllenAI’s OLMoE-1B-7B [117], which provides a comprehensive training recipe together with open-source data, code, and training logs. The model has vocabulary size 50,304 and hidden dimension 2048, making the embedding and LM head matrices considerably large. Relative to the original pre-training setup, we remove the auxiliary load-balancing loss [138] and the router z-loss [180] in order to reduce confounding effects from auxiliary objectives and isolate the effect of optimizer geometry. We also reduce the number of hidden layers from 16 to 12 and the number of experts from 64 to 32, yielding a total of 2,824,177,664 trainable parameters.

For matrix-valued parameters other than the hidden and attention matrices, we compare four optimizer assignments:

- (i) ROWNORMM for embeddings, LM head, and routers,
- (ii) ROWNORMM for embeddings and LM head, and LEFTPOLARGRADM for routers,
- (iii) ROWNORMM for embeddings and LM head, and ADAMW for routers,
- (iv) ADAMW for embeddings, LM head, and routers.

We choose ROWNORMM for the embedding and LM head matrices because it adds minimal computational overhead relative to ADAMW while preserving a symmetry-compatible geometry for vocabulary-indexed matrices. By contrast, RIGHTPOLARGRADM and HYBRIDPOLARGRADM require numerical polar decomposition, which is computationally more demanding and may require higher numerical precision for large vocabulary matrices. We leave broader ablations over these alternatives to future work.

We also expect the relative behavior of MUON and HYBRIDPOLARGRADM on SwiGLU MLP projections to depend on matrix aspect ratio. When $d_{\text{ff}} \gg d_{\text{model}}$, as in the dense Qwen3-0.6B-style and Gemma 3 1B-style models, the gate and up projections are tall-skinny and row-scale imbalance across intermediate neurons can be substantial. In this regime, the row-normalization step in HYBRIDPOLARGRADM can be beneficial. In the MOE experiments below, however, d_{ff} and d_{model} are much closer, so pure MUON-style right-spectral updates may already capture much of the relevant matrix geometry. For this reason, in the MOE experiments we use MUON-style right-spectral updates for the SwiGLU MLP projection tensors and focus our ablations on the vocabulary-indexed matrices and MOE routers.

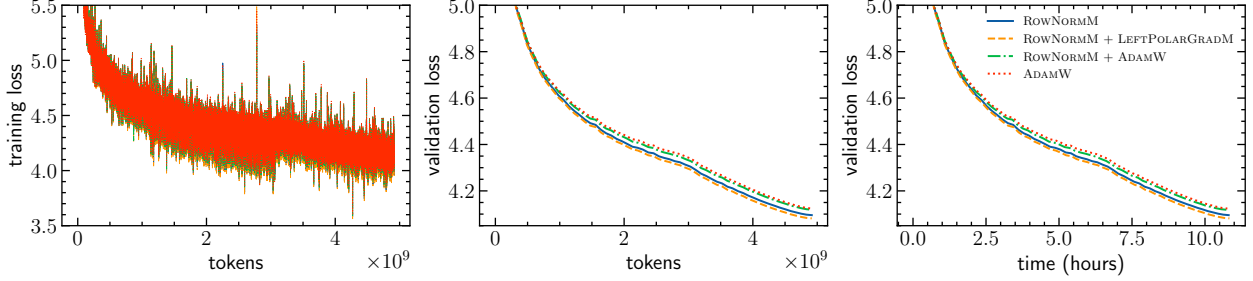


Figure 5: Training and validation losses for OLMoE-1B-7B-style pre-training. The configurations differ in the optimizers assigned to the embedding, LM head, and router matrices.

The final validation losses for configurations (i)–(iv) are 4.0955, 4.0815, 4.1187, and 4.1240 respectively. As shown in Figure 5, configuration (iv), which uses ADAMW for the embedding, LM head, and router matrices, makes faster initial progress over roughly the first 500 steps, but is eventually overtaken by configurations (i)–(iii). This reversal occurs before the onset of learning rate decay, which linearly decreases to zero for the last 40% of the training tokens. The validation loss gaps continue to widen during the decay phase.

Configurations (i) and (ii) use symmetry-compatible updates for all three special matrix classes considered here: embeddings, LM head, and routers. Configuration (iii) retains symmetry-compatible updates for the embedding and LM head matrices, but introduces a geometry mismatch in the router updates by using coordinate-wise ADAMW. Configuration (iv) applies ADAMW to all three classes and therefore departs most strongly from the proposed symmetry-compatible optimizer design. Empirically, both (i) and (ii) outperform (iii), while (iv) performs worst overall. These results are consistent with our theoretical perspective that respecting parameter symmetry and matrix geometry can matter for optimizer design, especially in large sparse architectures where router dynamics play a central role.

The performance gap between configurations (i) and (ii) is relatively small, suggesting that the row-norm and left-spectral router updates behave similarly in this setting. This small gap may reflect suboptimal learning-rate tuning for ROWNORMM, the effect of inexact polar oracles, or that left-spectral normalization is able to further capture the relevant router geometry in this experiment. Finally, we observe that configuration (iv) might exhibit slightly more pronounced training-loss spikes at around 2.1B seen training tokens than the other configurations, despite using MUON for the hidden and attention matrices. This suggests that geometry-matched optimizer choices for embeddings, LM heads, and routers may also improve training stability in practice.

4.4 Downsized gpt-oss Pre-Training

We finally pre-train a downsized variant of gpt-oss-20b [122]. This architecture differs from OLMoE in several important ways. For example, it uses QKV bias terms in its GQA modules and includes bias vectors in its MoE router networks. The model has vocabulary size 201,088, making the embedding and LM head matrices substantially larger than those in OLMoE. To obtain a tractable experimental variant, we downsize gpt-oss-20b by reducing the number of hidden layers from 24 to 12, the hidden and intermediate dimensions from 2880 to 2048, and the number of experts from 32 to 16. This yields a total of 3,467,779,008 trainable parameters. We use the same loss function and the same four optimizer assignments as in the OLMoE-1B-7B experiment.

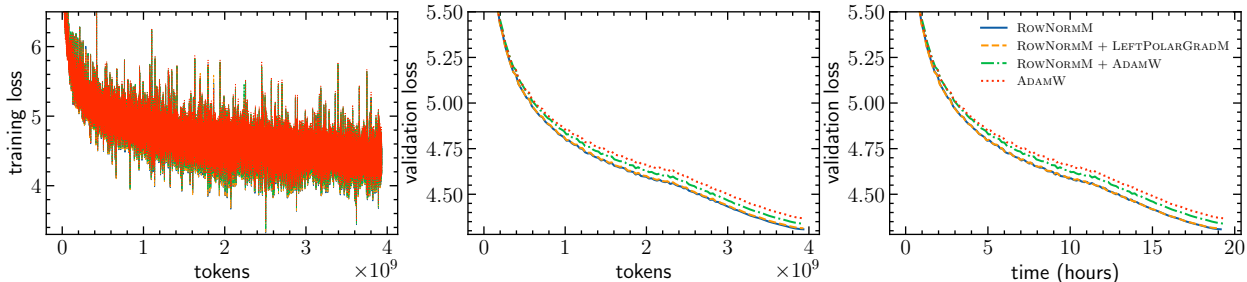


Figure 6: Training and validation losses for downsized gpt-oss pre-training. The configurations differ in the optimizers assigned to the embedding, LM head, and router matrices.

The final validation losses for configurations (i)–(iv) are 4.3080, 4.3136, 4.3388, and 4.3687, respectively. As in the OLMoE experiment, the fully coordinate-wise baseline in configuration (iv), which uses ADAMW for the embedding, LM head, and router matrices, obtains the worst final validation loss. The three configurations using ROWNORMM for the embedding and LM head matrices all substantially improve over this baseline, suggesting that the benefit of symmetry-compatible updates for vocabulary-indexed matrices persists in a distinct sparse MoE architecture.

Among configurations (i)–(iii), the differences are smaller. As shown in Figure 6, configuration (i), which uses ROWNORMM for embeddings, LM head, and routers, achieves the lowest final validation loss, followed closely by configuration (ii), which uses LEFTPOLARGRADM for routers. Configuration (iii), which keeps ROWNORMM for embeddings and LM head but uses ADAMW for routers, is slightly worse than both geometry-compatible router variants. This ordering is consistent with the view that router geometry can matter, although the smaller gap between configurations (i)–(iii) suggests that the dominant improvement in this setting comes from replacing ADAMW on the large vocabulary-indexed matrices.

Additional router diagnostics in Section G.4.1 show that symmetry-compatible router updates reduce load imbalance and router z-loss relative to the ADAMW router baseline, both with and without auxiliary router losses.

Overall, the downsized gpt-oss experiment provides an additional architecture check for our optimizer design principle. Despite architectural differences from OLMoE, including QKV biases and router bias terms, the same qualitative pattern holds: geometry-compatible optimizer assignments for special matrix-valued parameters improve the final validation loss relative to using coordinate-wise ADAMW for those parameters.

4.5 Cross-Model Comparison

We compare the results across Sections 4.1 to 4.4. These experiments span dense and sparse language models with increasing numbers of trainable parameters, from the Qwen3-0.6B-style dense model to the downsized gpt-oss sparse MoE model. Across these models, the vocabulary sizes differ substantially, while the hidden dimensions are relatively close. Consequently, the embedding and LM head matrices have comparable column dimensions, determined by d_{model} , but very different numbers of rows, determined by the vocabulary size v . Thus, as v grows, these vocabulary-indexed matrices become increasingly tall. This changes both their computational cost and the conditioning properties of their matrix gradients, and makes them a natural testbed for row-aware and one-sided spectral optimizer design.

The same row-versus-column distinction is also important for SwiGLU MLP projection matrices. For a dense SwiGLU block, the gate and up projections have shape $d_{\text{ff}} \times d_{\text{model}}$, so their rows correspond to intermediate neurons, whereas the down projection has shape $d_{\text{model}} \times d_{\text{ff}}$, so the same intermediate-neuron geometry appears along the column axis. In the dense Qwen3-0.6B-style and Gemma 3 1B-style models, d_{ff} is substantially larger than d_{model} , placing the gate and up projections in a tall-skinny regime. This is precisely the setting where row-normalization before a right-spectral step can be useful: HYBRIDPOLARGRADM can correct row-scale imbalance across intermediate neurons while retaining spectral geometry. In the MoE experiments, by contrast, the hidden and intermediate dimensions are closer in our downsized settings, so a pure MUON-style right-spectral update may already capture much of the relevant matrix geometry for expert SwiGLU projection tensors.

Across all experiments, replacing ADAMW on the large vocabulary-indexed matrices with symmetry-compatible optimizers consistently improves final validation loss. The gains are modest but visible for the smaller dense model, become more pronounced in the larger Gemma 3 1B-style experiment, and persist in both sparse MoE experiments. This trend is consistent with our matrix-geometry perspective: as embedding and LM head matrices grow in their row dimension, coordinate-wise updates increasingly operate in a parameterization-dependent augmented space, whereas row-norm and spectral updates preserve the relevant vocabulary-indexed matrix geometry.

The comparison should not be interpreted as a scaling law, since the models differ in architecture, vocabulary size, training length, and sparsity structure. Nevertheless, the consistent ordering across dense and sparse MoE models provides evidence that the benefit of symmetry-compatible optimizer assignments is not restricted to a single model family or architecture. In particular, the results suggest that large vocabulary-indexed matrices are a robust setting in which geometry-compatible updates can improve optimization. The sparse MoE experiments further show that router matrices and expert SwiGLU projection tensors provide additional layer types where symmetry-aware optimizer design can matter.

Taken together, these experiments support the view that symmetry-compatible optimizer design is most naturally applied as a layerwise optimizer stack, rather than as a single global replacement for ADAMW.

5 Discussion and Outlook

This work suggests a different view of deep learning optimization: optimizer design should be layerwise, geometry-aware, and symmetry-compatible. Popular coordinate-wise adaptive methods such as ADAM and ADAMW remain strong default optimizers because of their robustness, efficiency, and practical momentum. However, when applied indiscriminately to matrix-valued parameters, they treat matrices and tensors as collections of independent coordinates and therefore ignore the

intrinsic geometry of the parameter blocks they update. This mismatch is especially apparent in modern architectures, where different modules play different algebraic and semantic roles.

Our main contribution is a symmetry-compatible equivariance principle for designing optimizers for matrix-valued neural network parameters. For ordinary matrix layers, this principle recovers spectral optimizers as natural bi-orthogonally equivariant update maps. For embedding and LM head matrices, it leads to left-permutation/right-orthogonal equivariant updates. For SwiGLU MLP projections, it motivates row- and column-aware updates aligned with intermediate-neuron permutation geometry. For MoE router weights, it yields expert-permutation equivariant and shared-logit-shift invariant updates. Together, these examples support an architecture-optimizer co-design principle: different parameter classes should be updated by optimizers whose equivariance matches their layerwise symmetry.

This perspective connects recent progress in matrix-gradient optimization to a broader transition from generic coordinate-wise methods toward *module-aware* and *geometry-aware* optimization. Layerwise training itself is not new: classical examples include LARS and LAMB [166, 167], layerwise hyperparameter prescriptions such as those arising in μ P [163], and block-coordinate views of neural network training [170, 92]. What is emerging, however, is a more refined class of layerwise optimizers that account for the geometry of each parameter block. Recent methods such as SHAMPOO [64], MUON [78], SOAP [153], SCION [124], GLUON [134], and POLARGRAD [91] can be viewed as part of this trend.

The case for geometry-aware optimizer design becomes stronger as foundation models become more heterogeneous. Large language models [151, 132, 133, 19], vision transformers [41], multimodal models [131, 18], diffusion language models [112, 57, 121], MOEs [138, 93], and state space models [62, 31, 89] all contain parameter blocks with distinct natural symmetries. From this viewpoint, it is increasingly unnatural to optimize all such layers with a single coordinate-wise rule. Instead, optimizer design for modern deep learning systems should be treated as an architecture-aware problem, rather than as the selection of one universal update rule.

This viewpoint also offers a possible interpretation of some training stabilization practices. As model scale increases, coordinate-wise adaptive optimizers are rarely used in isolation; they are typically accompanied by stabilization tricks, modified variants such as STABLEADAMW [156], and many recipe-level heuristics [72]. While such techniques are practically important, some of them may be understood as partial corrections for a mismatch between optimizer geometry and model geometry. Geometry-aware optimizers provide a complementary path by encoding more appropriate invariance, normalization, and scaling structure directly into the update rule.

Several challenges remain. First, geometry-aware optimizers depend on efficient numerical linear algebra. The practical success of MUON and related methods was enabled in part by making polar decomposition or matrix orthogonalization efficient at scale, especially through Newton–Schulz iterations. Further progress will likely depend on fast, stable, and GPU-friendly matrix decomposition routines, including inexact polar oracles based on Newton–Schulz iteration [81], QDWH [91], POLAR EXPRESS [5], CANS [61], PRISM [165], Turbo-MUON [17], Flash-MUON [105], and Gram Newton–Schulz [172]. Second, non-elementwise optimizers raise new distributed-systems challenges, including communication costs, synchronization, memory sharding, and compatibility with tensor, pipeline, sequence, and data parallelism. Recent work on Distributed MUON [107, 47, 126], DION [3, 2], DISCO [49], and Parallel MUON [103] suggests that these challenges can be addressed in practice.

Encouragingly, matrix-aware optimizers have already begun to appear in industry-scale model training, including work by Moonshot AI [107, 82], Essential AI [48], Prime Intellect [126], Zhipu AI [54, 55], Zephyra [8, 154], Motif Technologies [103], Arcee AI [143], StepFun [144], and DeepSeek-AI [34]. These developments suggest that the question is no longer whether matrix-aware optimizers can be scaled, but how far they can be pushed once algorithmic geometry, numerical linear algebra,

and distributed systems are designed in concert.

Recent work by Du et al. [42] provides a complementary perspective, showing through a layer-peeled optimization model that symmetries in next-token distributions can transfer to learned LLM weights, logits, and context embeddings. The Newton–MUON optimizer [43] provides a complementary example of symmetry-aware matrix-gradient optimization. By deriving a MUON-like update from a quadratic surrogate involving the layer input matrix, Newton–MUON shows that the polar update can be combined with right preconditioning by the input second moment. In this sense, it extends the geometry of MUON from a purely weight-gradient update to one that also reflects data geometry.

Overall, our results suggest that much of modern deep learning still relies on optimizer updates whose geometry is mismatched to the layers they train. This mismatch may affect robustness, stability, scalability, and interpretability, since coordinate-wise adaptivity is sensitive to arbitrary parameterizations and can be viewed as operating in a pathological diagonal lifting of the original matrix space. By contrast, symmetry-compatible updates offer the prospect of more stable, parameterization-consistent, and theoretically grounded training procedures. Introducing symmetry and equivariance into optimizer design therefore opens a path toward a more principled science of large-scale model pre-training.

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Organization. In the appendix, we provide discussion on further related work (Section A), and supplementary technical background materials on amatrix analysis (Section B). Furthermore, we illustrate the geometric misalignment of coordinate-wise adaptive gradient methods in Section C. We then provide omitted proofs from the main text in Section D, as well as details of the implementation of practical optimizers Section E. We then give the convergence analysis in Section F. We also provide details of numerical experiments in Section G.

A Further Related Work

We further discuss connections between our framework and other existing paradigms for designing matrix-gradient optimizers in modern deep learning.

A.1 Non-Euclidean Norm-Based Steepest Descent and LMO-Based Frameworks

A prominent line of recent work interprets MUON and related methods through the lens of non-Euclidean steepest descent, trust-region methods, and linear minimization oracle (LMO) frameworks [78, 11, 13, 14, 124, 85, 134]. In these views, MUON can be understood as a normalized steepest descent method with respect to a non-Euclidean matrix norm—most notably the spectral norm—and this perspective has also been used to connect MUON to optimizers such as SIGNSGD [15] and SHAMPOO [64]. Related work has further explored alternative norm choices and generalized preconditioned steepest descent constructions [29, 152, 87, 162].

While these frameworks provide useful algorithmic interpretations of MUON and its variants, they leave open a fundamental question: what is the *principled* criterion for choosing the “correct” norm for a given layer? In practice, existing prescriptions are not always fully aligned with actual usage. For example, Large et al. [90] suggest using the maximum column ℓ_2 -norm for embedding layers, whereas in practice ADAMW is often used for embeddings while MUON is applied to other matrix parameters in modded-nanogpt speedrunning [77]. More broadly, several works suggest that one may use various $\ell_p \rightarrow \ell_q$ operator norms for steepest descent or LMO for embedding and LM head layers (see e.g., [124]). Our theoretical development suggests that such norm choices might generally not be geometrically principled for matrix-valued parameters.

In contrast, our symmetry-based analysis provides a concrete design principle. For matrix parameters whose geometry should respect orthogonal changes of basis, only unitarily invariant matrix norms are appropriate, since they are precisely the norms compatible with orthogonal invariance. By comparison, the $\ell_p \rightarrow \ell_q$ operator norm is generally *not* unitarily invariant, except in the special case $p = q = 2$, which reduces to the spectral norm. Accordingly, non-unitarily invariant norms typically induce an incorrect matrix geometry. This offers a possible explanation for the gap between certain norm-based theoretical prescriptions and empirical practice.

A.2 Modular Norm Theory

Our right-spectral viewpoint is closely related in spirit to modular norm theory [14, 90]: both seek architecture-aware optimizer geometries derived from structural properties of the module rather than from an ambient coordinate system. The two approaches are nevertheless conceptually distinct.

Modular norm theory derives updates from steepest descent under module-adapted operator norms, and is primarily motivated by scale transfer across width and depth through recursively constructed global norms. By contrast, right-spectral optimizers are derived directly from left-permutation and right-orthogonal equivariance, leading to update rules of the form (3), namely spectral transformations determined by the right Gram matrix $D^\top D$. In this sense, right-spectral

optimizers form a symmetry-derived subclass of LPRO-equivariant updates, whereas modular-norm-based methods may also include coordinate-dependent row-wise or column-wise transformations that are compatible with only part of the underlying symmetry.

From this perspective, the two frameworks may be viewed as complementary. Modular norm theory aims to provide the correct *scale invariance*, namely how learning rates and normalized updates should behave as architecture width and depth vary. Our spectral framework instead aims to provide the correct *directional geometry*, namely how update directions should respect the invariance structure of different layer types. Their combination suggests the possibility of a fully geometry-aware optimizer that uses modular-norm-based global scaling together with layerwise spectral, right-spectral, or left-spectral update directions dictated by symmetry.

A.3 Rotation-Based Optimizers

The recent work [58] is perhaps the closest in spirit to our approach, but there are several important differences. First, we do not assume that the layerwise loss function itself is rotationally, or more generally bi-orthogonally, invariant. Rather, we derive optimizer classes from the symmetry structure of the parameter geometry and the corresponding equivariance properties of update rules. This distinction matters, since exact rotational invariance of the layerwise loss may be difficult to verify and can fail in practice.

Second, we do not advocate a single update rule for all matrix-valued parameters. Instead, our framework emphasizes that different layer types admit different symmetry groups and therefore should, in principle, use different optimizer geometries. In particular, embedding and LM head matrices possess different equivariance properties from weight matrices in linear and attention layers, which leads naturally to an architecture-optimizer co-design perspective.

Third, the update rules in ARO for matrix parameters are based only on left multiplication by a rotation matrix. By contrast, our framework includes full spectral, right-spectral, left-spectral, row-norm-based, and hybrid classes, depending on the relevant symmetry structure of the layer. In this sense, our theory provides a broader symmetry-based taxonomy of matrix-gradient optimizers.

B Supplemental Technical Background on Matrix Analysis

We include several technical definitions arising in matrix analysis, which is closely related to the derivation of spectral optimizers via steepest descent. Details of these materials can be found in [97, 71, 70, 16].

Let us recall the notion of unitarily invariant norms, which provide the norm-level analogue of left-right orthogonal symmetry. Since we work over the field of real numbers $\mathbb{R}$, “unitarily invariant” here is equivalent to invariance under left and right orthogonal transformations.

Definition B.1 (Unitarily invariant norms). A norm $\|\cdot\|: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}_+$ is said to be *unitarily invariant* if $\|A\| = \|UAV\|$ for all $A \in \mathbb{R}^{m \times n}$ and all orthogonal matrices $U \in \mathbb{O}^m$ and $V \in \mathbb{O}^n$. A *unitarily invariant matrix norm* is a unitarily invariant norm on $\mathbb{R}^{m \times n}$ that is also submultiplicative.

Unitarily invariant norms can be completely characterized through symmetric gauge functions acting on singular values.

Definition B.2 (Symmetric gauge functions). A function $\psi: \mathbb{R}^r \rightarrow \mathbb{R}_+$ is a *symmetric gauge function* if: (i) ψ is a norm; (ii) $\psi(|x|) = \psi(x)$ for all $x \in \mathbb{R}^r$, where $|\cdot|$ is understood coordinatewise; and (iii) $\psi(Px) = \psi(x)$ for all permutation matrices $P \in \mathbb{P}^r$ and all $x \in \mathbb{R}^r$.

Proposition B.1. *A norm $\|\cdot\|: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}_+$ is unitarily invariant if and only if there exists a symmetric gauge function ψ such that $\|W\| = \psi(\sigma(W)) = (\psi \circ \sigma)(W)$, where $\sigma(W)$ is the vector of singular values of W arranged in descending order.*

Example B.1. Important examples of unitarily invariant matrix norms include the Schatten p -norms $\|\cdot\|_p$ for $p \in [1, \infty]$, including the nuclear norm $\|\cdot\|_{\text{nuc}}$, Frobenius norm $\|\cdot\|_F$, and the spectral norm $\|\cdot\|_S$. Another notable example is the Ky Fan k -norm.

C Geometric Misalignment of Coordinate-wise Adaptive Gradient Methods

If an optimizer for matrix parameters in linear layers does not respect the intrinsic geometry of the parameter space through bi-orthogonal equivariance, several issues arise. In particular, for coordinate-wise adaptive gradient methods, the optimizer iterates depend on arbitrary coordinate choices: rotating the input space can change the optimizer itself, leading to different training dynamics under equivalent reparameterizations.

For a vector parameter $w \in \mathbb{R}^n$, the general form of a coordinate-wise adaptive gradient method is

$$(\forall k \in \mathbb{N}) \quad w_{k+1} = w_k - \gamma_k \mathcal{U}(d_k),$$

where d_k is an update direction and $\mathcal{U}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a vector-valued map. A simple but important observation is that optimizers built only from linear operations, such as additions, subtractions, scalar multiplications, and their linear combinations, behave analogously for vector and matrix parameters. By contrast, coordinate-wise adaptive methods such as ADAM rely on elementwise nonlinear operations, including division, squaring, and square roots. These operations distort the geometry of the original update direction, whether it is formed from the gradient itself or from a momentum term.

While spectral methods (e.g., spectral descent and MUON; formally introduced in Section 3.8) respect bi-orthogonal equivariance, coordinate-wise methods are typically equivariant only under the much smaller signed permutation group. As a result, they fail to respect the intrinsic matrix geometry of the optimization problem. This mismatch is especially problematic in deep learning, where the optimization landscape often exhibits an intrinsically low-dimensional structure. Spectral optimizers are designed to adapt to this geometry, whereas coordinate-wise methods largely ignore it. Moreover, as model size increases, the advantage of such geometry-aware methods typically becomes more pronounced.

Sign descent. For a matrix gradient G , sign descent or SIGNSGD [15] uses the coordinate-wise signs $\text{sgn}(G)$ of the gradient as the update direction, which satisfies $\langle\langle G, \text{sgn}(G) \rangle\rangle_F = \|G\|_1$. Thus, sign descent is governed by the duality associated with the coordinate-wise ℓ_∞ -norm, rather than by any unitarily invariant matrix geometry (see Definition B.1).

Adam. More generally, ADAM [83] can be viewed as a smoothed version of sign descent, since its update is dominated by coordinate-wise normalization. In particular, ADAM applies the coordinate-wise scaling (omitting bias corrections for simplicity)

$$W_{ij} \leftarrow W_{ij} - \gamma \cdot \frac{M_{ij}}{\sqrt{V_{ij} + \varepsilon}},$$

where M_{ij} and V_{ij} denote the first- and second-moment statistics (with bias corrections) at coordinate (i, j) , $\gamma > 0$ is the learning rate, and $\varepsilon > 0$ is a small constant. This update does not transform equivariantly under bi-orthogonal reparameterizations $W \mapsto \tilde{W} := PWQ^\top$ for $P \in \mathbb{O}^m$ and $Q \in \mathbb{O}^n$. Consequently, the update direction of ADAM depends on how the entries of W are indexed, rather than only on the intrinsic geometry of W as a matrix.

Furthermore, coordinate-wise methods tend to inject high-rank coordinate noise even when the underlying gradient G is low rank. In contrast, spectral methods act directly on the matrix structure of G , and therefore remain aligned with the low-dimensional geometry that frequently governs deep network optimization.

Sign descent as a special case of spectral descent. Following similar arguments in [91], we further interpret sign descent as spectral descent applied to the diagonal matricization of the vectorized matrix parameter. Define the *diagonal matricization of the vectorization* of G by $\tilde{G} := \text{Diag}(\text{vec}(G)) \in \mathbb{R}^{mn \times mn}$. The following lemma makes this connection precise.

Lemma C.1. *Let $G := \nabla_W f(W) \in \mathbb{R}^{m \times n}$ contain no exact zero entries, and define $\tilde{G} := \text{Diag}(\text{vec}(G)) \in \mathbb{R}^{mn \times mn}$. Then the orthogonal polar factor of $\tilde{G}$ is $\text{polar}(\tilde{G}) = \text{Diag}(\text{sgn}(\text{vec}(G)))$. Consequently, $\text{sgn}(G) = \text{reshape}(\text{diag}(\text{polar}(\tilde{G})), m, n)$.*

Proof. Since $\tilde{G} = \text{Diag}(\text{vec}(G))$ is diagonal, its singular values are given by the absolute values of its diagonal entries. Hence we may write its singular value decomposition as

$$\tilde{G} = I_{mn} \text{Diag}(|\text{vec}(G)|) \text{Diag}(\text{sgn}(\text{vec}(G))),$$

or equivalently,

$$\tilde{G} = \text{Diag}(\text{sgn}(\text{vec}(G))) \text{Diag}(|\text{vec}(G)|).$$

Therefore, by the definition of the orthogonal polar factor, we have $\text{polar}(\tilde{G}) = \text{Diag}(\text{sgn}(\text{vec}(G)))$. Taking the diagonal of $\text{polar}(\tilde{G})$ and reshaping it back to an $m \times n$ matrix yields

$$\text{reshape}(\text{diag}(\text{polar}(\tilde{G})), m, n) = \text{reshape}(\text{sgn}(\text{vec}(G)), m, n) = \text{sgn}(G).$$

□

In other words, coordinate-wise sign descent can be viewed as spectral gradient descent applied to the highly degenerate diagonal lifting $\tilde{G} = \text{Diag}(\text{vec}(G)) \in \mathbb{R}^{mn \times mn}$. Thus, sign-based methods inherit only the trivial geometry of this pathological diagonal representation, rather than the intrinsic matrix geometry of G itself. This highlights a *fundamental geometric mismatch* in coordinate-wise updates, which is further worsened when the parameter dimensions m and n and hence model sizes grow.

D Proofs of Main Text

Proof of Proposition 3.1. We also state the desired results for Polyak and Nesterov momentum.

For EMA momentum, we define $M_k = \beta M_{k-1} + (1 - \beta)G_k$ with $M_{-1} = 0$, and, for the transformed sequence, $\tilde{M}_k = \beta \tilde{M}_{k-1} + (1 - \beta)\tilde{G}_k$ with $\tilde{M}_{-1} = 0$. Then we have $\tilde{M}_k = PM_kQ^\top$ and $\text{polar}(\tilde{M}_k) = P \text{polar}(M_k)Q^\top$.

For Polyak heavy-ball momentum, we define $M_k = \beta M_{k-1} + G_k$ with $M_{-1} = 0$, and $\tilde{M}_k = \beta \tilde{M}_{k-1} + \tilde{G}_k$ with $\tilde{M}_{-1} = 0$. Then we have $\tilde{M}_k = PM_kQ^\top$ and $\text{polar}(\tilde{M}_k) = P \text{polar}(M_k)Q^\top$.

We first prove the stated transformation for the momentum sequences. For the EMA recursion, assume inductively that $\widetilde{M}_{k-1} = PM_{k-1}Q^\top$. Using the bi-orthogonal equivariance of the gradient, we have

$$\widetilde{G}_k = \nabla_W f(\widetilde{W}_k) = \nabla_W f(PW_kQ^\top) = P\nabla_W f(W_k)Q^\top = PG_kQ^\top.$$

Hence

$$\begin{aligned}\widetilde{M}_k &= \beta\widetilde{M}_{k-1} + (1-\beta)\widetilde{G}_k = \beta PM_{k-1}Q^\top + (1-\beta)PG_kQ^\top \\ &= P(\beta M_{k-1} + (1-\beta)G_k)Q^\top = PM_kQ^\top.\end{aligned}$$

Thus the EMA momentum is bi-orthogonally equivariant. The Polyak case is identical, replacing $(1-\beta)$ by 1, which gives $\widetilde{M}_k = PM_kQ^\top$.

For Nesterov momentum, the momentum recursion is the same as in the Polyak case, so $\widetilde{M}_k = PM_kQ^\top$. Therefore,

$$\widetilde{N}_k = \widetilde{G}_k + \beta\widetilde{M}_k = PG_kQ^\top + \beta PM_kQ^\top = P(G_k + \beta M_k)Q^\top = PN_kQ^\top.$$

All claims about orthogonal polar factors follow from (1). For example,

$$\text{polar}(\widetilde{N}_k) = \text{polar}(PN_kQ^\top) = P \text{polar}(N_k)Q^\top.$$

The proofs for the momentum polar factors are identical. $\square$

Proof of Theorem 3.2. Let $P \in \mathbb{P}^v$, $R \in \mathbb{O}^d$, and $D \in \mathbb{R}^{v \times d}$. Since P is a permutation matrix, we have $P^\top P = I_v$. Therefore,

$$(PDR^\top)^\top(PDR^\top) = RD^\top P^\top PDR^\top = RD^\top DR^\top.$$

Using the orthogonal equivariance of Φ , it follows that

$$\Phi((PDR^\top)^\top(PDR^\top)) = \Phi(RD^\top DR^\top) = R\Phi(D^\top D)R^\top.$$

Hence we obtain

$$\begin{aligned}\mathcal{U}_R(PDR^\top) &= PDR^\top \Phi((PDR^\top)^\top(PDR^\top)) = PDR^\top (R\Phi(D^\top D)R^\top) \\ &= PD\Phi(D^\top D)R^\top = P\mathcal{U}_R(D)R^\top.\end{aligned}$$

$\square$

Proof of Proposition 3.3. For any $D \in \mathbb{R}^{v \times d}$, permutation matrix $P \in \mathbb{P}^v$, and orthogonal matrix $R \in \mathbb{O}^d$,

$$(\mathcal{U}_2 \circ \mathcal{U}_1)(PDR^\top) = \mathcal{U}_2(\mathcal{U}_1(PDR^\top)) = \mathcal{U}_2(P\mathcal{U}_1(D)R^\top) = P\mathcal{U}_2(\mathcal{U}_1(D))R^\top.$$

Hence $\mathcal{U}_2 \circ \mathcal{U}_1$ is left-permutation and right-orthogonal equivariant. $\square$

Proof of Proposition 3.4. Since σ is applied coordinatewise and P is a permutation matrix, $\sigma(PW_{\text{gate}}x) = P\sigma(W_{\text{gate}}x)$. Moreover, $(Pu) \odot (Pv) = P(u \odot v)$ for all $u, v \in \mathbb{R}^{d_{\text{ff}}}$. Hence

$$\begin{aligned}\text{SwiGLU}(x; \widetilde{W}_{\text{gate}}, \widetilde{W}_{\text{up}}, \widetilde{W}_{\text{down}}) &= W_{\text{down}}P^\top(\sigma(PW_{\text{gate}}x) \odot (PW_{\text{up}}x)) \\ &= W_{\text{down}}P^\top P(\sigma(W_{\text{gate}}x) \odot (W_{\text{up}}x)) \\ &= W_{\text{down}}(\sigma(W_{\text{gate}}x) \odot (W_{\text{up}}x)).\end{aligned}$$

$\square$

Proof of Proposition 3.5. We first record two elementary identities. Since $P \in \mathbb{P}^e$ satisfies $P\mathbf{1}_e = \mathbf{1}_e$ and $P^\top \mathbf{1}_e = \mathbf{1}_e$, the centering projector $\Pi_\perp = I_e - \frac{1}{e}\mathbf{1}_e\mathbf{1}_e^\top$ commutes with P , i.e., $\Pi_\perp P = P\Pi_\perp$. Moreover, for any $a \in \mathbb{R}^d$, $\Pi_\perp \mathbf{1}_e a^\top = 0$. Therefore, if $\tilde{D} = PD + \mathbf{1}_e a^\top$, then its centered direction is

$$\tilde{D}_c = \Pi_\perp \tilde{D} = \Pi_\perp (PD + \mathbf{1}_e a^\top) = P\Pi_\perp D = PD_c.$$

We first consider the centered left-spectral update. Since $\tilde{D}_c = PD_c$, we have $\tilde{D}_c \tilde{D}_c^\top = PD_c D_c^\top P^\top$. Using the permutation equivariance of Ψ ,

$$\Psi(\tilde{D}_c \tilde{D}_c^\top) = \Psi(PD_c D_c^\top P^\top) = P\Psi(D_c D_c^\top)P^\top.$$

Hence

$$\mathcal{U}_L^{\text{router}}(\tilde{D}) = \Psi(\tilde{D}_c \tilde{D}_c^\top) \tilde{D}_c = P\Psi(D_c D_c^\top) P^\top P D_c = P\Psi(D_c D_c^\top) D_c = P\mathcal{U}_L^{\text{router}}(D).$$

This proves expert-permutation equivariance and shared-row-shift invariance for $\mathcal{U}_L^{\text{router}}$.

It remains to verify horizontality. Since $D_c = \Pi_\perp D$, every column of D_c lies in $\mathbf{1}_e^\perp$. By assumption, $\Psi(D_c D_c^\top)$ maps the centered expert subspace to itself. Therefore every column of $\Psi(D_c D_c^\top) D_c$ also lies in $\mathbf{1}_e^\perp$, and hence

$$\mathbf{1}_e^\top \mathcal{U}_L^{\text{router}}(D) = 0.$$

We next consider the projected centered row-norm update. Define

$$A(D_c) := \text{Diag}(\eta(\|D_{c,1:}\|_2), \dots, \eta(\|D_{c,e:}\|_2)).$$

Since $\tilde{D}_c = PD_c$, the row norms of $\tilde{D}_c$ are the corresponding permutation of the row norms of D_c . Hence

$$A(\tilde{D}_c) = PA(D_c)P^\top.$$

Using again that $\Pi_\perp P = P\Pi_\perp$, we obtain

$$\mathcal{U}_{\text{row}}^{\text{router}}(\tilde{D}) = \Pi_\perp A(\tilde{D}_c) \tilde{D}_c = \Pi_\perp PA(D_c) P^\top P D_c = P\Pi_\perp A(D_c) D_c = P\mathcal{U}_{\text{row}}^{\text{router}}(D).$$

Thus the projected centered row-norm update is also expert-permutation equivariant and shared-row-shift invariant.

Finally, horizontality of the row-norm update follows directly from the final projection:

$$\mathbf{1}_e^\top \mathcal{U}_{\text{row}}^{\text{router}}(D) = \mathbf{1}_e^\top \Pi_\perp A(D_c) D_c = 0.$$

Therefore both update families are router-compatible. $\square$

Proof of Theorem 3.7. Suppose first that $\mathcal{U}$ is of the stated spectral form. Let $P \in \mathbb{O}^m$ and $Q \in \mathbb{O}^n$ be orthogonal. If $D = U \text{Diag}(\sigma(D)) V^\top$, then $PDQ^\top = (PU) \text{Diag}(\sigma(D)) (QV)^\top$ is a singular value decomposition of $PDQ^\top$. Therefore, we have

$$\mathcal{U}(PDQ^\top) = (PU) \text{Diag}(\psi(\sigma(D))) (QV)^\top = P\mathcal{U}(D)Q^\top,$$

so $\mathcal{U}$ is bi-orthogonally equivariant.

Conversely, suppose that $\mathcal{U}$ is bi-orthogonally equivariant. Let $D = U \text{Diag}(\sigma(D)) V^\top$ be a singular value decomposition of D . Then we have

$$\mathcal{U}(D) = \mathcal{U}(U \text{Diag}(\sigma(D)) V^\top) = U\mathcal{U}(\text{Diag}(\sigma(D))) V^\top.$$

Thus the action of $\mathcal{U}$ is completely determined by its action on diagonal matrices of singular values. Since bi-orthogonal equivariance ensures $\mathcal{U}$ commutes with arbitrary diagonal sign matrices, forcing all off-diagonal elements to be zero, let us define $\text{Diag}(\psi(s)) := \mathcal{U}(\text{Diag}(s))$, where $s \in \mathbb{R}_+^r$. The bi-orthogonal equivariance of $\mathcal{U}$ further implies that this definition is independent of the particular singular value decomposition and that ψ is absolutely symmetric with respect to permutations and sign changes compatible with singular-value representations. Hence we have $\mathcal{U}(D) = U \text{Diag}(\psi(\sigma(D)))V^\top$, which is exactly the claimed spectral form. $\square$

E Implementation Details of Practical Optimizers

We provide the implementation details of our proposed practical optimizers in this section.

E.1 Numerical Algorithm for Matrix Inverse Square Root via Polar Express

We illustrate below the algorithm for computing matrix inverse square root via POLAR EXPRESS, which is based on the intrinsic connection between polynomial iterations for computing the orthogonal polar factor and those for computing the inverse square root of a square matrix, stated in the following theorem.

Theorem E.1 (Higham [67]). *Let $A \in \mathbb{R}^{n \times n}$ be a square matrix with nonnegative eigenvalues. Consider any iteration of the form $X_{k+1} = X_k h(X_k^2)$ that converges to $\text{msgn}(X_0)$ for $X_0 = \begin{pmatrix} 0 & A \\ I_n & 0 \end{pmatrix}$ with order of convergence q . Then in the coupled iteration $X_{k+1} = X_k h(Y_k X_k)$, $Y_{k+1} = h(Y_k X_k) Y_k$, with $X_0 = A$ and $Y_0 = I_n$, we have $X_k \rightarrow A^{1/2}$ and $Y_k \rightarrow A^{-1/2}$, both with order of convergence q . Here msgn denotes the matrix sign function.*

Algorithm E.1 Matrix Inverse Square Root via POLAR EXPRESS [5]

Input: $A \in \mathbb{R}^{n \times n}$, $\varepsilon > 0$, $K \in \mathbb{N}^*$, sequence of POLAR EXPRESS coefficients $\{(a_k, b_k, c_k)\}_{k=1}^K$

- 1: $A = (A + A^\top)/2 + \varepsilon I_n$
- 2: $\alpha = 1.02 \|A\|_F + \varepsilon$
- 3: $Y_1 = A/\alpha$
- 4: $Z_1 = I_n$
- 5: **for** $k = 1, \dots, K$ **do**
- 6: $(\bar{a}_k, \bar{b}_k, \bar{c}_k) = (a_k + b_k + c_k, -b_k - 2c_k, c_k)$
- 7: $R_k = I_n - Z_k Y_k$
- 8: $R_k = (R_k + R_k^\top)/2$ (optional matrix symmetrization)
- 9: $M_k = \bar{a}_k I_n + \bar{b}_k R_k + \bar{c}_k R_k^2$
- 10: $Y_{k+1} = Y_k M_k$
- 11: $Z_{k+1} = M_k Z_k$
- 12: **end for**

Return: $Z_{K+1}/\sqrt{\alpha}$

F Convergence Analysis of Symmetry-Compatible Optimizers

In this section, we study the convergence of several symmetry-compatible optimizer classes introduced above, including full spectral, one-sided spectral, row-norm-based, and hybrid optimizers. To present the analysis in a unified way, we begin from a general first-order iteration and state the basic assumptions once. The subsequent subsections then specialize these assumptions to each optimizer class.

F.1 General Update Scheme and Standing Assumptions

We consider the generic iteration

$$(\forall k \in \mathbb{N}) \quad W_{k+1} = W_k - \gamma_k \mathcal{T}(G_k), \quad G_k := \nabla f(W_k), \quad (\text{F.1})$$

where $W_k \in \mathbb{R}^{m \times n}$, $\gamma_k > 0$ is the learning rate, and $\mathcal{T}: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ is an update map whose precise form depends on the optimizer class under consideration. We do not consider momentum in our analysis for simplicity and leave it to future work. The layerwise loss function f is assumed to satisfy the following standard regularity conditions.

Assumption F.1 (L -smoothness). Let $f: \mathbb{R}^{m \times n} \rightarrow \bar{\mathbb{R}}$ be differentiable and L -smooth with respect to the Frobenius norm, i.e., there exists $L \in (0, \infty)$ such that

$$(\forall X, Y \in \mathbb{R}^{m \times n}) \quad \|\nabla f(X) - \nabla f(Y)\|_{\text{F}} \leq L \|X - Y\|_{\text{F}}.$$

Assumption F.2 (μ -Polyak–Łojasiewicz). Let $f: \mathbb{R}^{m \times n} \rightarrow \bar{\mathbb{R}}$ satisfy the μ -Polyak–Łojasiewicz (PL) inequality:

$$(\forall X \in \mathbb{R}^{m \times n}) \quad \|\nabla f(X)\|_{\text{F}}^2 \geq 2\mu(f(X) - f^*),$$

where $f^* := \inf f$.

The PL condition will only be needed to obtain linear convergence. Under smoothness alone, we will still obtain monotonic descent and sublinear convergence to stationarity.

Remark F.1 (On the stochastic setting). In large-scale deep learning, one typically replaces the full gradient G by a stochastic gradient estimator $\hat{G}$. However, for the symmetry-compatible update maps considered here, including spectral, one-sided spectral, row-norm-based, and hybrid updates, the map $\mathcal{T}$ is generally nonlinear. As a result, unbiasedness of the stochastic gradient does not imply unbiasedness of the resulting update direction. A rigorous stochastic analysis therefore requires additional assumptions directly on the expected alignment and norm of $\mathcal{T}(\hat{G})$, and is left for future work.

Throughout the section, the basic descent estimate follows from the standard smoothness inequality:

$$f(W_{k+1}) \leq f(W_k) - \gamma_k \langle G_k, \mathcal{T}(G_k) \rangle_{\text{F}} + \frac{L\gamma_k^2}{2} \|\mathcal{T}(G_k)\|_{\text{F}}^2.$$

Thus, the convergence analysis for each optimizer class reduces to controlling two quantities $\langle G, \mathcal{T}(G) \rangle_{\text{F}}$ and $\|\mathcal{T}(G)\|_{\text{F}}^2$. The relevant bounds depend on the geometry of the update map $\mathcal{T}$, and are stated separately in the subsections below.

We now specialize (F.1) to each symmetry-compatible optimizer class introduced earlier. In each case, the key step is to identify the geometry-dependent alignment term $\langle G, \mathcal{T}(G) \rangle_{\text{F}}$ and the corresponding update norm $\|\mathcal{T}(G)\|_{\text{F}}^2$, which together determine the admissible learning rates and convergence rates.

The optimizer classes considered in this section fall into two broad regimes. The first consists of scale-compatible updates, for which $\|\mathcal{T}(G)\|_F$ scales proportionally to $\|G\|_F$. This includes standard spectral, one-sided spectral, and bounded row-norm-based updates. The second consists of fully normalized updates, such as polar or row-normalized directions, whose magnitude is controlled by rank or support rather than gradient norm. The former admit convergence bounds through uniform alignment and norm-control constants, whereas the latter are more naturally analyzed through geometry-dependent ratio quantities.

F.2 Full Spectral Optimizers

We begin with full spectral optimizers. The following assumption captures the two structural properties needed for convergence: positive alignment with the gradient and control of the update norm. We specialize (F.1) to the full spectral case, where $\mathcal{T}$ is a spectral operator.

Assumption F.3 (Singular-value alignment and boundedness). Let $\mathcal{T}: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ be the spectral update map defined by $\mathcal{T}(G) = U \text{Diag}(\psi(\sigma(G)))V^\top$ whenever $G = U \text{Diag}(\sigma(G))V^\top$ is a singular value decomposition of G , for some absolutely symmetric map $\psi: \mathbb{R}_+^r \rightarrow \mathbb{R}^r$. Assume there exist constants $0 < c_1 \leq c_2 < \infty$ such that for all $s \in \mathbb{R}_+^r$,

$$\sum_{i=1}^r s_i \psi_i(s) \geq c_1 \sum_{i=1}^r s_i^2 \quad \text{and} \quad \sum_{i=1}^r \psi_i(s)^2 \leq c_2 \sum_{i=1}^r s_i^2,$$

where $r = \min\{m, n\}$.

Lemma F.1 (Alignment and norm bounds). Under Assumption F.3, for all $G \in \mathbb{R}^{m \times n}$, we have $\langle G, \mathcal{T}(G) \rangle_F \geq c_1 \|G\|_F^2$ and $\|\mathcal{T}(G)\|_F^2 \leq c_2 \|G\|_F^2$.

Proof. If we write $s = \sigma(G)$, then we have $\mathcal{T}(G) = U \text{Diag}(\psi(s))V^\top$. Using orthogonal invariance of the Frobenius inner product, we have $\langle G, \mathcal{T}(G) \rangle_F = \langle \text{Diag}(s), \text{Diag}(\psi(s)) \rangle_F = \sum_{i=1}^r s_i \psi_i(s)$. By Assumption F.3, we have $\langle G, \mathcal{T}(G) \rangle_F \geq c_1 \sum_{i=1}^r s_i^2 = c_1 \|G\|_F^2$. Similarly, we also have $\|\mathcal{T}(G)\|_F^2 = \sum_{i=1}^r \psi_i(s)^2 \leq c_2 \sum_{i=1}^r s_i^2 = c_2 \|G\|_F^2$. $\square$

Theorem F.2 (Descent lemma for spectral optimizers). Suppose Assumptions F.1 and F.3 hold. Then the iteration (F.1) satisfies

$$f(W_{k+1}) \leq f(W_k) - \gamma_k \langle G_k, \mathcal{T}(G_k) \rangle_F + \frac{L\gamma_k^2}{2} \|\mathcal{T}(G_k)\|_F^2. \quad (\text{F.2})$$

Consequently, we have $f(W_{k+1}) \leq f(W_k) - (c_1\gamma_k - Lc_2\gamma_k^2/2) \|G_k\|_F^2$. In particular, if $\gamma_k \in (0, 2c_1/(Lc_2))$, then $f(W_{k+1}) \leq f(W_k)$.

Proof. By L -smoothness of f , we have

$$f(W_{k+1}) \leq f(W_k) + \langle \nabla f(W_k), W_{k+1} - W_k \rangle_F + \frac{L}{2} \|W_{k+1} - W_k\|_F^2.$$

Using $W_{k+1} - W_k = -\gamma_k \mathcal{T}(G_k)$ and $G_k = \nabla f(W_k)$, we obtain (F.2). Now apply Lemma F.1 to obtain $\langle G_k, \mathcal{T}(G_k) \rangle_F \geq c_1 \|G_k\|_F^2$ and $\|\mathcal{T}(G_k)\|_F^2 \leq c_2 \|G_k\|_F^2$, which yields $f(W_{k+1}) \leq f(W_k) - (c_1\gamma_k - Lc_2\gamma_k^2/2) \|G_k\|_F^2$. $\square$

Theorem F.3 (Sublinear convergence to stationarity). Suppose Assumptions F.1 and F.3 hold. If the learning rate is constant and satisfies $\gamma \in (0, 2c_1/(Lc_2))$, then

$$\sum_{k=0}^{T-1} \|G_k\|_F^2 \leq \frac{f(W_0) - f^*}{\gamma(c_1 - Lc_2\gamma/2)}, \quad \text{and therefore} \quad \min_{0 \leq k < T} \|G_k\|_F^2 \leq \frac{f(W_0) - f^*}{T\gamma(c_1 - Lc_2\gamma/2)}.$$

Proof. By Theorem F.2, $f(W_{k+1}) \leq f(W_k) - \gamma(c_1 - Lc_2\gamma/2)\|G_k\|_F^2$. Summing from $k = 0$ to $T - 1$ yields

$$f(W_T) \leq f(W_0) - \gamma\left(c_1 - \frac{Lc_2}{2}\gamma\right) \sum_{k=0}^{T-1} \|G_k\|_F^2.$$

Since $f(W_T) \geq f^*$, rearranging gives $\sum_{k=0}^{T-1} \|G_k\|_F^2 \leq (f(W_0) - f^*)/(\gamma(c_1 - Lc_2\gamma/2))$. The second inequality follows from $\min_{0 \leq k < T} \|G_k\|_F^2 \leq \frac{1}{T} \sum_{k=0}^{T-1} \|G_k\|_F^2$. $\square$

Theorem F.4 (Linear convergence under the PŁ condition). *Suppose Assumptions F.1 to F.3 hold. If the learning rate is constant and satisfies $\gamma \in (0, 2c_1/(Lc_2))$, then the spectral iteration (F.1) obeys*

$$f(W_{k+1}) - f^* \leq \left(1 - 2\mu\gamma\left(c_1 - \frac{Lc_2}{2}\gamma\right)\right)(f(W_k) - f^*).$$

Hence $f(W_k) - f^* \leq \rho^k(f(W_0) - f^*)$, where $\rho := 1 - 2\mu\gamma(c_1 - Lc_2\gamma/2) \in (0, 1)$.

Proof. By Theorem F.2, subtracting f^* from both sides gives

$$f(W_{k+1}) - f^* \leq f(W_k) - f^* - \left(c_1\gamma - \frac{Lc_2}{2}\gamma^2\right)\|G_k\|_F^2.$$

Using the PŁ inequality, we obtain the desired inequality. Since $0 < \gamma < 2c_1/(Lc_2)$, the factor $c_1 - Lc_2\gamma/2$ is positive, so $\rho \in (0, 1)$. Iterating the recursion yields $f(W_k) - f^* \leq \rho^k(f(W_0) - f^*)$. $\square$

Corollary F.5 (Optimal constant learning rate within this bound). *Under the assumptions of Theorem F.4, the contraction factor in Theorem F.4 is minimized over constant learning rates $\gamma > 0$ by $\gamma^* = c_1/(Lc_2)$, for which $\rho^* = 1 - \mu c_1^2/(Lc_2)$.*

Proof. The contraction factor $\rho(\gamma) = 1 - 2\mu\gamma(c_1 - Lc_2\gamma/2)$ is minimized when the quadratic $-2\mu c_1\gamma + \mu Lc_2\gamma^2$ is minimized, equivalently when $c_1\gamma - Lc_2\gamma^2/2$ is maximized. This occurs at $\gamma^* = c_1/(Lc_2)$. Substituting into $\rho(\gamma)$ gives $\rho^* = 1 - 2\mu \frac{c_1}{Lc_2} \left(c_1 - \frac{Lc_2}{2} \frac{c_1}{Lc_2}\right) = 1 - \mu c_1^2/(Lc_2)$. $\square$

F.2.1 Specialization to Normalized Polar-Type Spectral Methods

The abstract convergence result above assumes that the spectral update $\mathcal{T}(G)$ grows proportionally to the gradient norm. This covers many spectral maps, but excludes fully normalized polar updates such as $\mathcal{T}(G) = \text{polar}(G) = UV^\top$, for which $\|\mathcal{T}(G)\|_F^2 = \text{rank}(G)$ does not vanish as $G \rightarrow 0$ (cf. *null-gradient consistency* defined in [91]). To analyze such methods, it is more natural to work with a geometry-dependent ratio between the update norm and its alignment with the gradient, defined below.

Assumption F.4 (Positive alignment). For every $G \in \mathbb{R}^{m \times n} \setminus \{0\}$, we assume that $\langle G, \mathcal{T}(G) \rangle_F > 0$.

Definition F.1 (Spectral advantage ratio). For a spectral update map $\mathcal{T}$, define its *spectral advantage ratio* at $G \neq 0$ by

$$\mathfrak{R}_{\mathcal{T}}(G) := \frac{\langle G, \mathcal{T}(G) \rangle_F}{\|\mathcal{T}(G)\|_F^2}.$$

Equivalently, its reciprocal is $\mathfrak{R}_{\mathcal{T}}^{-1}(G) = \|\mathcal{T}(G)\|_F^2 / \langle G, \mathcal{T}(G) \rangle_F$.

The quantity $\mathfrak{R}_{\mathcal{T}}(G)$ measures how much descent is obtained per unit squared update norm. Larger values correspond to more favorable geometry.

Lemma F.6 (Descent lemma in ratio form). *Suppose Assumptions F.1 and F.4 hold. Then the iteration (F.1) satisfies*

$$f(W_{k+1}) \leq f(W_k) - \left(\gamma_k - \frac{L\gamma_k^2}{2\mathfrak{R}_{\mathcal{T}}(G_k)} \right) \langle\langle G_k, \mathcal{T}(G_k) \rangle\rangle_{\mathbb{F}}.$$

In particular, if $0 < \gamma_k < 2\mathfrak{R}_{\mathcal{T}}(G_k)/L$, then $f(W_{k+1}) \leq f(W_k)$.

Proof. The inequality follows from L -smoothness of f and the definition of $\mathfrak{R}_{\mathcal{T}}(G_k)$. The final claim follows immediately. $\square$

The preceding lemma shows that normalized spectral methods are governed by two distinct quantities: the ratio term $\mathfrak{R}_{\mathcal{T}}(G)$, which controls the admissible learning rate through the smoothness bound, and the alignment term $\langle\langle G, \mathcal{T}(G) \rangle\rangle_{\mathbb{F}}$, which must still be related to $\|G\|_{\mathbb{F}}^2$ in order to combine the descent estimate with the PL inequality.

Theorem F.7 (Linear convergence under PL and ratio/alignment bounds). *Suppose Assumptions F.1, F.2 and F.4 hold. Assume there exists a constant $\underline{\mathfrak{R}} > 0$ such that $\mathfrak{R}_{\mathcal{T}}(G) \geq \underline{\mathfrak{R}}$ for all $G \neq 0$, and that there exists $\alpha > 0$ such that $\langle\langle G, \mathcal{T}(G) \rangle\rangle_{\mathbb{F}} \geq \alpha \|G\|_{\mathbb{F}}^2$ for all $G \in \mathbb{R}^{m \times n}$. If the learning rate is constant and satisfies $0 < \gamma < 2\underline{\mathfrak{R}}/L$, then*

$$f(W_{k+1}) - f^* \leq \left(1 - 2\mu\alpha\gamma \left(1 - \frac{L\gamma}{2\underline{\mathfrak{R}}} \right) \right) (f(W_k) - f^*).$$

Hence $f(W_k) - f^ \leq \rho^k (f(W_0) - f^*)$, where $\rho := 1 - 2\mu\alpha\gamma(1 - L\gamma/(2\underline{\mathfrak{R}})) \in (0, 1)$.*

Proof. By Lemma F.6, $f(W_{k+1}) \leq f(W_k) - \left(\gamma - \frac{L\gamma^2}{2\mathfrak{R}_{\mathcal{T}}(G_k)} \right) \langle\langle G_k, \mathcal{T}(G_k) \rangle\rangle_{\mathbb{F}}$. Using $\mathfrak{R}_{\mathcal{T}}(G) \geq \underline{\mathfrak{R}}$ and $\langle\langle G, \mathcal{T}(G) \rangle\rangle_{\mathbb{F}} \geq \alpha \|G\|_{\mathbb{F}}^2$, we obtain

$$f(W_{k+1}) \leq f(W_k) - \alpha\gamma \left(1 - \frac{L\gamma}{2\underline{\mathfrak{R}}} \right) \|G_k\|_{\mathbb{F}}^2.$$

Applying the PL inequality yields the desired inequality. Since $0 < \gamma < 2\underline{\mathfrak{R}}/L$, the factor $1 - L\gamma/(2\underline{\mathfrak{R}})$ is positive, so $\rho \in (0, 1)$. Iterating the recursion proves the claim. $\square$

Specialization to the polar update. For the normalized polar update $\mathcal{T}(G) = \text{polar}(G) = UV^\top$ when $G = U\Sigma V^\top$, one has $\langle\langle G, \text{polar}(G) \rangle\rangle_{\mathbb{F}} = \|G\|_{\text{nuc}}$ and $\|\text{polar}(G)\|_{\mathbb{F}}^2 = \text{rank}(G)$, and therefore $\mathfrak{R}_{\text{polar}}(G) = \|G\|_{\text{nuc}}/\text{rank}(G)$. This is the average nonzero singular value of G . In particular, the descent condition becomes

$$0 < \gamma_k < \frac{2\|G_k\|_{\text{nuc}}}{L \text{rank}(G_k)}.$$

Moreover, $\langle\langle G, \text{polar}(G) \rangle\rangle_{\mathbb{F}} = \|G\|_{\text{nuc}} \geq \|G\|_{\mathbb{F}}$, but in general one does not have a uniform lower bound of the form

$$\langle\langle G, \text{polar}(G) \rangle\rangle_{\mathbb{F}} \geq \alpha \|G\|_{\mathbb{F}}^2$$

without additional scale control. Thus, for normalized polar updates, the ratio $\mathfrak{R}_{\text{polar}}(G)$ naturally governs the admissible learning rate, whereas stronger convergence rates require an additional lower bound relating $\|G\|_{\text{nuc}}$ to $\|G\|_{\mathbb{F}}^2$. Using the stable rank defined by $\text{srnk}(G) := \|G\|_{\mathbb{F}}^2/\|G\|_{\mathbb{S}}^2$, one has

$$\|G\|_{\text{nuc}} \geq \frac{\|G\|_{\mathbb{F}}^2}{\|G\|_{\mathbb{S}}} = \text{srnk}(G) \|G\|_{\mathbb{S}},$$

and hence

$$\mathfrak{R}_{\text{polar}}(G) = \frac{\|G\|_{\text{nuc}}}{\text{rank}(G)} \geq \frac{\text{srank}(G)}{\text{rank}(G)} \|G\|_{\text{S}}.$$

This lower bound shows explicitly how the descent margin depends on the spectral spread of the gradient: flatter spectra, corresponding to larger stable rank, lead to more favorable ratio bounds for the polar update.

F.2.2 Specialization to PolarGrad with Nuclear-Norm Scaling

For POLARGRAD with nuclear-norm scaling [91], the normalized polar direction is rescaled in such a way that both its alignment with the gradient and its squared Frobenius norm admit exact closed-form expressions. This removes the scale ambiguity present in fully normalized polar updates and leads to a particularly transparent descent analysis in terms of gradient rank and stable rank.

Given a gradient matrix $G = U \text{Diag}(\sigma(G))V^\top$, define the update map

$$\mathcal{T}_{\text{PG}}(G) := \|G\|_{\text{nuc}} \text{polar}(G) = \|G\|_{\text{nuc}} UV^\top.$$

The corresponding iteration is

$$(\forall k \in \mathbb{N}) \quad W_{k+1} = W_k - \gamma_k \|G_k\|_{\text{nuc}} \text{polar}(G_k), \quad G_k = \nabla f(W_k). \quad (\text{F.3})$$

In what follows, we write $r_k := \text{rank}(G_k)$.

Lemma F.8 (Alignment and norm identities for PolarGrad). *For every $G \in \mathbb{R}^{m \times n}$, we have $\langle G, \mathcal{T}_{\text{PG}}(G) \rangle_{\text{F}} = \|G\|_{\text{nuc}}^2$ and $\|\mathcal{T}_{\text{PG}}(G)\|_{\text{F}}^2 = \text{rank}(G) \|G\|_{\text{nuc}}^2$. Consequently,*

$$\mathfrak{R}_{\mathcal{T}_{\text{PG}}}(G) = \frac{\langle G, \mathcal{T}_{\text{PG}}(G) \rangle_{\text{F}}}{\|\mathcal{T}_{\text{PG}}(G)\|_{\text{F}}^2} = \frac{1}{\text{rank}(G)}.$$

Proof. Using orthogonal invariance of the Frobenius inner product,

$$\langle G, \mathcal{T}_{\text{PG}}(G) \rangle_{\text{F}} = \|G\|_{\text{nuc}} \langle U \text{Diag}(\sigma(G))V^\top, UV^\top \rangle_{\text{F}} = \|G\|_{\text{nuc}} \sum_{i=1}^r \sigma_i(G) = \|G\|_{\text{nuc}}^2.$$

Also, $\|\mathcal{T}_{\text{PG}}(G)\|_{\text{F}}^2 = \|G\|_{\text{nuc}}^2 \|UV^\top\|_{\text{F}}^2 = \|G\|_{\text{nuc}}^2 \text{rank}(G)$. The ratio identity follows immediately. $\square$

Theorem F.9 (Descent lemma for nuclear-norm-scaled PolarGrad). *Suppose f satisfies Assumption F.1. Then the iteration (F.3) satisfies*

$$f(W_{k+1}) \leq f(W_k) - \gamma_k \|G_k\|_{\text{nuc}}^2 + \frac{L\gamma_k^2}{2} r_k \|G_k\|_{\text{nuc}}^2.$$

Equivalently,

$$f(W_{k+1}) \leq f(W_k) - \gamma_k \left(1 - \frac{L\gamma_k}{2} r_k\right) \|G_k\|_{\text{nuc}}^2.$$

In particular, if $\gamma_k < 2/(Lr_k)$, then $f(W_{k+1}) \leq f(W_k)$.

Proof. By L -smoothness of f and $W_{k+1} - W_k = -\gamma_k \mathcal{T}_{\text{PG}}(G_k)$, we obtain

$$f(W_{k+1}) \leq f(W_k) - \gamma_k \langle G_k, \mathcal{T}_{\text{PG}}(G_k) \rangle_{\text{F}} + \frac{L\gamma_k^2}{2} \|\mathcal{T}_{\text{PG}}(G_k)\|_{\text{F}}^2.$$

Now apply Lemma F.8. $\square$

Corollary F.10 (Stable-rank improvement). *Under the assumptions of Theorem F.9,*

$$f(W_{k+1}) \leq f(W_k) - \gamma_k \left(1 - \frac{L\gamma_k}{2} r_k\right) \|G_k\|_{\text{nuc}}^2.$$

Since $\|G_k\|_{\text{nuc}}^2 \geq \text{srnk}(G_k) \|G_k\|_{\text{F}}^2$ with $\text{srnk}(G_k) := \|G_k\|_{\text{F}}^2 / \|G_k\|_{\text{S}}^2$, it follows that

$$f(W_{k+1}) \leq f(W_k) - \gamma_k \left(1 - \frac{L\gamma_k}{2} r_k\right) \text{srnk}(G_k) \|G_k\|_{\text{F}}^2.$$

Thus, compared with vanilla gradient descent, nuclear-norm-scaled POLARGRAD admits a potentially stronger one-step decrease when the gradient has nontrivial spectral spread, measured by its stable rank.

Theorem F.11 (Sublinear convergence to stationarity for nuclear-norm-scaled PolarGrad). *Suppose f satisfies Assumption F.1. Consider the iteration (F.3) with constant learning rate $\gamma > 0$. Assume there exists $\bar{r} \in \mathbb{N}^*$ such that $r_k \leq \bar{r}$ for all $k \in \mathbb{N}$, and that $\gamma \in (0, 2/(L\bar{r}))$. Then*

$$f(W_{k+1}) \leq f(W_k) - \gamma \left(1 - \frac{L\gamma}{2} \bar{r}\right) \|G_k\|_{\text{nuc}}^2.$$

Consequently, we have

$$\sum_{k=0}^{T-1} \|G_k\|_{\text{F}}^2 \leq \sum_{k=0}^{T-1} \|G_k\|_{\text{nuc}}^2 \leq \frac{f(W_0) - f^*}{\gamma(1 - L\gamma\bar{r}/2)} \quad \text{and} \quad \min_{0 \leq k < T} \|G_k\|_{\text{F}}^2 \leq \frac{f(W_0) - f^*}{T\gamma(1 - L\gamma\bar{r}/2)}.$$

Hence the method converges to stationarity at the standard $\mathcal{O}(1/T)$ rate in the minimum gradient norm.

Proof. By Theorem F.9, $f(W_{k+1}) \leq f(W_k) - \gamma \left(1 - \frac{L\gamma}{2} r_k\right) \|G_k\|_{\text{nuc}}^2$. Using $r_k \leq \bar{r}$, we obtain $f(W_{k+1}) \leq f(W_k) - \gamma \left(1 - \frac{L\gamma}{2} \bar{r}\right) \|G_k\|_{\text{nuc}}^2$. Summing from $k = 0$ to $T - 1$ yields

$$f(W_T) \leq f(W_0) - \gamma \left(1 - \frac{L\gamma}{2} \bar{r}\right) \sum_{k=0}^{T-1} \|G_k\|_{\text{nuc}}^2.$$

Since $f(W_T) \geq f^*$ and $\|G_k\|_{\text{nuc}} \geq \|G_k\|_{\text{F}}$,

$$\sum_{k=0}^{T-1} \|G_k\|_{\text{F}}^2 \leq \sum_{k=0}^{T-1} \|G_k\|_{\text{nuc}}^2 \leq \frac{f(W_0) - f^*}{\gamma(1 - L\gamma\bar{r}/2)}.$$

Finally, $\min_{0 \leq k < T} \|G_k\|_{\text{F}}^2 \leq \frac{1}{T} \sum_{k=0}^{T-1} \|G_k\|_{\text{F}}^2$ proves the stated $\mathcal{O}(1/T)$ bound. $\square$

Theorem F.12 (Linear convergence under PŁ for nuclear-norm-scaled PolarGrad). *Suppose f satisfies Assumptions F.1 and F.2. Assume there exists $\bar{r} \in \mathbb{N}^*$ such that $r_k \leq \bar{r}$ for all $k \in \mathbb{N}$. Then any constant learning rate satisfying $\gamma \in (0, 2/(L\bar{r}))$ yields*

$$f(W_{k+1}) - f^* \leq \left(1 - 2\mu\gamma \left(1 - \frac{L\gamma}{2} \bar{r}\right)\right) (f(W_k) - f^*).$$

Hence $f(W_k) - f^* \leq \rho^k (f(W_0) - f^*)$, where $\rho = 1 - 2\mu\gamma(1 - L\gamma\bar{r}/2) \in (0, 1)$.

Proof. By Theorem F.9, and using $r_k \leq \bar{r}$ and $\|G_k\|_{\text{nuc}}^2 \geq \|G_k\|_{\text{F}}^2$, we obtain

$$f(W_{k+1}) \leq f(W_k) - \gamma \left(1 - \frac{L\gamma}{2} \bar{r}\right) \|G_k\|_{\text{F}}^2.$$

Applying the PŁ inequality gives

$$f(W_{k+1}) - f^* \leq \left(1 - 2\mu\gamma \left(1 - \frac{L\gamma}{2} \bar{r}\right)\right) (f(W_k) - f^*).$$

Since $0 < \gamma < 2/(L\bar{r})$, the factor $1 - L\gamma\bar{r}/2$ is positive, hence $\rho \in (0, 1)$. $\square$

Remark F.2 (Relaxing the global PŁ condition). The global PŁ condition is used here only to obtain global linear convergence. Under smoothness alone, nuclear-norm-scaled POLARGRAD still enjoys monotonic descent and an $\mathcal{O}(1/T)$ convergence rate to stationarity in the minimum gradient norm. If the PŁ inequality holds only locally in a neighborhood of a limit point, then the same descent argument yields eventual linear convergence once the iterates enter that neighborhood. More generally, one may replace the PŁ condition by a Kurdyka–Łojasiewicz (KŁ) inequality [110, 88], which provides a broader convergence framework and recovers linear or sublinear rates depending on the local KŁ exponent.

F.3 One-Sided Spectral Optimizers

We now specialize the preceding convergence analysis to one-sided spectral optimizers. For one-sided spectral optimizers, the convergence analysis follows the same smoothness-based template as in the full spectral case. The only additional ingredients are alignment and norm-control conditions adapted to the relevant one-sided Gram operator. These yield standard descent, sublinear stationarity, and linear convergence under the PŁ condition.

Recall that right-spectral updates take the form $\mathcal{T}_{\text{R}}(G) = G\Psi(G^{\top}G)$, while left-spectral updates take the form $\mathcal{T}_{\text{L}}(G) = \Phi(GG^{\top})G$, where Ψ and Φ are orthogonally equivariant spectral operators on the corresponding Gram matrices.

F.3.1 Right-Spectral Optimizers

Consider the iteration

$$(\forall k \in \mathbb{N}) \quad W_{k+1} = W_k - \gamma_k \mathcal{T}_{\text{R}}(G_k), \quad G_k := \nabla f(W_k), \quad (\text{F.4})$$

with $\mathcal{T}_{\text{R}}(G) = G\Psi(G^{\top}G)$.

Assumption F.5 (Right-spectral eigenvalue alignment and boundedness). For every $G \in \mathbb{R}^{m \times n}$, let $G^{\top}G = V \text{Diag}(\lambda(G^{\top}G))V^{\top}$ and $\Psi(G^{\top}G) = V \text{Diag}(\psi(\lambda(G^{\top}G)))V^{\top}$, where $\lambda(G^{\top}G) = (\lambda_1, \dots, \lambda_n) \in \mathbb{R}_+^n$. Assume there exist constants $0 < c_{\text{R},1} \leq c_{\text{R},2} < \infty$ such that for all $\lambda \in \mathbb{R}_+^n$,

$$\sum_{i=1}^n \lambda_i \psi_i(\lambda) \geq c_{\text{R},1} \sum_{i=1}^n \lambda_i \quad \text{and} \quad \sum_{i=1}^n \lambda_i \psi_i(\lambda)^2 \leq c_{\text{R},2} \sum_{i=1}^n \lambda_i.$$

Lemma F.13 (Right-spectral alignment identities). Under Assumption F.5, for all $G \in \mathbb{R}^{m \times n}$, $\langle\langle G, \mathcal{T}_{\text{R}}(G) \rangle\rangle_{\text{F}} = \sum_{i=1}^n \lambda_i \psi_i(\lambda)$ and $\|\mathcal{T}_{\text{R}}(G)\|_{\text{F}}^2 = \sum_{i=1}^n \lambda_i \psi_i(\lambda)^2$, where $\lambda = \lambda(G^{\top}G)$. Consequently, $\langle\langle G, \mathcal{T}_{\text{R}}(G) \rangle\rangle_{\text{F}} \geq c_{\text{R},1} \|G\|_{\text{F}}^2$ and $\|\mathcal{T}_{\text{R}}(G)\|_{\text{F}}^2 \leq c_{\text{R},2} \|G\|_{\text{F}}^2$.

Proof. Let $G = U \text{Diag}(\sigma(G))V^\top$ be a singular value decomposition of G . Since $\lambda_i(G^\top G) = \sigma_i(G)^2$, we may write $G^\top G = V \text{Diag}(\lambda)V^\top$ and $\Psi(G^\top G) = V \text{Diag}(\psi(\lambda))V^\top$. Therefore,

$$\mathcal{T}_R(G) = G\Psi(G^\top G) = U \text{Diag}(\sigma(G))V^\top V \text{Diag}(\psi(\lambda))V^\top = U \text{Diag}(\sigma_i(G)\psi_i(\lambda))V^\top.$$

Hence $\langle\langle G, \mathcal{T}_R(G) \rangle\rangle_F = \sum_{i=1}^n \sigma_i(G)^2 \psi_i(\lambda) = \sum_{i=1}^n \lambda_i \psi_i(\lambda)$, and $\|\mathcal{T}_R(G)\|_F^2 = \sum_{i=1}^n \sigma_i(G)^2 \psi_i(\lambda)^2 = \sum_{i=1}^n \lambda_i \psi_i(\lambda)^2$. The final inequalities follow from Assumption F.5 and $\sum_{i=1}^n \lambda_i = \|G\|_F^2$. $\square$

Theorem F.14 (Right-spectral descent and convergence). *Suppose Assumptions F.1 and F.5 hold. Then the iteration (F.4) satisfies*

$$f(W_{k+1}) \leq f(W_k) - \left(c_{R,1}\gamma_k - \frac{L_{CR,2}}{2}\gamma_k^2 \right) \|G_k\|_F^2.$$

In particular, if $\gamma_k \in (0, 2c_{R,1}/(L_{CR,2}))$, then $f(W_{k+1}) \leq f(W_k)$. If, in addition, f satisfies Assumption F.2 and $\gamma_k \equiv \gamma$ is constant with $\gamma \in (0, 2c_{R,1}/(L_{CR,2}))$, then

$$f(W_{k+1}) - f^* \leq \left(1 - 2\mu\gamma \left(c_{R,1} - \frac{L_{CR,2}}{2}\gamma \right) \right) (f(W_k) - f^*),$$

and therefore $f(W_k) - f^ \leq \rho_R^k (f(W_0) - f^*)$, where $\rho_R = 1 - 2\mu\gamma(c_{R,1} - L_{CR,2}\gamma/2) \in (0, 1)$. Moreover, without the PL condition, if $\gamma_k \equiv \gamma$ is constant, then*

$$\min_{0 \leq k < T} \|G_k\|_F^2 \leq \frac{f(W_0) - f^*}{T\gamma(c_{R,1} - L_{CR,2}\gamma/2)}.$$

Proof. Combine the smoothness inequality with Lemma F.13, as in the proof of Theorem F.2, and then use either the PL inequality or summation of the descent bound. $\square$

F.3.2 Left-Spectral Optimizers

Consider the iteration

$$(\forall k \in \mathbb{N}) \quad W_{k+1} = W_k - \gamma_k \mathcal{T}_L(G_k), \quad G_k := \nabla f(W_k), \quad (\text{F.5})$$

with $\mathcal{T}_L(G) = \Phi(GG^\top)G$.

Assumption F.6 (Left-spectral eigenvalue alignment and boundedness). For every $G \in \mathbb{R}^{m \times n}$, let $GG^\top = U \text{Diag}(\lambda(GG^\top))U^\top$ and $\Phi(GG^\top) = U \text{Diag}(\phi(\lambda(GG^\top)))U^\top$, where $\lambda(GG^\top) = (\lambda_1, \dots, \lambda_m) \in \mathbb{R}_+^m$. Assume there exist constants $0 < c_{L,1} \leq c_{L,2} < \infty$ such that for all $\lambda \in \mathbb{R}_+^m$,

$$\sum_{i=1}^m \lambda_i \phi_i(\lambda) \geq c_{L,1} \sum_{i=1}^m \lambda_i \quad \text{and} \quad \sum_{i=1}^m \lambda_i \phi_i(\lambda)^2 \leq c_{L,2} \sum_{i=1}^m \lambda_i.$$

Lemma F.15 (Left-spectral alignment identities). *Under Assumption F.6, for all $G \in \mathbb{R}^{m \times n}$, $\langle\langle G, \mathcal{T}_L(G) \rangle\rangle_F = \sum_{i=1}^m \lambda_i \phi_i(\lambda)$ and $\|\mathcal{T}_L(G)\|_F^2 = \sum_{i=1}^m \lambda_i \phi_i(\lambda)^2$, where $\lambda = \lambda(GG^\top)$. Consequently, $\langle\langle G, \mathcal{T}_L(G) \rangle\rangle_F \geq c_{L,1} \|G\|_F^2$ and $\|\mathcal{T}_L(G)\|_F^2 \leq c_{L,2} \|G\|_F^2$.*

Proof. Let $G = U \text{Diag}(\sigma(G))V^\top$ be a singular value decomposition of G . Since $\lambda_i(GG^\top) = \sigma_i(G)^2$, we may write $GG^\top = U \text{Diag}(\lambda)U^\top$ and $\Phi(GG^\top) = U \text{Diag}(\phi(\lambda))U^\top$. Therefore,

$$\mathcal{T}_L(G) = \Phi(GG^\top)G = U \text{Diag}(\phi(\lambda))U^\top U \text{Diag}(\sigma(G))V^\top = U \text{Diag}(\phi_i(\lambda)\sigma_i(G))V^\top.$$

Hence $\langle\langle G, \mathcal{T}_L(G) \rangle\rangle_F = \sum_{i=1}^m \sigma_i(G)^2 \phi_i(\lambda) = \sum_{i=1}^m \lambda_i \phi_i(\lambda)$, and $\|\mathcal{T}_L(G)\|_F^2 = \sum_{i=1}^m \sigma_i(G)^2 \phi_i(\lambda)^2 = \sum_{i=1}^m \lambda_i \phi_i(\lambda)^2$. The final inequalities follow from Assumption F.6 and $\sum_{i=1}^m \lambda_i = \|G\|_F^2$. $\square$

Theorem F.16 (Left-spectral descent and convergence). *Suppose Assumptions F.1 and F.6 hold. Then the iteration (F.5) satisfies*

$$f(W_{k+1}) \leq f(W_k) - \left(c_{L,1} \gamma_k - \frac{L c_{L,2}}{2} \gamma_k^2 \right) \|G_k\|_F^2.$$

In particular, if $\gamma_k \in (0, 2c_{L,1}/(L c_{L,2}))$, then $f(W_{k+1}) \leq f(W_k)$. If, in addition, f satisfies Assumption F.2 and $\gamma_k \equiv \gamma$ is constant with $\gamma \in (0, 2c_{L,1}/(L c_{L,2}))$, then $f(W_{k+1}) \leq f(W_k)$ then

$$f(W_{k+1}) - f^* \leq \left(1 - 2\mu\gamma \left(c_{L,1} - \frac{L c_{L,2}}{2} \gamma \right) \right) (f(W_k) - f^*),$$

and therefore $f(W_k) - f^ \leq \rho_L^k (f(W_0) - f^*)$, where $\rho_L = 1 - 2\mu\gamma(c_{L,1} - L c_{L,2}\gamma/2) \in (0, 1)$. Moreover, without the PL condition, if $\gamma_k \equiv \gamma$ is constant, then*

$$\min_{0 \leq k < T} \|G_k\|_F^2 \leq \frac{f(W_0) - f^*}{T\gamma(c_{L,1} - L c_{L,2}\gamma/2)}.$$

Proof. Identical to the proof of Theorem F.14, replacing $\Psi(G^\top G)$ by $\Phi(GG^\top)$. $\square$

For the canonical one-sided polar updates, the abstract (c_1, c_2) -based analysis is less natural, and is better replaced by the same ratio-style viewpoint used for normalized POLARGRAD. After nuclear-norm scaling, however, both one-sided variants recover the same closed-form alignment and norm identities as full nuclear-norm-scaled POLARGRAD.

Theorem F.17 (Convergence of nuclear-norm-scaled one-sided PolarGrad). *Consider either the right-sided update*

$$\mathcal{T}_{\text{PG,R}}(G) = \|G\|_{\text{nuc}} G(G^\top G)^{\dagger/2}$$

or the left-sided update

$$\mathcal{T}_{\text{PG,L}}(G) = \|G\|_{\text{nuc}} (GG^\top)^{\dagger/2} G.$$

Then for both updates, $\langle\langle G, \mathcal{T}(G) \rangle\rangle_F = \|G\|_{\text{nuc}}^2$ and $\|\mathcal{T}(G)\|_F^2 = \text{rank}(G) \|G\|_{\text{nuc}}^2$. Hence the identities in Lemma F.8 remain valid for both one-sided nuclear-norm-scaled variants. Therefore, the descent, stationarity, and PL linear convergence results of Theorems F.9, F.11 and F.12 apply verbatim.

Proof. We first consider the right-sided update $\mathcal{T}_{\text{PG,R}}(G)$. Let $G = U \text{Diag}(\sigma(G)) V^\top$ be a singular value decomposition of G . Then

$$G(G^\top G)^{\dagger/2} = U \text{Diag}(\sigma(G)) V^\top \left(V \text{Diag}(\sigma(G)^2) V^\top \right)^{\dagger/2} = UV^\top = \text{polar}(G),$$

so $\mathcal{T}_{\text{PG,R}}(G) = \|G\|_{\text{nuc}} \text{polar}(G)$. Therefore, $\langle\langle G, \mathcal{T}_{\text{PG,R}}(G) \rangle\rangle_F = \|G\|_{\text{nuc}} \langle\langle G, \text{polar}(G) \rangle\rangle_F = \|G\|_{\text{nuc}}^2$, and $\|\mathcal{T}_{\text{PG,R}}(G)\|_F^2 = \|G\|_{\text{nuc}}^2 \|\text{polar}(G)\|_F^2 = \text{rank}(G) \|G\|_{\text{nuc}}^2$. The left-sided update follows similarly. The final claim follows immediately by invoking Theorems F.9, F.11 and F.12. $\square$

F.4 Row-Norm-Based Optimizers

We next study row-norm-based optimizers, whose update maps act locally on each row of the gradient matrix. Such methods are especially natural for parameter matrices whose row axis carries the relevant structural symmetry, as in embeddings, LM heads, and MoE routers.

Consider the iteration

$$(\forall k \in \mathbb{N}) \quad W_{k+1} = W_k - \gamma_k \mathcal{T}_{\text{row}}(G_k), \quad G_k := \nabla f(W_k), \quad (\text{F.6})$$

where $\mathcal{T}_{\text{row}}(G) = D_\eta(G) G$ and $D_\eta(G) := \text{Diag}(\eta(\|G_{1\cdot}\|_2), \dots, \eta(\|G_{v\cdot}\|_2))$, for some scalar function $\eta: \mathbb{R}_+ \rightarrow \mathbb{R}$.

Assumption F.7 (Uniform row-scaling bounds). There exist constants $0 < \underline{\eta} \leq \bar{\eta} < \infty$ such that $\underline{\eta} \leq \eta(t) \leq \bar{\eta}$ for all $t \geq 0$.

Lemma F.18 (Alignment and norm bounds for row-norm updates). Under Assumption F.7, for all $G \in \mathbb{R}^{v \times d}$,

$$\langle\langle G, \mathcal{T}_{\text{row}}(G) \rangle\rangle_{\text{F}} = \sum_{i=1}^v \eta(\|G_{i:}\|_2) \|G_{i:}\|_2^2 \geq \underline{\eta} \|G\|_{\text{F}}^2,$$

and

$$\|\mathcal{T}_{\text{row}}(G)\|_{\text{F}}^2 = \sum_{i=1}^v \eta(\|G_{i:}\|_2)^2 \|G_{i:}\|_2^2 \leq \bar{\eta}^2 \|G\|_{\text{F}}^2.$$

Proof. By definition, we have

$$\langle\langle G, \mathcal{T}_{\text{row}}(G) \rangle\rangle_{\text{F}} = \sum_{i=1}^v \langle\langle G_{i:}, \eta(\|G_{i:}\|_2) G_{i:} \rangle\rangle_{\text{F}} = \sum_{i=1}^v \eta(\|G_{i:}\|_2) \|G_{i:}\|_2^2.$$

Using $\eta(\|G_{i:}\|_2) \geq \underline{\eta}$, we obtain $\langle\langle G, \mathcal{T}_{\text{row}}(G) \rangle\rangle_{\text{F}} \geq \underline{\eta} \sum_{i=1}^v \|G_{i:}\|_2^2 = \underline{\eta} \|G\|_{\text{F}}^2$. Similarly, we also have

$$\|\mathcal{T}_{\text{row}}(G)\|_{\text{F}}^2 = \sum_{i=1}^v \eta(\|G_{i:}\|_2)^2 \|G_{i:}\|_2^2 \leq \bar{\eta}^2 \sum_{i=1}^v \|G_{i:}\|_2^2 = \bar{\eta}^2 \|G\|_{\text{F}}^2.$$

□

Theorem F.19 (One-step descent for row-norm-based optimizers). Suppose Assumptions F.1 and F.7 hold. Then the iteration (F.6) satisfies

$$f(W_{k+1}) \leq f(W_k) - \left(\underline{\eta} \gamma_k - \frac{L \bar{\eta}^2}{2} \gamma_k^2 \right) \|G_k\|_{\text{F}}^2.$$

In particular, if $\gamma_k \in (0, 2\underline{\eta}/(L\bar{\eta}^2))$, then $f(W_{k+1}) \leq f(W_k)$.

Proof. By L -smoothness of f and $W_{k+1} - W_k = -\gamma_k \mathcal{T}_{\text{row}}(G_k)$, we obtain

$$f(W_{k+1}) \leq f(W_k) - \gamma_k \langle\langle G_k, \mathcal{T}_{\text{row}}(G_k) \rangle\rangle_{\text{F}} + \frac{L \gamma_k^2}{2} \|\mathcal{T}_{\text{row}}(G_k)\|_{\text{F}}^2.$$

Apply Lemma F.18. □

Theorem F.20 (Sublinear convergence to stationarity). Suppose Assumptions F.1 and F.7 hold, and let $\gamma > 0$ be constant with $\gamma \in (0, 2\underline{\eta}/(L\bar{\eta}^2))$. Then

$$\sum_{k=0}^{T-1} \|G_k\|_{\text{F}}^2 \leq \frac{f(W_0) - f^*}{\gamma(\underline{\eta} - L\bar{\eta}^2\gamma/2)}, \quad \text{and therefore} \quad \min_{0 \leq k < T} \|G_k\|_{\text{F}}^2 \leq \frac{f(W_0) - f^*}{T\gamma(\underline{\eta} - L\bar{\eta}^2\gamma/2)}.$$

Proof. Sum the descent inequality in Theorem F.19. □

Theorem F.21 (Linear convergence under the PL condition). Suppose Assumptions F.1, F.2 and F.7 hold, and let $\gamma > 0$ be constant with $\gamma \in (0, 2\underline{\eta}/(L\bar{\eta}^2))$. Then

$$f(W_{k+1}) - f^* \leq \left(1 - 2\mu\gamma \left(\underline{\eta} - \frac{L\bar{\eta}^2}{2} \gamma \right) \right) (f(W_k) - f^*).$$

Hence $f(W_k) - f^* \leq \rho_{\text{row}}^k (f(W_0) - f^*)$, where $\rho_{\text{row}} = 1 - 2\mu\gamma(\underline{\eta} - L\bar{\eta}^2\gamma/2) \in (0, 1)$.

Proof. Combine Theorem F.19 with the PL inequality. □

F.4.1 Specialization to Smoothed Row Normalization

A useful smoothed variant of row normalization is obtained by taking $\eta(t) = 1/(t + \varepsilon)$ for some $\varepsilon > 0$. The corresponding update map is

$$\mathcal{T}_{\text{row},\varepsilon}(G) = \text{Diag}\left(\frac{1}{\|G_{1:}\|_2 + \varepsilon}, \dots, \frac{1}{\|G_{v:}\|_2 + \varepsilon}\right)G.$$

Unlike the fully normalized choice $\eta(t) = 1/t$, this smoothed row-norm-based map remains bounded at zero and therefore fits naturally into the preceding bounded-scaling framework.

Assumption F.8 (Uniform row-norm upper bound). There exists $M > 0$ such that $\|G_{k,i}\|_2 \leq M$ for all $k \in \mathbb{N}$, $i \in \llbracket v \rrbracket := \{1, \dots, v\}$.

Corollary F.22 (Descent and convergence for smoothed row normalization). *Suppose Assumptions F.1 and F.8 hold, and let $\eta(t) = 1/(t + \varepsilon)$ for some $\varepsilon > 0$. Then $1/(M + \varepsilon) \leq \eta(\|G_{k,i}\|_2) \leq 1/\varepsilon$ for all $k \in \mathbb{N}$ and $i \in \llbracket v \rrbracket$. Hence Assumption F.7 holds with $\underline{\eta} = 1/(M + \varepsilon)$ and $\bar{\eta} = 1/\varepsilon$. Consequently, the descent, stationarity, and PL linear-convergence results of Theorems F.19 to F.21 apply directly. In particular, any constant learning rate satisfying $\gamma \in (0, 2\varepsilon^2/(L(M + \varepsilon)))$ guarantees monotonic descent, and under Assumption F.2 one obtains linear convergence with contraction factor*

$$\rho_{\text{row},\varepsilon} = 1 - 2\mu\gamma\left(\frac{1}{M + \varepsilon} - \frac{L\gamma}{2\varepsilon^2}\right).$$

Proof. For $t \geq 0$, the map $t \mapsto 1/(t + \varepsilon)$ is decreasing, so $1/(M + \varepsilon) \leq 1/(\|G_{k,i}\|_2 + \varepsilon) \leq 1/\varepsilon$. Thus Assumption F.7 holds with the stated constants. The conclusions then follow immediately from Theorems F.19 to F.21. $\square$

Thus, the choice $\eta(t) = 1/(t + \varepsilon)$ interpolates between the bounded row-norm regime and the fully normalized regime: it preserves the local row-adaptive flavor of normalization while avoiding the singular behavior of $\eta(t) = 1/t$ at small row norms.

Having established convergence guarantees for right-spectral and row-norm-based optimizers separately, we now turn to their finite compositions. The resulting hybrid methods inherit the geometric structure of the right polar factor together with a local row-wise normalization, and their analyses are naturally expressed through the preserved alignment ratio and the active row support.

F.5 Nuclear-Norm-Scaled Right-Spectral/Row-Norm Hybrid Optimizers

We now study the hybrid optimizer obtained by composing the right polar factor with row-wise normalization. In this construction, the update is first normalized with respect to the feature geometry through the right polar factor, and then normalized locally across rows. After an additional nuclear-norm scaling, both the alignment and the squared Frobenius norm of the resulting update admit explicit expressions, leading to a clean descent analysis in terms of two interpretable quantities: the preserved alignment ratio $\mathfrak{A}_{\text{hyb}}(G)$ and the active row support $s_{\text{row}}(G)$.

Define the right polar factor

$$Z(G) := G(G^\top G)^{\dagger/2},$$

and let the row-normalized hybrid map $\mathcal{T}_{\text{hyb}}: \mathbb{R}^{v \times d} \rightarrow \mathbb{R}^{v \times d}$ be given row-wise by

$$\mathcal{T}_{\text{hyb}}(G)_{i:} = \begin{cases} \frac{Z(G)_{i:}}{\|Z(G)_{i:}\|_2}, & Z(G)_{i:} \neq 0, \\ 0, & Z(G)_{i:} = 0. \end{cases}$$

Its nuclear-norm-scaled version is defined by $\mathcal{T}_{\text{hyb,nuc}}(G) := \|G\|_{\text{nuc}} \mathcal{T}_{\text{hyb}}(G)$. The corresponding iteration is

$$(\forall k \in \mathbb{N}) \quad W_{k+1} = W_k - \gamma_k \mathcal{T}_{\text{hyb,nuc}}(G_k), \quad G_k := \nabla f(W_k). \quad (\text{F.7})$$

Definition F.2 (Active row support of the right polar factor). For $G \in \mathbb{R}^{v \times d}$, define

$$s_{\text{row}}(G) := \#\{i \in [v] : Z(G)_{i,:} \neq 0\}, \quad Z(G) = G(G^\top G)^{\dagger/2}.$$

Definition F.3 (Hybrid row-polar alignment ratio). For $G \neq 0$, define

$$\mathfrak{A}_{\text{hyb}}(G) := \frac{\langle\langle G, \mathcal{T}_{\text{hyb}}(G) \rangle\rangle_{\text{F}}}{\|G\|_{\text{nuc}}}.$$

The quantity $\mathfrak{A}_{\text{hyb}}(G)$ measures how much of the nuclear-norm alignment of the right polar factor is preserved after row normalization.

Lemma F.23 (Norm and alignment identities). For every $G \in \mathbb{R}^{v \times d}$, we have $\|\mathcal{T}_{\text{hyb}}(G)\|_{\text{F}}^2 = s_{\text{row}}(G)$, $\|\mathcal{T}_{\text{hyb,nuc}}(G)\|_{\text{F}}^2 = \|G\|_{\text{nuc}}^2 s_{\text{row}}(G)$, and $\langle\langle G, \mathcal{T}_{\text{hyb,nuc}}(G) \rangle\rangle_{\text{F}} = \|G\|_{\text{nuc}} \langle\langle G, \mathcal{T}_{\text{hyb}}(G) \rangle\rangle_{\text{F}} = \mathfrak{A}_{\text{hyb}}(G) \|G\|_{\text{nuc}}^2$.

Proof. By construction, every nonzero row of $\mathcal{T}_{\text{hyb}}(G)$ has Euclidean norm 1, while every zero row remains zero. Hence $\|\mathcal{T}_{\text{hyb}}(G)\|_{\text{F}}^2 = s_{\text{row}}(G)$. Multiplying by $\|G\|_{\text{nuc}}$ yields $\|\mathcal{T}_{\text{hyb,nuc}}(G)\|_{\text{F}}^2 = \|G\|_{\text{nuc}}^2 \|\mathcal{T}_{\text{hyb}}(G)\|_{\text{F}}^2 = \|G\|_{\text{nuc}}^2 s_{\text{row}}(G)$. Similarly, $\langle\langle G, \mathcal{T}_{\text{hyb,nuc}}(G) \rangle\rangle_{\text{F}} = \|G\|_{\text{nuc}} \langle\langle G, \mathcal{T}_{\text{hyb}}(G) \rangle\rangle_{\text{F}} = \mathfrak{A}_{\text{hyb}}(G) \|G\|_{\text{nuc}}^2$. $\square$

Theorem F.24 (Descent lemma for the nuclear-norm-scaled hybrid update). Suppose f satisfies Assumption F.1. Then the iteration (F.7) satisfies

$$f(W_{k+1}) \leq f(W_k) - \gamma_k \mathfrak{A}_{\text{hyb}}(G_k) \|G_k\|_{\text{nuc}}^2 + \frac{L\gamma_k^2}{2} s_{\text{row}}(G_k) \|G_k\|_{\text{nuc}}^2.$$

Equivalently,

$$f(W_{k+1}) \leq f(W_k) - \gamma_k \left(\mathfrak{A}_{\text{hyb}}(G_k) - \frac{L\gamma_k}{2} s_{\text{row}}(G_k) \right) \|G_k\|_{\text{nuc}}^2.$$

In particular, if $\gamma_k \in (0, 2\mathfrak{A}_{\text{hyb}}(G_k)/(Ls_{\text{row}}(G_k)))$, then $f(W_{k+1}) \leq f(W_k)$.

Proof. By L -smoothness of f and (F.7), we obtain

$$f(W_{k+1}) \leq f(W_k) - \gamma_k \langle\langle G_k, \mathcal{T}_{\text{hyb,nuc}}(G_k) \rangle\rangle_{\text{F}} + \frac{L\gamma_k^2}{2} \|\mathcal{T}_{\text{hyb,nuc}}(G_k)\|_{\text{F}}^2.$$

Now apply Lemma F.23. $\square$

Assumption F.9 (Uniform hybrid alignment and row-support bounds). There exist constants $\underline{a} > 0$ and $\bar{s} \in \mathbb{N}^*$ such that for all $k \in \mathbb{N}$, $\mathfrak{A}_{\text{hyb}}(G_k) \geq \underline{a}$ and $s_{\text{row}}(G_k) \leq \bar{s}$.

Theorem F.25 (Sublinear convergence to stationarity). Suppose Assumptions F.1 and F.9 hold. If the learning rate γ is constant and satisfies $\gamma \in (0, 2\underline{a}/(L\bar{s}))$, then

$$f(W_{k+1}) \leq f(W_k) - \gamma \left(\underline{a} - \frac{L\gamma}{2} \bar{s} \right) \|G_k\|_{\text{nuc}}^2.$$

Consequently,

$$\sum_{k=0}^{T-1} \|G_k\|_{\text{F}}^2 \leq \sum_{k=0}^{T-1} \|G_k\|_{\text{nuc}}^2 \leq \frac{f(W_0) - f^*}{\gamma(\underline{a} - L\gamma\bar{s}/2)} \quad \text{and} \quad \min_{0 \leq k < T} \|G_k\|_{\text{F}}^2 \leq \frac{f(W_0) - f^*}{T\gamma(\underline{a} - L\gamma\bar{s}/2)}.$$

Proof. By Theorem F.24 and Assumption F.9,

$$f(W_{k+1}) \leq f(W_k) - \gamma \left(\underline{a} - \frac{L\gamma\bar{s}}{2} \right) \|G_k\|_{\text{nuc}}^2.$$

Summing from $k = 0$ to $T - 1$ gives

$$f(W_T) \leq f(W_0) - \gamma \left(\underline{a} - \frac{L\gamma\bar{s}}{2} \right) \sum_{k=0}^{T-1} \|G_k\|_{\text{nuc}}^2.$$

Using $f(W_T) \geq f^*$ yields

$$\sum_{k=0}^{T-1} \|G_k\|_{\text{nuc}}^2 \leq \frac{f(W_0) - f^*}{\gamma(\underline{a} - L\gamma\bar{s}/2)}.$$

The Frobenius-norm and minimum-gradient-norm bounds follow from $\|G_k\|_{\text{nuc}} \geq \|G_k\|_{\text{F}}$. $\square$

Theorem F.26 (Linear convergence under the PL condition). *Suppose Assumptions F.1, F.2 and F.9 hold. If the learning rate is constant and satisfies $\gamma \in (0, 2\underline{a}/(L\bar{s}))$, then*

$$f(W_{k+1}) - f^* \leq \left(1 - 2\mu\gamma \left(\underline{a} - \frac{L\gamma\bar{s}}{2} \right) \right) (f(W_k) - f^*).$$

Hence $f(W_k) - f^* \leq \rho_{\text{hyb}}^k (f(W_0) - f^*)$, where $\rho_{\text{hyb}} = 1 - 2\mu\gamma(\underline{a} - L\gamma\bar{s}/2) \in (0, 1)$.

Proof. By Theorem F.24 and Assumption F.9,

$$f(W_{k+1}) \leq f(W_k) - \gamma \left(\underline{a} - \frac{L\gamma\bar{s}}{2} \right) \|G_k\|_{\text{nuc}}^2.$$

Applying $\|G_k\|_{\text{nuc}}^2 \geq \|G_k\|_{\text{F}}^2$ and the PL inequality gives the desired inequality. $\square$

Switching the order of the two normalizations leads to a genuinely different optimizer. If row normalization is applied first, then the subsequent right-spectral step is computed from the modified Gram matrix $\tilde{G}^\top \tilde{G} = G^\top D_\eta(G)^2 G$, rather than from the original gradient Gram matrix $G^\top G$. Thus, right-polar-first preserves the original feature geometry before applying local row-wise normalization, whereas row-normalize-first alters that geometry prior to the spectral step.

F.6 Nuclear-Norm-Scaled Row-Norm/Right-Spectral Hybrid Optimizers

We now turn to the hybrid optimizer obtained by reversing the order of the two normalizations. Instead of first applying the right polar factor and then normalizing rows, we first apply row-wise normalization to the gradient and then compute a right-spectral update from the resulting row-normalized matrix. This construction is genuinely different from the right-spectral/row-norm hybrid in Section F.5. Indeed, the spectral step is now computed from the modified Gram matrix

$$\tilde{G}^\top \tilde{G} = G^\top D_\eta(G)^2 G,$$

rather than from the original feature Gram matrix $G^\top G$.

Let $\eta: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be a row-scaling function, and define

$$D_\eta(G) := \text{Diag}(\eta(\|G_{1\cdot}\|_2), \dots, \eta(\|G_{v\cdot}\|_2)).$$

The row-normalized gradient is defined by $\tilde{G} := D_\eta(G)G$. For example, the normalized row-norm choice corresponds to

$$\eta(t) = \begin{cases} 1/t, & t > 0, \\ 0, & t = 0, \end{cases}$$

or, in practical implementations, the smoothed variant $\eta(t) = 1/(t + \varepsilon)$.

Define the row-norm/right-spectral hybrid map by $\mathcal{T}_{\text{rowhyb}}(G) := Z(\tilde{G}) = \tilde{G}(\tilde{G}^\top \tilde{G})^{\dagger/2}$. Its nuclear-norm-scaled version is

$$\mathcal{T}_{\text{rowhyb,nuc}}(G) := \|\tilde{G}\|_{\text{nuc}} \mathcal{T}_{\text{rowhyb}}(G).$$

The corresponding iteration is

$$W_{k+1} = W_k - \gamma_k \mathcal{T}_{\text{rowhyb,nuc}}(G_k), \quad G_k = \nabla f(W_k). \quad (\text{F.8})$$

Definition F.4 (Row-normalized effective rank). For $G \in \mathbb{R}^{v \times d}$, define $r_{\text{rowhyb}}(G) := \text{rank}(\tilde{G})$, where $\tilde{G} = D_\eta(G)G$.

Definition F.5 (Row-norm/right-spectral alignment ratio). For $G \neq 0$, define

$$\mathfrak{A}_{\text{rowhyb}}(G) := \frac{\langle\langle G, \mathcal{T}_{\text{rowhyb}}(G) \rangle\rangle_{\text{F}}}{\|\tilde{G}\|_{\text{nuc}}} \quad \text{with} \quad \tilde{G} = D_\eta(G)G.$$

The quantity $\mathfrak{A}_{\text{rowhyb}}(G)$ measures the alignment between the original gradient G and the right polar factor of its row-normalized version $\tilde{G}$. Unlike the polar-first hybrid, this alignment is not automatically tied to the nuclear norm of G , because the spectral step is computed after row-wise reweighting.

Lemma F.27 (Norm and alignment identities). For every $G \in \mathbb{R}^{v \times d}$, let $\tilde{G} = D_\eta(G)G$. Then $\|\mathcal{T}_{\text{rowhyb}}(G)\|_{\text{F}}^2 = r_{\text{rowhyb}}(G)$, $\|\mathcal{T}_{\text{rowhyb,nuc}}(G)\|_{\text{F}}^2 = \|\tilde{G}\|_{\text{nuc}}^2 r_{\text{rowhyb}}(G)$, and $\langle\langle G, \mathcal{T}_{\text{rowhyb,nuc}}(G) \rangle\rangle_{\text{F}} = \mathfrak{A}_{\text{rowhyb}}(G) \|\tilde{G}\|_{\text{nuc}}^2$.

Proof. Since $\mathcal{T}_{\text{rowhyb}}(G)$ is the right polar factor of $\tilde{G}$, its Frobenius norm squared equals $\text{rank}(\tilde{G}) = r_{\text{rowhyb}}(G)$. Multiplying by $\|\tilde{G}\|_{\text{nuc}}$ gives

$$\|\mathcal{T}_{\text{rowhyb,nuc}}(G)\|_{\text{F}}^2 = \|\tilde{G}\|_{\text{nuc}}^2 \|\mathcal{T}_{\text{rowhyb}}(G)\|_{\text{F}}^2 = \|\tilde{G}\|_{\text{nuc}}^2 r_{\text{rowhyb}}(G).$$

The alignment identity follows directly from the definition of $\mathfrak{A}_{\text{rowhyb}}(G)$. $\square$

Theorem F.28 (Descent lemma for the nuclear-norm-scaled row-norm/right-spectral update). Suppose f satisfies Assumption F.1. Then the iteration (F.8) satisfies

$$f(W_{k+1}) \leq f(W_k) - \gamma_k \mathfrak{A}_{\text{rowhyb}}(G_k) \|\tilde{G}_k\|_{\text{nuc}}^2 + \frac{L\gamma_k^2}{2} r_{\text{rowhyb}}(G_k) \|\tilde{G}_k\|_{\text{nuc}}^2,$$

where $\tilde{G}_k = D_\eta(G_k)G_k$. Equivalently,

$$f(W_{k+1}) \leq f(W_k) - \gamma_k \left(\mathfrak{A}_{\text{rowhyb}}(G_k) - \frac{L\gamma_k}{2} r_{\text{rowhyb}}(G_k) \right) \|\tilde{G}_k\|_{\text{nuc}}^2.$$

In particular, if

$$\gamma_k \in \left(0, \frac{2\mathfrak{A}_{\text{rowhyb}}(G_k)}{Lr_{\text{rowhyb}}(G_k)} \right),$$

then $f(W_{k+1}) \leq f(W_k)$.

Proof. By L -smoothness and (F.8),

$$f(W_{k+1}) \leq (W_k) - \gamma_k \langle G_k, \mathcal{T}_{\text{rowhyb,nuc}}(G_k) \rangle_F + \frac{L\gamma_k^2}{2} \|\mathcal{T}_{\text{rowhyb,nuc}}(G_k)\|_F^2.$$

Applying Lemma F.27 gives the claim. $\square$

Assumption F.10 (Uniform row-norm/right-spectral alignment and rank bounds). There exist constants $\underline{a}_{\text{row}} > 0$ and $\bar{r}_{\text{row}} \in \mathbb{N}^*$ such that for all $k \in \mathbb{N}$, $\mathfrak{A}_{\text{rowhyb}}(G_k) \geq \underline{a}_{\text{row}}$ and $r_{\text{rowhyb}}(G_k) \leq \bar{r}_{\text{row}}$.

We need to make the following extra comparability assumption, which is necessary if we want convergence to stationarity in terms of $\|G_k\|_F^2$, instead of $\|\tilde{G}_k\|_{\text{nuc}}^2$.

Assumption F.11 (Row-normalization comparability). There exists a constant $\kappa_{\text{row}} > 0$ such that for all $k \in \mathbb{N}$, $\|\tilde{G}_k\|_{\text{nuc}} \geq \kappa_{\text{row}} \|G_k\|_F$, where $\tilde{G}_k = D_\eta(G_k)G_k$.

Theorem F.29 (Sublinear convergence to stationarity). Suppose Assumptions F.1, F.10 and F.11 hold. If the learning rate γ is constant and satisfies $\gamma \in (0, 2\underline{a}_{\text{row}}/(L\bar{r}_{\text{row}}))$, then

$$f(W_{k+1}) \leq f(W_k) - \gamma \left(\underline{a}_{\text{row}} - \frac{L\gamma}{2} \bar{r}_{\text{row}} \right) \|\tilde{G}_k\|_{\text{nuc}}^2.$$

Consequently,

$$\sum_{k=0}^{T-1} \|G_k\|_F^2 \leq \frac{f(W_0) - f^*}{\kappa_{\text{row}}^2 \gamma (\underline{a}_{\text{row}} - L\gamma \bar{r}_{\text{row}}/2)},$$

and hence

$$\min_{0 \leq k < T} \|G_k\|_F^2 \leq \frac{f(W_0) - f^*}{T \kappa_{\text{row}}^2 \gamma (\underline{a}_{\text{row}} - L\gamma \bar{r}_{\text{row}}/2)}.$$

Proof. By Theorem F.28 and Assumption F.10,

$$f(W_{k+1}) \leq f(W_k) - \gamma \left(\underline{a}_{\text{row}} - \frac{L\gamma}{2} \bar{r}_{\text{row}} \right) \|\tilde{G}_k\|_{\text{nuc}}^2.$$

Summing from $k = 0$ to $T - 1$ and using $f(W_T) \geq f^*$ yields

$$\sum_{k=0}^{T-1} \|\tilde{G}_k\|_{\text{nuc}}^2 \leq \frac{f(W_0) - f^*}{\gamma (\underline{a}_{\text{row}} - L\gamma \bar{r}_{\text{row}}/2)}.$$

The comparability assumption gives $\|\tilde{G}_k\|_{\text{nuc}}^2 \geq \kappa_{\text{row}}^2 \|G_k\|_F^2$, which proves the claim. $\square$

Theorem F.30 (Linear convergence under the PŁ condition). Suppose Assumptions F.1, F.2, F.10 and F.11 hold. If the learning rate γ is constant and satisfies $\gamma \in (0, 2\underline{a}_{\text{row}}/(L\bar{r}_{\text{row}}))$, then

$$f(W_{k+1}) - f^* \leq \left(1 - 2\mu\kappa_{\text{row}}^2 \gamma \left(\underline{a}_{\text{row}} - \frac{L\gamma}{2} \bar{r}_{\text{row}} \right) \right) (f(W_k) - f^*).$$

Hence

$$f(W_k) - f^* \leq \rho_{\text{rowhyb}}^k (f(W_0) - f^*),$$

where

$$\rho_{\text{rowhyb}} = 1 - 2\mu\kappa_{\text{row}}^2 \gamma (\underline{a}_{\text{row}} - L\gamma \bar{r}_{\text{row}}/2) \in (0, 1).$$

Proof. By Theorem F.28 and Assumption F.10,

$$f(W_{k+1}) \leq f(W_k) - \gamma \left(\underline{a}_{\text{row}} - \frac{L\gamma}{2} \bar{r}_{\text{row}} \right) \|\tilde{G}_k\|_{\text{nuc}}^2.$$

Using Assumption F.11,

$$\|\tilde{G}_k\|_{\text{nuc}}^2 \geq \kappa_{\text{row}}^2 \|G_k\|_{\text{F}}^2.$$

The PL inequality then proves the claimed recursion. Iterating the recursion yields the linear rate. $\square$

Remark F.3 (Comparison with the right-spectral/row-norm hybrid). The right-spectral/row-norm hybrid in Section F.5 first computes the right polar factor from the original gradient Gram matrix $G^\top G$ and then applies row-wise normalization. In contrast, the row-norm/right-spectral hybrid first replaces G by $\tilde{G} = D_\eta(G)G$ and then computes the right polar factor from $\tilde{G}^\top \tilde{G}$. Thus, the former preserves the original feature geometry before applying local row-wise normalization, while the latter modifies the feature geometry before the spectral step. This distinction is important in practice: row-normalize-first can better suppress row-scale imbalance before the spectral computation, whereas polar-first preserves more of the original right singular geometry.

Now, we specialize the convergence results when $\eta(t) = 1/(t + \varepsilon)$ for $\varepsilon > 0$, further assuming a uniform row-norm bound.

Assumption F.12 (Uniform row-norm bound). There exists $R > 0$ such that, for all $k \in \mathbb{N}$, $\max_{i \in [v]} \|G_{k,i}\|_2 \leq R$.

Lemma F.31 (Verification for $\eta(t) = 1/(t + \varepsilon)$). Let $\varepsilon > 0$ and define

$$D_\varepsilon(G) = \text{Diag}\left(\frac{1}{\|G_{1:}\|_2 + \varepsilon}, \dots, \frac{1}{\|G_{v:}\|_2 + \varepsilon}\right),$$

and $\tilde{G} = D_\varepsilon(G)G$. Then, for every $G \neq 0$,

$$\mathfrak{A}_{\text{rowhyb}}(G) = \frac{\langle\langle G, \text{polar}(\tilde{G}) \rangle\rangle_{\text{F}}}{\|\tilde{G}\|_{\text{nuc}}} \geq \varepsilon.$$

Moreover, if $\max_{i \in [v]} \|G_{i:}\|_2 \leq R$, then

$$\|\tilde{G}\|_{\text{nuc}} \geq \|\tilde{G}\|_{\text{F}} \geq \frac{1}{R + \varepsilon} \|G\|_{\text{F}}.$$

Consequently, along any sequence satisfying Assumption F.12, Assumption F.10 and Assumption F.11 hold with $\underline{a}_{\text{row}} = \varepsilon$, $\bar{r}_{\text{row}} = d$, and $\kappa_{\text{row}} = \frac{1}{R + \varepsilon}$.

Proof. Let $\tilde{G} = D_\varepsilon(G)G$ and write $\tilde{U}_{\text{p}} = \text{polar}(\tilde{G})$. Since $G = D_\varepsilon(G)^{-1}\tilde{G}$, we have $\langle\langle G, \tilde{U}_{\text{p}} \rangle\rangle_{\text{F}} = \langle\langle D_\varepsilon(G)^{-1}\tilde{G}, \tilde{U}_{\text{p}} \rangle\rangle_{\text{F}}$. Let $\tilde{G} = \tilde{U}\tilde{\Sigma}\tilde{V}^\top$ be a compact singular value decomposition. Then $\tilde{U}_{\text{p}} = \tilde{U}\tilde{V}^\top$, and therefore

$$\langle\langle D_\varepsilon(G)^{-1}\tilde{G}, \tilde{U}_{\text{p}} \rangle\rangle_{\text{F}} = \text{tr}(\tilde{\Sigma}\tilde{U}^\top D_\varepsilon(G)^{-1}\tilde{U}).$$

Because $D_\varepsilon(G)^{-1} = \text{Diag}(\|G_{1:}\|_2 + \varepsilon, \dots, \|G_{v:}\|_2 + \varepsilon) \succcurlyeq \varepsilon I_v$, we obtain

$$\text{tr}(\tilde{\Sigma}\tilde{U}^\top D_\varepsilon(G)^{-1}\tilde{U}) \geq \varepsilon \text{tr}(\tilde{\Sigma}) = \varepsilon \|\tilde{G}\|_{\text{nuc}}.$$

This proves $\mathfrak{A}_{\text{rowhyb}}(G) \geq \varepsilon$.

For the comparability bound, if $\max_{i \in \llbracket v \rrbracket} \|G_{i:}\|_2 \leq R$, then

$$\frac{1}{\|G_{i:}\|_2 + \varepsilon} \geq \frac{1}{R + \varepsilon}.$$

Hence

$$\|\tilde{G}\|_{\text{F}}^2 = \sum_{i=1}^v \frac{\|G_{i:}\|_2^2}{(\|G_{i:}\|_2 + \varepsilon)^2} \geq \frac{1}{(R + \varepsilon)^2} \|G\|_{\text{F}}^2.$$

Since $\|\tilde{G}\|_{\text{nuc}} \geq \|\tilde{G}\|_{\text{F}}$, the claimed comparability bound follows. Finally, $r_{\text{rowhyb}}(G) = \text{rank}(\tilde{G}) \leq d$, so we may take $\bar{r}_{\text{row}} = d$. $\square$

Corollary F.32 (Convergence for $\eta(t) = 1/(t + \varepsilon)$). *Suppose Assumptions F.1 and F.12 hold and consider the row-norm/right-spectral hybrid optimizer with $\eta(t) = 1/(t + \varepsilon)$. If the learning rate γ is constant and satisfies $\gamma \in (0, 2\varepsilon/(Ld))$, then*

$$f(W_{k+1}) \leq f(W_k) - \gamma \left(\varepsilon - \frac{L\gamma}{2} d \right) \|\tilde{G}_k\|_{\text{nuc}}^2.$$

Moreover,

$$\min_{0 \leq k < T} \|G_k\|_{\text{F}}^2 \leq \frac{(R + \varepsilon)^2 (f(W_0) - f^*)}{\gamma(\varepsilon - L\gamma d/2)}.$$

If, in addition, f satisfies the μ -PL condition, then

$$f(W_k) - f^* \leq \left(1 - \frac{2\mu\gamma(\varepsilon - L\gamma d/2)}{(R + \varepsilon)^2} \right)^k (f(W_0) - f^*).$$

G Experimental Details

Note that we reduce the number of hidden layers and the number of experts from the original architecture. We give the main modified model architecture specifications of the model in Table G.1. Their detailed designs can be found in their corresponding technical reports [129, 51, 117, 122]. We initialize all 2D weights by Gaussian random numbers with zero mean and standard deviation 0.02.

Table G.1: Modified model architectures of language models.

	Model	Qwen3-0.6B	Gemma 3 1B	OLMoE-1B-7B	gpt-oss
# trainable parameters		625,784,832	1,087,138,944	2,824,177,664	3,467,779,008
d_{model}		1024	1152	2048	2048
d_{ff}		3072	6912	1024	2048
n_{layers}		20	18	12	12
n_{heads} (Q / KV)		16 / 8	4 / 1	16 / 16	64 / 8
d_{heads}		128	256	128	64
n_{experts}		1	1	8	4
n_{experts} activated		N/A	N/A	32	16
vocabulary size		151,936	262,144	50,304	201,088
layer norm		RMSNorm	RMSNorm	RMSNorm	RMSNorm
activation function		SwiGLU	GeGLU with tanh	SwiGLU	SwiGLU ¹

In the following experiments, we use POLAR EXPRESS [5] for computing the matrix inverse square root in LEFTPOLARGRADM and Gram Newton–Schulz [172] for computing the orthogonal polar factor directly in HYBRIDPOLARGRADM, respectively. Unless otherwise specified, for POLAR EXPRESS and Gram Newton–Schulz, we use 5 inner steps with $\varepsilon_{\text{NS}} = 10^{-7}$. We use $\varepsilon = 10^{-8}$ for all of ROWNORMM, LEFTPOLARGRADM and HYBRIDPOLARGRADM. The row-scaling rule in ROWNORMM and HYBRIDPOLARGRADM is chosen to be the *smoothed row normalization*, i.e., $\eta(t) = 1/(t + \varepsilon)$ with $\varepsilon = 10^{-8}$.

For ADAMW on scalar and vector parameters, we use a linear warmup with cosine decay learning rate schedule with 100 warmup steps and a half-cosine decay to 0. For all other parameters, regardless of the choice of the optimizers, we use a stable-decay schedule with an initial learning rate γ_0 for the first 60% of training steps and linear decay to 0 for the last 40% training steps. We use the fused implementation of ADAMW in PyTorch, while the implementations of ROWNORMM, LEFTPOLARGRADM, RIGHTPOLARGRADM and HYBRIDPOLARGRADM are not optimized with customized kernels, except for the usage of Gram Newton–Schulz [172].

We give the training configurations and optimizer hyperparameters of all four model pre-training experiments in the following subsections.

¹The SwiGLU implementation in gpt-oss is unconventional as it includes clamping and residual connection.

G.1 Qwen3-0.6B-Style Pre-Training

In terms of wall-clock training time, for (a), configurations (i)–(iii) take 7.360 hours, 7.498 hours and 7.369 hours respectively, while for (b), configurations (i)–(iii) take 7.765 hours, 7.886 hours and 7.771 hours respectively. The time difference between (a) and (b) is expected as HYBRIDPOLARGRADM for SwiGLU MLP projection matrices have additional computational overheads than MUON.

Table G.2: Training configurations for Qwen3-0.6B-style pre-training.

Model	Qwen3-0.6B on FineWeb-Edu-10B
Context length	1024 tokens
Per-device batch size	28 sequences
Training steps	30,000
Training tokens	6,881,280,000
Validation steps	46
Validation tokens	10,551,296
Precision	bf16
Data-parallel size	8 (Nvidia H200)

Table G.3: Optimizer hyperparameters for Qwen3-0.6B-style pre-training.

Configuration	Parameter type	Optimizer	γ_0	momentum β	weight decay λ
(a)(i) ADAMW + MUON + MUON + RowNORMM	scalar / vector	ADAMW	0.05	(0.9, 0.95)	0.001
	linear / attention	MUON	0.02	0.95	0.001
	SwiGLU MLP	MUON	0.02	0.95	0.001
	embedding	RowNORMM	0.50	0.95	0
	head	RowNORMM	0.005	0.95	0
(a)(ii) ADAMW + MUON + MUON + HYBRIDPOLARGRADM (row-norm/right-spectral; $\alpha = 1$)	scalar / vector	ADAMW	0.05	(0.9, 0.95)	0.001
	linear / attention	MUON	0.02	0.95	0.001
	SwiGLU MLP	MUON	0.02	0.95	0.001
	embedding	HYBRIDPOLARGRADM	1.00	0.95	0
	head	HYBRIDPOLARGRADM	0.01	0.95	0
(a)(iii) ADAMW + MUON + MUON + ADAMW	scalar / vector	ADAMW	0.05	(0.9, 0.95)	0.001
	linear / attention	MUON	0.02	0.95	0.001
	SwiGLU MLP	MUON	0.02	0.95	0.001
	embedding	ADAMW	0.10	(0.9, 0.95)	0
	head	ADAMW	0.001	(0.9, 0.95)	0
(b)(i) ADAMW + MUON + HYBRIDPOLARGRADM (row-norm/right-spectral; $\alpha = 1$) + RowNORMM	scalar / vector	ADAMW	0.05	(0.9, 0.95)	0.001
	linear / attention	MUON	0.02	0.95	0.001
	SwiGLU MLP	HYBRIDPOLARGRADM	0.02	0.95	0.001
	embedding	RowNORMM	0.50	0.95	0
	head	RowNORMM	0.005	0.95	0
(b)(ii) ADAMW + MUON + HYBRIDPOLARGRADM + HYBRIDPOLARGRADM (row-norm/right-spectral; $\alpha = 1$)	scalar / vector	ADAMW	0.05	(0.9, 0.95)	0.001
	linear / attention	MUON	0.02	0.95	0.001
	SwiGLU MLP	HYBRIDPOLARGRADM	0.02	0.95	0.001
	embedding	HYBRIDPOLARGRADM	1.00	0.95	0
	head	HYBRIDPOLARGRADM	0.01	0.95	0
(b)(iii) ADAMW + MUON + HYBRIDPOLARGRADM (row-norm/right-spectral; $\alpha = 1$) + ADAMW	scalar / vector	ADAMW	0.05	(0.9, 0.95)	0.001
	linear / attention	MUON	0.02	0.95	0.001
	SwiGLU MLP	HYBRIDPOLARGRADM	0.02	0.95	0.001
	embedding	ADAMW	0.10	(0.9, 0.95)	0
	head	ADAMW	0.001	(0.9, 0.95)	0

G.2 Gemma 3 1B-Style Pre-Training

In terms of wall-clock training time, for (a), configurations (i)–(iii) take 8.341 hours, 8.807 hours and 8.132 hours respectively, whereas for (b), configurations (i)–(iii) take 8.675 hours, 9.101 hours and 8.469 hours respectively.

Table G.4: Training configurations for Gemma 3 1B-style pre-training.

Model	Gemma 3 1B on FineWeb-Edu-10B
Context length	1024 tokens
Per-device batch size	18 sequences
Training steps	50,000
Training tokens	7,372,800,000
Validation steps	71
Validation tokens	10,469,376
Precision	bfloat16
Data-parallel size	8 (Nvidia H200)

Table G.5: Optimizer hyperparameters for Gemma 3 1B-style pre-training.

Configuration	Parameter type	Optimizer	γ_0	momentum β	weight decay λ
(a)(i) ADAMW + MUON + MUON + RowNORMM	scalar / vector	ADAMW	0.05	(0.9, 0.95)	0.001
	linear / attention	MUON	0.02	0.95	0.001
	SwiGLU MLP	MUON	0.02	0.95	0.001
	embedding	RowNORMM	0.0025	0.95	0
	head	RowNORMM	0.0025	0.95	0
(a)(ii) ADAMW + MUON + MUON + HYBRIDPOLARGRADM (row-norm/right-spectral; $\alpha = 1$)	scalar / vector	ADAMW	0.05	(0.9, 0.95)	0.001
	linear / attention	MUON	0.02	0.95	0.001
	SwiGLU MLP	MUON	0.02	0.95	0.001
	embedding	HYBRIDPOLARGRADM	0.0025	0.95	0
	head	HYBRIDPOLARGRADM	0.0025	0.95	0
(a)(iii) ADAMW + MUON + MUON + ADAMW	scalar / vector	ADAMW	0.05	(0.9, 0.95)	0.001
	linear / attention	MUON	0.02	0.95	0.001
	SwiGLU MLP	MUON	0.02	0.95	0.001
	embedding	ADAMW	0.0005	(0.9, 0.95)	0
	head	ADAMW	0.0005	(0.9, 0.95)	0
(b)(i) ADAMW + MUON + HYBRIDPOLARGRADM (row-norm/right-spectral; $\alpha = 1$) + RowNORMM	scalar / vector	ADAMW	0.05	(0.9, 0.95)	0.001
	linear / attention	MUON	0.02	0.95	0.001
	SwiGLU MLP	HYBRIDPOLARGRADM	0.02	0.95	0.001
	embedding	RowNORMM	0.0025	0.95	0
	head	RowNORMM	0.0025	0.95	0
(b)(ii) ADAMW + MUON + HYBRIDPOLARGRADM + HYBRIDPOLARGRADM (row-norm/right-spectral; $\alpha = 1$)	scalar / vector	ADAMW	0.05	(0.9, 0.95)	0.001
	linear / attention	MUON	0.02	0.95	0.001
	SwiGLU MLP	HYBRIDPOLARGRADM	0.02	0.95	0.001
	embedding	HYBRIDPOLARGRADM	0.001	0.95	0
	head	HYBRIDPOLARGRADM	0.001	0.95	0
(b)(iii) ADAMW + MUON + HYBRIDPOLARGRADM (row-norm/right-spectral; $\alpha = 1$) + ADAMW	scalar / vector	ADAMW	0.05	(0.9, 0.95)	0.001
	linear / attention	MUON	0.02	0.95	0.001
	SwiGLU MLP	HYBRIDPOLARGRADM	0.02	0.95	0.001
	embedding	ADAMW	0.0005	(0.9, 0.95)	0
	head	ADAMW	0.0005	(0.9, 0.95)	0

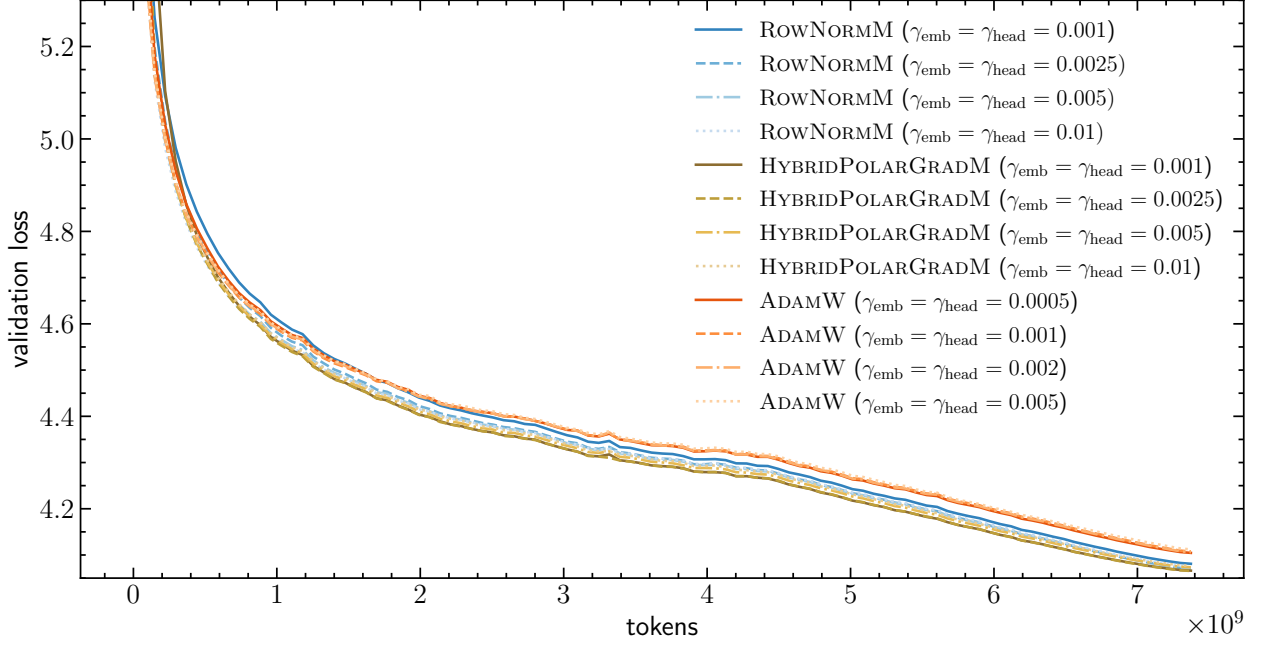
G.2.1 Gemma 3 1B-Style Pre-Training Learning Rate Sweep

For this pre-training experiment, we also perform a base learning rate sweep for the embedding and LM head matrices, keeping the learning rates for scalars/vector and matrices fixed. For simplicity, we keep both base learning rates for the embedding and LM head matrices to be equal, although more delicate tuning is possible.

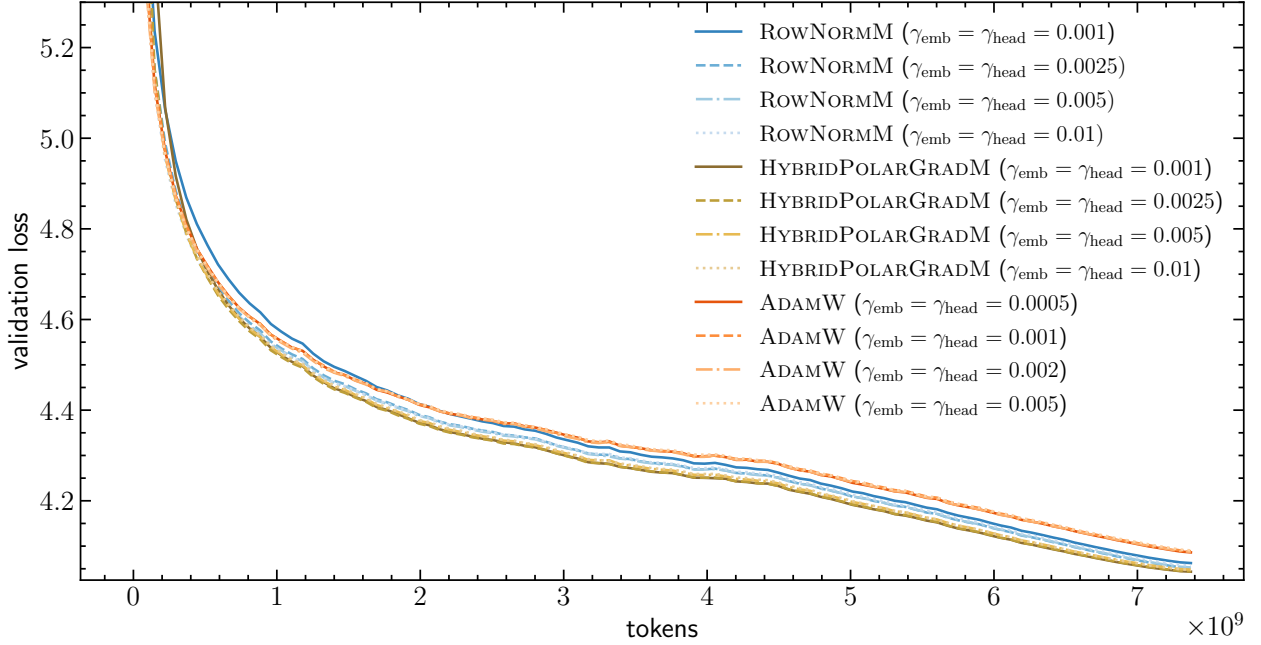
Table G.6: Base learning-rate sweep for the input embedding and LM head matrices in Gemma 3 1B-style pre-training. We sweep only $\gamma_{0,\text{emb}} = \gamma_{0,\text{head}}$. In setting (a), SwiGLU MLP projection matrices use MUON; in setting (b), they use HYBRIDPOLARGRADM with a row-norm/right-spectral composition.

Setting	Embedding/LM head optimizer	$\gamma_{0,\text{emb}} = \gamma_{0,\text{head}}$	Final validation loss
(a) SwiGLU MLP: MUON	RowNORMM	0.001	4.0810
		0.0025	4.0702
		0.005	4.0703
		0.01	4.0735
	HYBRIDPOLARGRADM (row-norm/right-spectral, $\alpha = 1$)	0.001	4.0662
		0.0025	4.0655
		0.005	4.0734
		0.01	4.0765
	ADAMW	0.0005	4.1046
		0.001	4.1075
		0.002	4.1060
		0.005	4.1120
	RowNORMM	0.001	4.0626
		0.0025	4.0516
		0.005	4.0528
		0.01	4.0572
(b) SwiGLU MLP: HYBRIDPOLARGRADM	HYBRIDPOLARGRADM (row-norm/right-spectral, $\alpha = 1$)	0.001	4.0435
		0.0025	4.0441
		0.005	4.0481
		0.01	4.0477
	ADAMW	0.0005	4.0862
		0.001	4.0886
		0.002	4.0872
		0.005	4.0905

We observe from Table G.6 that the validation loss gaps between configuration (iii) and configurations (i)–(ii) remain substantial across the swept base learning rates. As shown in Figure G.1, the separation between the ADAMW embedding/LM head baselines and the symmetry-compatible alternatives is not explained by a single learning-rate choice. Across the swept values of $\gamma_{0,\text{emb}} = \gamma_{0,\text{head}}$, the ADAMW curves make comparable or slightly faster initial progress, but remain consistently above the RowNORMM and HYBRIDPOLARGRADM curves later in training. This qualitative ordering holds both when the SwiGLU MLP projection matrices use MUON and when they use HYBRIDPOLARGRADM. Thus, the improvement from symmetry-compatible vocabulary-indexed updates is robust to reasonable base learning-rate variation and does not depend on simultaneously replacing MUON on the SwiGLU MLP projections.



(a) SwiGLU MLP projection matrices use MUON.



(b) SwiGLU MLP projection matrices use HYBRIDPOLARGRADM with a row-norm/right-spectral composition.

Figure G.1: Validation loss curves for the Gemma 3 1B-style embedding/LM head learning-rate sweep. The swept learning rate is $\gamma_{0,\text{emb}} = \gamma_{0,\text{head}}$. The two panels differ only in the optimizer assigned to the SwiGLU MLP projection matrices: MUON in Figure G.1a and HYBRIDPOLARGRADM in Figure G.1b. Across both settings, ROWNORM and HYBRIDPOLARGRADM remain consistently better than ADAMW for the input embedding and LM head matrices.

G.2.2 Gemma 3 1B-Style Pre-Training Across Random Seeds

To assess the robustness of the observed optimizer ordering, we repeat the Gemma 3 1B-style pre-training experiment with two additional random seeds. As in Section 4.2, we consider the same three optimizer assignments for the input embedding and LM head matrices: (i) ROWNORMM, (ii) HYBRIDPOLARGRADM, and (iii) ADAMW. In all runs in this subsection, the SwiGLU MLP projection matrices use HYBRIDPOLARGRADM with a row-norm/right-spectral composition.

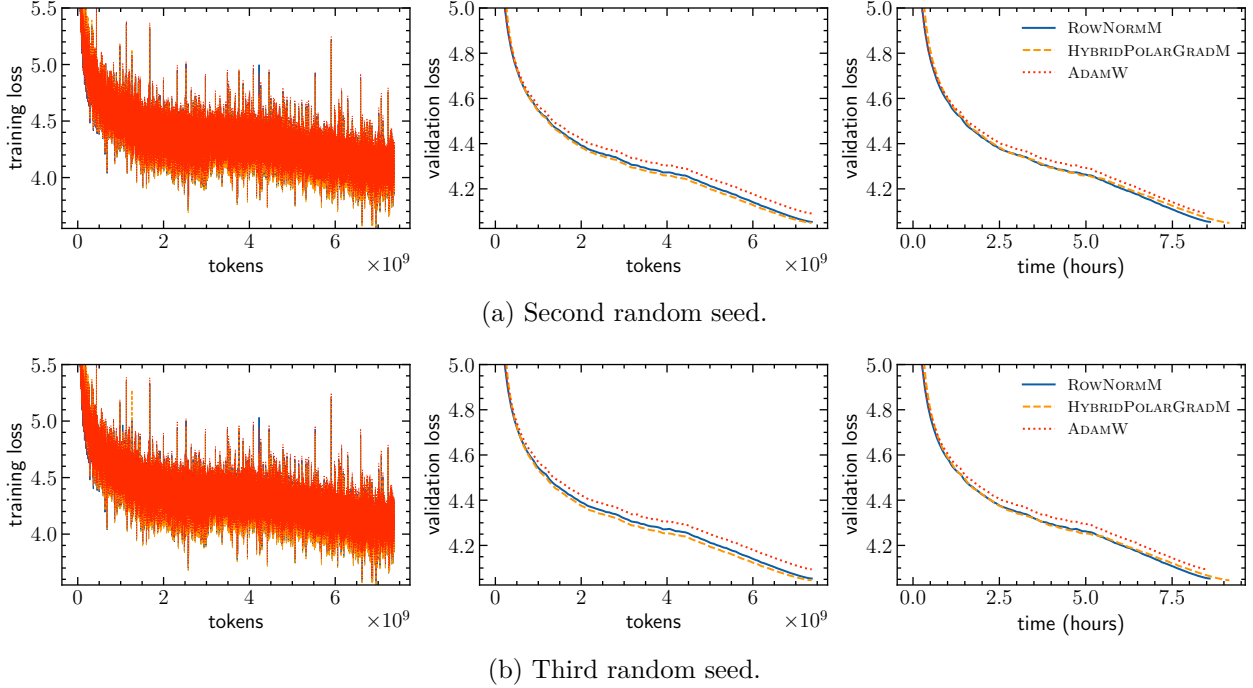


Figure G.2: Training and validation losses for Gemma 3 1B-style pre-training under two additional random seeds. In each subfigure, the three configurations differ only in the optimizer assigned to the input embedding and LM head matrices: ROWNORMM, HYBRIDPOLARGRADM, or ADAMW. The SwiGLU MLP projection matrices use HYBRIDPOLARGRADM in all runs.

Table G.7: Final validation losses across three random seeds for Gemma 3 1B-style pre-training. The optimizer assignment varies only for the input embedding and LM head matrices; SwiGLU MLP projection matrices use HYBRIDPOLARGRADM throughout.

Embedding/LM head optimizer	Seed 1	Seed 2	Seed 3	Mean loss $\pm$ std
ROWNORMM	4.0516	4.0539	4.0535	4.0530 ± 0.0012
HYBRIDPOLARGRADM	4.0435	4.0497	4.0459	4.0464 ± 0.0031
ADAMW	4.0862	4.0912	4.0941	4.0905 ± 0.0040

As shown in Table G.7, the qualitative ordering is stable across random seeds. Both symmetry-compatible optimizer assignments, ROWNORMM and HYBRIDPOLARGRADM, consistently outperform the ADAMW baseline on the vocabulary-indexed matrices. HYBRIDPOLARGRADM achieves the lowest mean final validation loss, improving over ADAMW by approximately 0.0441, while ROWNORMM improves over ADAMW by approximately 0.0375. These results indicate that the gains observed in the main Gemma 3 1B-style experiment are not an artifact of a single random initialization.

G.2.3 Vocabulary-Logit Growth and Final Logit Soft-Capping

We additionally study whether symmetry-compatible LM head updates affect the growth of final vocabulary logits in dense pre-training. This diagnostic is motivated by prior work on training instabilities in large Transformer models. Chowdhery et al. [28] introduce an auxiliary output z-loss, proportional to $\log^2 Z$, to keep the softmax normalizer $\log Z$ close to zero and improve stability. Wortsman et al. [157] study small-scale proxies for large-scale Transformer instabilities, including attention-logit growth and divergence between output logits and log probabilities. Gemma 2 [52] instead uses explicit logit soft-capping in both attention layers and the final vocabulary layer, applying $z \leftarrow c \tanh(z/c)$ with soft-cap values $c = 50$ for attention logits and $c = 30$ for final logits.

Our goal is to test whether optimizer geometry provides a complementary mechanism for controlling final vocabulary logits. For an LM head $W_{\text{out}} \in \mathbb{R}^{v \times d}$, the final logits are $z_t = W_{\text{out}} h_t \in \mathbb{R}^v$. Since the softmax distribution is invariant under shared shifts $z_t \mapsto z_t + c \mathbf{1}_v$, the LM head has a natural shared-logit-shift quotient geometry. The projected LM head updates proposed in this paper remove this symmetry-redundant direction from the actual parameter update, and may therefore reduce unnecessary or unstable growth of the final vocabulary logits.

We run Gemma 3 1B-style pre-training while varying the LM head optimizer and the final-logit soft-capping setting. Attention-logit soft-capping is disabled in all runs, so the diagnostic isolates the final vocabulary logits. In addition to validation cross-entropy, we log final-logit diagnostics on sampled validation token positions. Let $\mathcal{S}$ denote the sampled positions. For each $t \in \mathcal{S}$, let

$$z_t = (z_{t,i})_{1 \leq i \leq v} = W_{\text{out}} h_t \in \mathbb{R}^v, \quad \bar{z}_t := \frac{1}{v} \sum_{i=1}^v z_{t,i}, \quad \tilde{z}_t := z_t - \bar{z}_t \mathbf{1}_v.$$

We report the raw final-logit RMS, centered final-logit RMS, and maximum log-sum-exp given by

$$\left(\frac{1}{|\mathcal{S}|} \sum_{t \in \mathcal{S}} \frac{\|z_t\|_2^2}{v} \right)^{1/2}, \quad \left(\frac{1}{|\mathcal{S}|} \sum_{t \in \mathcal{S}} \frac{\|\tilde{z}_t\|_2^2}{v} \right)^{1/2}, \quad \max_{t \in \mathcal{S}} \log \sum_{i=1}^v \exp(z_{t,i}).$$

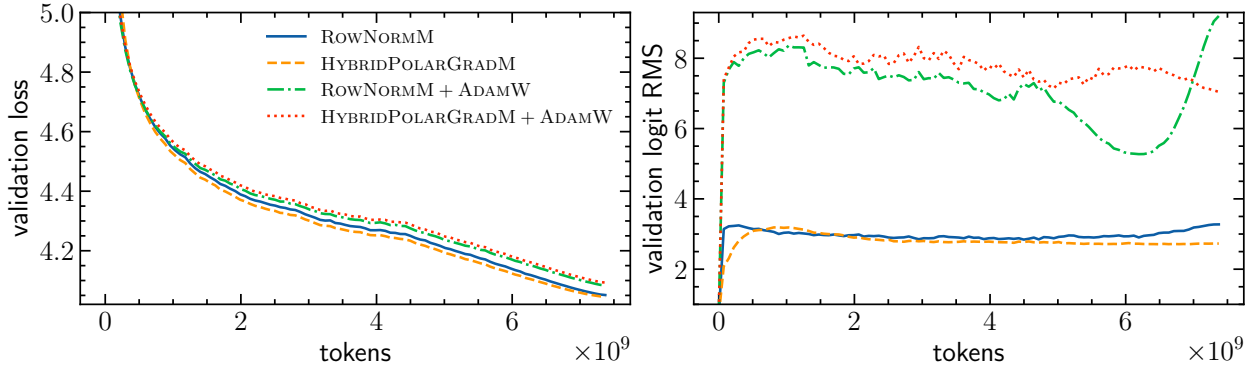
The raw RMS measures the overall logit scale, including the shared-logit-shift component. The centered RMS removes this shared component and measures logit growth on the LM head quotient space. The maximum log-sum-exp is a spike-sensitive diagnostic for unusually large final vocabulary logits. Under distributed training, this maximum is reduced across devices and therefore corresponds to the maximum over the union of sampled positions across ranks.

This experiment tests whether projected symmetry-compatible LM head updates provide an optimizer-side form of logit control. The quotient-geometry prediction is not that such updates shrink all logit variation, but that they suppress drift in the shared-logit-shift direction while preserving the centered logits that determine the softmax distribution. Thus, a positive outcome is that, without final logit soft-capping, projected ROWNORMM or HYBRIDPOLARGRADM LM head updates reduce raw logit RMS and log-sum-exp growth while maintaining comparable centered-logit diagnostics and comparable or improved validation cross-entropy relative to the corresponding ADAMW LM head baseline.

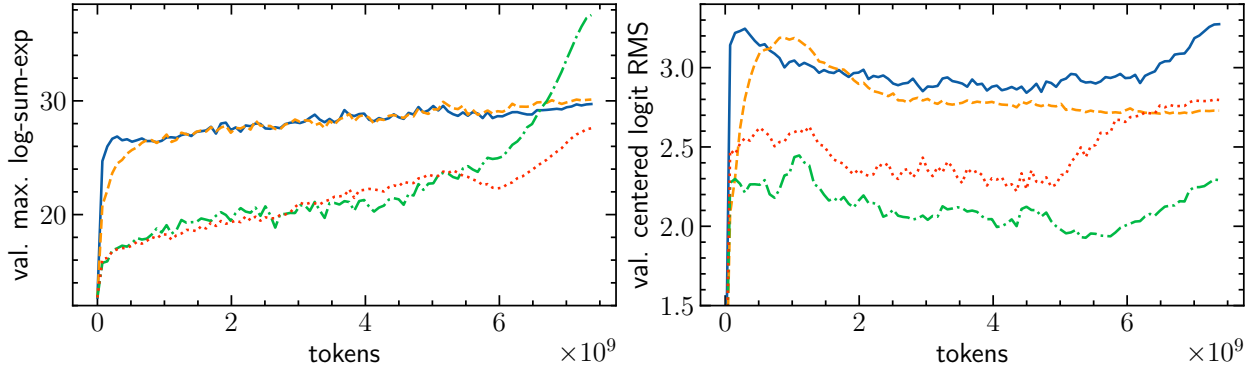
In the figures below, ROWNORMM and HYBRIDPOLARGRADM are configurations (b)(i) and (b)(ii) in Table G.5, respectively, where the same optimizer is used for both the embedding and LM head matrices. ROWNORMM + ADAMW and HYBRIDPOLARGRADM + ADAMW use the first optimizer for the embedding and ADAMW for the LM head. All other optimizer hyperparameters are kept as in the original Gemma 3 1B-style setting given in Table G.5.

Without final logit soft-capping. We first consider Gemma 3 1B-style pre-training without final logit soft-capping. As shown in Figure G.3, projected ROWNORMM and HYBRIDPOLARGRADM LM head updates achieve slightly lower validation loss than the corresponding configurations that use ADAMW on the LM head. The logit diagnostics reveal a sharper distinction. The ADAMW LM head variants exhibit substantially larger raw logit RMS, and the ROWNORMM + ADAMW configuration shows a pronounced late-stage increase in both raw logit RMS and maximum log-sum-exp. In contrast, the projected ROWNORMM and HYBRIDPOLARGRADM LM head updates keep the raw logit RMS much smaller and avoid this late-stage growth.

The centered diagnostics show that this growth is not primarily caused by larger relative differences among vocabulary logits. The centered logit RMS for the ADAMW LM head variants remains comparable to, and is often smaller than, that of the projected ROWNORMM and HYBRIDPOLARGRADM LM head updates. Thus, the raw logit growth under ADAMW is driven mainly by the shared-logit-shift component rather than by the centered component that determines the softmax distribution. This supports the quotient-geometry interpretation: projected LM-head updates suppress drift in a symmetry-redundant direction while preserving the softmax-relevant centered logits and slightly improving validation performance.



(a) Validation cross-entropy and raw final-logit RMS.

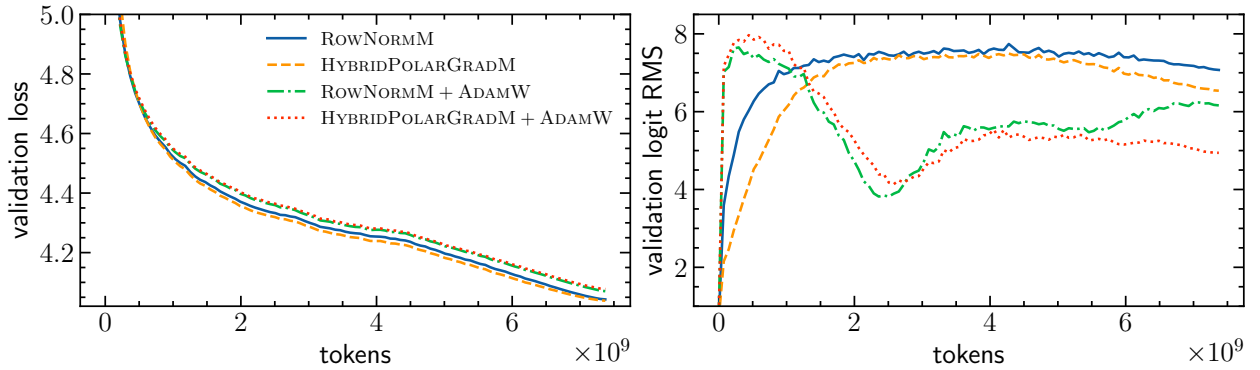


(b) Validation maximum log-sum-exp and centered final-logit RMS.

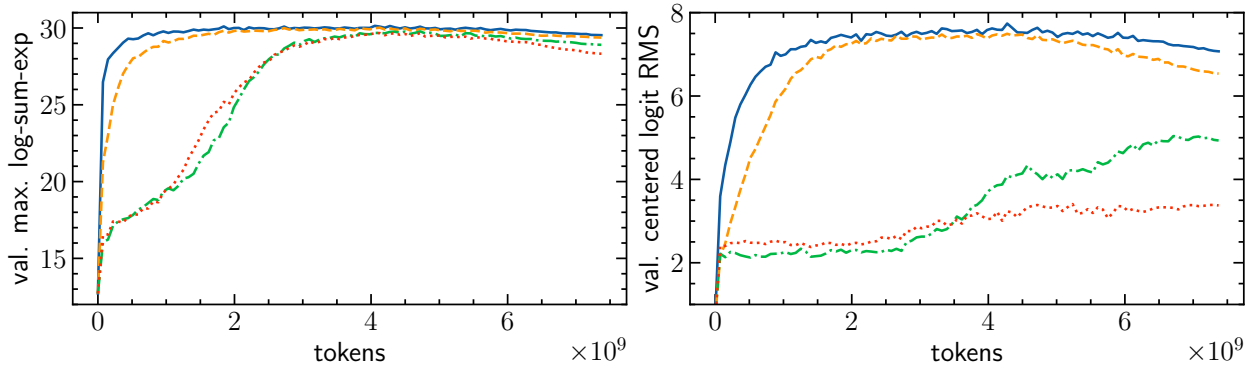
Figure G.3: Validation loss and final vocabulary-logit diagnostics for Gemma 3 1B-style pre-training without final logit soft-capping. The ADAMW LM head variants exhibit larger raw logit RMS and, for ROWNORMM + ADAMW, a late-stage increase in maximum log-sum-exp, while the centered logit RMS remains comparable. This indicates that the growth is mainly in the shared-logit-shift component.

With final logit soft-capping. We next repeat the same comparison with final logit soft-capping enabled. As shown in Figure G.4, final logit soft-capping largely suppresses the raw logit-scale growth of the ADAMW LM head variants. In contrast to the no-soft-capping setting, the raw logit RMS no longer exhibits sharp late-stage growth, and the maximum log-sum-exp curves are kept in a comparable range across methods. This confirms that final logit soft-capping acts as an effective direct stabilizer of the final vocabulary logits.

Nevertheless, optimizer geometry remains important. The projected ROWNORMM and HYBRIDPOLARGRADM LM head updates still achieve lower validation loss than the ADAMW LM head variants. Moreover, with soft-capping enabled, the projected methods have larger centered logit RMS than the ADAMW LM head variants while maintaining better validation loss. This suggests that the projected updates do not simply shrink all logit variation. Rather, they suppress redundant shared-shift drift while allowing useful centered vocabulary separation. Thus, final logit soft-capping and projected symmetry-compatible LM head updates are complementary: soft-capping directly bounds the effective final logits, while the optimizer geometry controls the quotient direction and improves language modeling performance.



(a) Validation cross-entropy and raw final-logit RMS.



(b) Validation maximum log-sum-exp and centered final-logit RMS.

Figure G.4: Validation loss and final vocabulary-logit diagnostics for Gemma 3 1B-style pre-training with final logit soft-capping enabled. Soft-capping suppresses the raw logit-scale growth of the ADAMW LM head variants, while projected ROWNORMM and HYBRIDPOLARGRADM LM head updates still achieve lower validation loss and stronger centered logit variation.

G.3 OLMoE-1B-7B-Style Pre-Training

In terms of wall-clock training time, configurations (i)–(iv) take 10.79 hours, 10.82 hours, 10.71 hours and 10.75 hours respectively.

Table G.8: Training configurations for OLMoE-1B-7B-style pre-training.

Model	OLMoE-1B-7B on FineWeb-Edu-10B
Context length	1024 tokens
Per-device batch size	20 sequences
Training steps	30,000
Training tokens	4,915,200,000
Validation steps	64
Validation tokens	10,485,760
Precision	bf16
Data-parallel size	8 (Nvidia H200)

Table G.9: Optimizer hyperparameters for OLMoE-1B-7B-style pre-training.

Configuration	Parameter type	Optimizer	γ_0	momentum β	weight decay λ
(i) ADAMW + MUON + RowNORMM	scalar / vector	ADAMW	0.01	(0.9, 0.95)	0.001
	matrix	MUON	0.005	0.95	0.001
	embedding	RowNORMM	0.5	0.95	0
	head	RowNORMM	0.005	0.95	0
	router	RowNORMM	0.00075	0.95	0
(ii) ADAMW + MUON + RowNORMM + LEFTPOLARGRADM ($\alpha = 1$)	scalar / vector	ADAMW	0.01	(0.9, 0.95)	0.001
	matrix	MUON	0.005	0.95	0.001
	embedding	RowNORMM	0.5	0.95	0
	head	RowNORMM	0.005	0.95	0
	router	LEFTPOLARGRADM	0.00075	0.95	0
(iii) ADAMW + MUON + RowNORMM + ADAMW	scalar / vector	ADAMW	0.01	(0.9, 0.95)	0.001
	matrix	MUON	0.005	0.95	0.001
	embedding	RowNORMM	0.5	0.95	0
	head	RowNORMM	0.005	0.95	0
	router	ADAMW	0.00075	0.95	0
(iv) ADAMW + MUON + ADAMW	scalar / vector	ADAMW	0.01	(0.9, 0.95)	0.001
	matrix	MUON	0.005	0.95	0.001
	embedding	ADAMW	0.0005	(0.9, 0.95)	0
	head	ADAMW	0.0005	(0.9, 0.95)	0
	router	ADAMW	0.00075	(0.9, 0.95)	0

G.4 Downsized gpt-oss Pre-Training

In terms of wall-clock training time, configurations (i)–(iv) take 19.08 hours, 18.91 hours, 19.34 hours and 19.22 hours respectively.

Table G.10: Training configurations for downsized gpt-oss pre-training.

Model	Downsized gpt-oss on FineWeb-Edu-10B
Context length	1024 tokens
Per-device batch size	8 sequences
Training steps	60,000
Training tokens	3,932,160,000
Validation steps	160
Validation tokens	10,485,760
Precision	bf16
Data-parallel size	8 (Nvidia H200)

Table G.11: Optimizer hyperparameters for downsized gpt-oss pre-training.

Configuration	Parameter type	Optimizer	γ_0	momentum β	weight decay λ
(i) ADAMW + MUON + ROWNORMM	scalar / vector	ADAMW	0.005	(0.9, 0.95)	0.001
	matrix	MUON	0.001	0.95	0.001
	embedding	ROWNORMM	0.1	0.95	0
	head	ROWNORMM	0.001	0.95	0
	router	ROWNORMM	0.00075	0.95	0
(ii) ADAMW + MUON + ROWNORMM + LEFTPOLARGRADM ($\alpha = 1$)	scalar / vector	ADAMW	0.005	(0.9, 0.95)	0.001
	matrix	MUON	0.001	0.95	0.001
	embedding	ROWNORMM	0.1	0.95	0
	head	ROWNORMM	0.001	0.95	0
	router	LEFTPOLARGRADM	0.00075	0.95	0
(iii) ADAMW + MUON + ROWNORMM + ADAMW	scalar / vector	ADAMW	0.005	(0.9, 0.95)	0.001
	matrix	MUON	0.001	0.95	0.001
	embedding	ROWNORMM	0.1	0.95	0
	head	ROWNORMM	0.001	0.95	0
	router	ADAMW	0.00075	0.95	0
(iv) ADAMW + MUON + ADAMW	scalar / vector	ADAMW	0.005	(0.9, 0.95)	0.001
	matrix	MUON	0.001	0.95	0.001
	embedding	ADAMW	0.0005	(0.9, 0.95)	0
	head	ADAMW	0.0005	(0.9, 0.95)	0
	router	ADAMW	0.00075	(0.9, 0.95)	0

G.4.1 Load-Balancing Loss and Router z-Loss Diagnostics

Following the diagnostic protocol of [117], we evaluate whether symmetry-compatible router updates improve expert load balancing and router stability. For each MOE layer, we log the fraction of routed assignments sent to each expert, the total router probability mass assigned to each expert, the auxiliary load-balancing loss [138], and the router z-loss [180]. We also track load entropy, load coefficient of variation, dead-expert fraction, maximum expert load, and router-logit scale. These diagnostics allow us to distinguish improvements in language model validation loss from improvements in routing dynamics.

Let E denote the number of experts and let k denote the number of activated experts per token. We measure the router load-balancing loss by $\mathcal{L}_{LB} = E \sum_{i=1}^E f_i P_i$, where f_i is the fraction of routed top- k assignments sent to expert i , and P_i is the average router probability mass assigned

to expert i . A perfectly balanced router has $\mathcal{L}_{\text{LB}}$ close to 1. We also measure the router z-loss $\mathcal{L}_z = \frac{1}{N} \sum_{t=1}^N \left(\log \sum_{i=1}^E \exp(z_{t,i}) \right)^2$, where $z_{t,i}$ is the router logit for token t and expert i and N is the batch size. This term penalizes excessive router-logit scale.

We compare ROWNORMM, LEFTPOLARGRADM, uncentered LEFTPOLARGRADM, HYBRIDPOLARGRADM, and ADAMW on router matrices while keeping all non-router optimizer assignments fixed: ADAMW for scalars and vectors, MUON for linear, attention, and SwiGLU MLP matrices, and ROWNORMM for embedding and LM head matrices. The uncentered LEFTPOLARGRADM baseline applies a Muon-style left-polar update directly to the router momentum without removing the shared-row component, and therefore ablates the centering operation induced by shared-logit-shift invariance. We keep the settings from Tables G.10 and G.11; the hyperparameters of uncentered LEFTPOLARGRADM and HYBRIDPOLARGRADM are set to be the same as those of ROWNORMM and LEFTPOLARGRADM.

Training objective without auxiliary losses. We first evaluate router dynamics when neither $\mathcal{L}_{\text{LB}}$ nor $\mathcal{L}_z$ is included in the training objective. As shown in Figures G.5 and G.6, the choice of router optimizer has a large effect on both expert load balancing and router-logit scale. Compared with ADAMW, all symmetry-compatible router updates achieve substantially lower measured load-balancing loss and router z-loss on both training and validation batches. Among them, LEFTPOLARGRADM and uncentered LEFTPOLARGRADM give the strongest load-balancing performance, while ROWNORMM also improves substantially over ADAMW. HYBRIDPOLARGRADM remains clearly better than ADAMW, but is less effective than the purely left-polar variants on the load-balancing diagnostics.

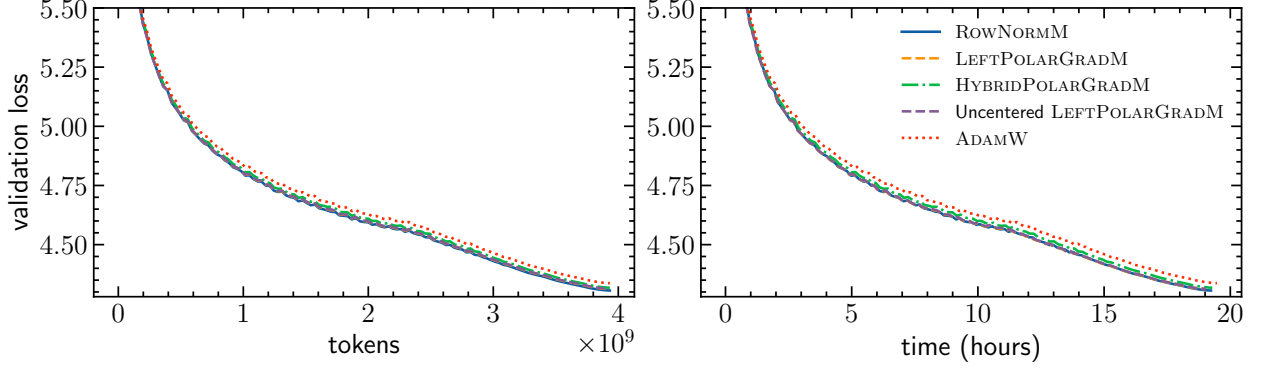
The expert-assignment summaries in Figure G.7 show the same pattern. ADAMW has lower load entropy, higher load coefficient of variation, a larger dead-expert fraction, and higher maximum expert load. In contrast, LEFTPOLARGRADM and uncentered LEFTPOLARGRADM nearly eliminate dead experts and achieve the most balanced expert assignment. ROWNORMM provides a moderate improvement, whereas HYBRIDPOLARGRADM is intermediate: it reduces dead experts and maximum load relative to ADAMW, but does not match the balance achieved by the left-polar updates. These results suggest that matrix-aware router updates can improve load balancing intrinsically, even without explicitly optimizing a load-balancing auxiliary objective.

The validation cross-entropy curves in Figure G.5 show that these routing improvements do not hurt language modeling performance and slightly improve over the ADAMW router baseline. The centered and uncentered LEFTPOLARGRADM variants behave similarly in this experiment, suggesting that the dominant empirical gain comes from the left-polar matrix geometry of the router update, while centering provides the symmetry-compatible quotient-space formulation induced by shared-logit-shift invariance.

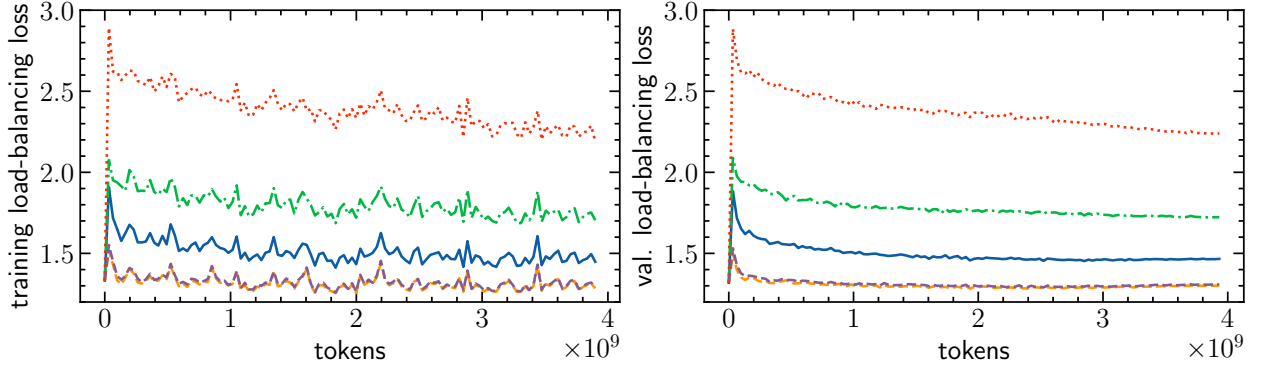
Training objective with auxiliary losses. We next include $\mathcal{L}_{\text{LB}}$ and $\mathcal{L}_z$ in the training objective with weights 0.01 and 0.001 respectively, following the choices of similar work [117? ?]. In this setting, we report validation cross-entropy as the primary language modeling metric, since the total training objective additionally contains auxiliary router penalties. As shown in Figures G.8 and G.9, auxiliary losses improve routing behavior for all methods: the measured load-balancing loss moves closer to its balanced value and the router z-loss decreases rapidly.

Nevertheless, optimizer geometry remains important. Compared with ADAMW, the symmetry-compatible router updates still yield lower measured load-balancing loss, lower router z-loss, and better expert-assignment statistics; see Figure G.10. With auxiliary losses enabled, ROWNORMM, LEFTPOLARGRADM, and uncentered LEFTPOLARGRADM are closely clustered and achieve the best

balance, while HYBRIDPOLARGRADM improves over ADAMW but remains somewhat less balanced than the strongest router-specific updates. Thus, auxiliary router losses and symmetry-compatible router updates are complementary: auxiliary losses improve all methods, but symmetry-compatible updates still yield better-controlled routing dynamics than coordinate-wise ADAMW.



(a) Validation cross-entropy loss.



(b) Training and validation load-balancing loss.

Figure G.5: Validation loss and measured load-balancing loss for downsized gpt-oss pre-training. The configurations differ only in the optimizer assigned to the MOE router matrices. Lower load-balancing loss indicates more balanced expert usage.

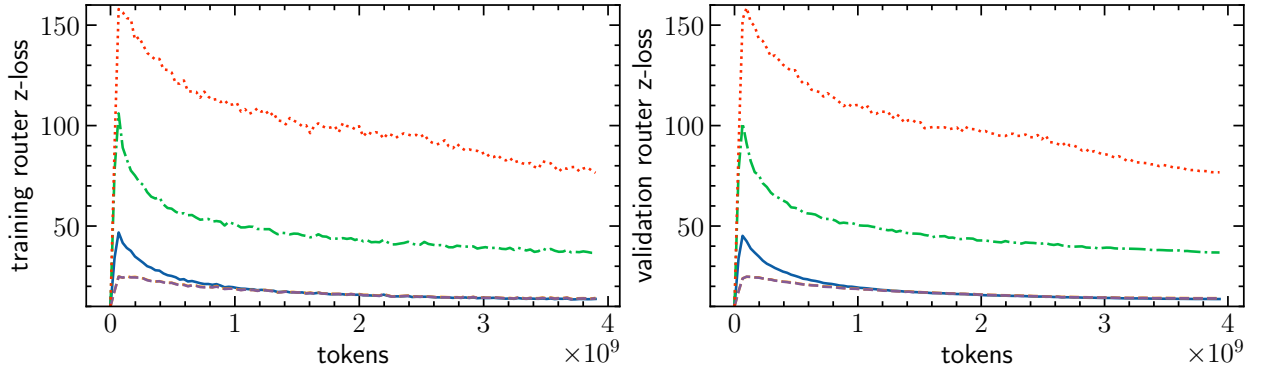
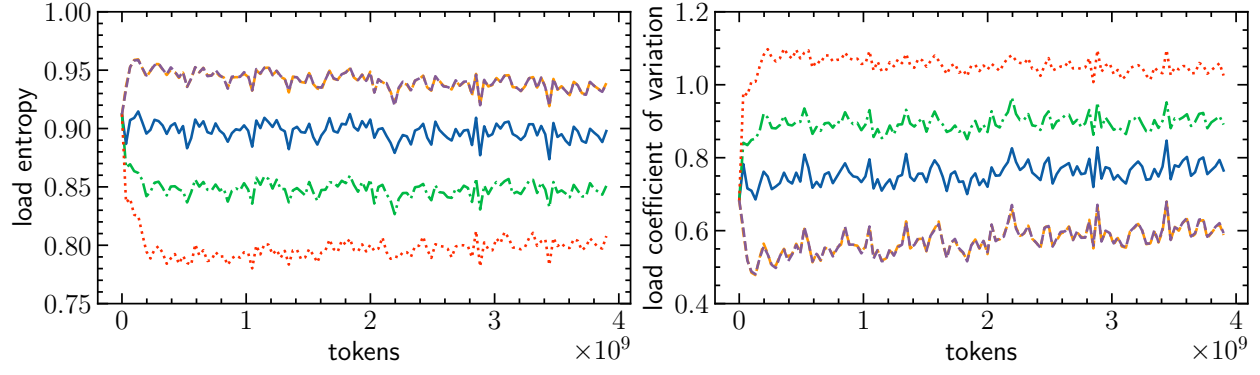
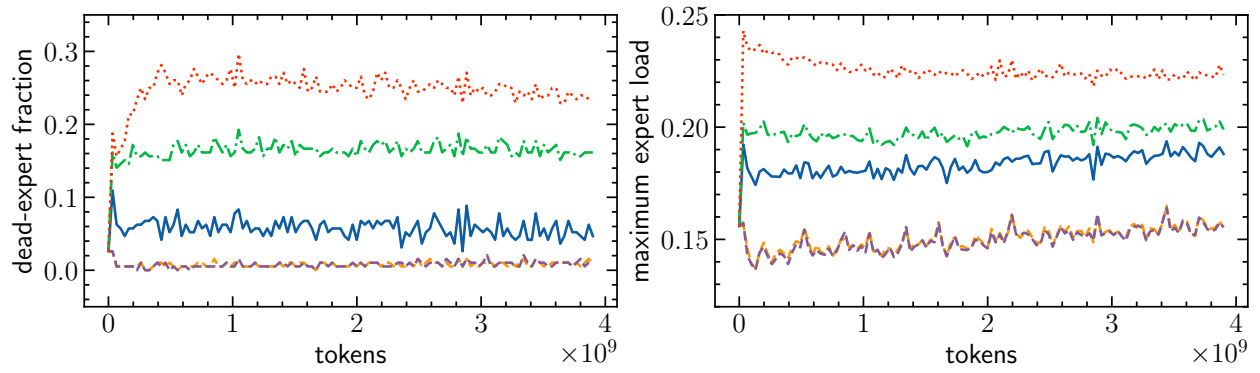


Figure G.6: Training and validation router z-loss for downsized gpt-oss pre-training. Lower router z-loss indicates better control of router-logit scale.

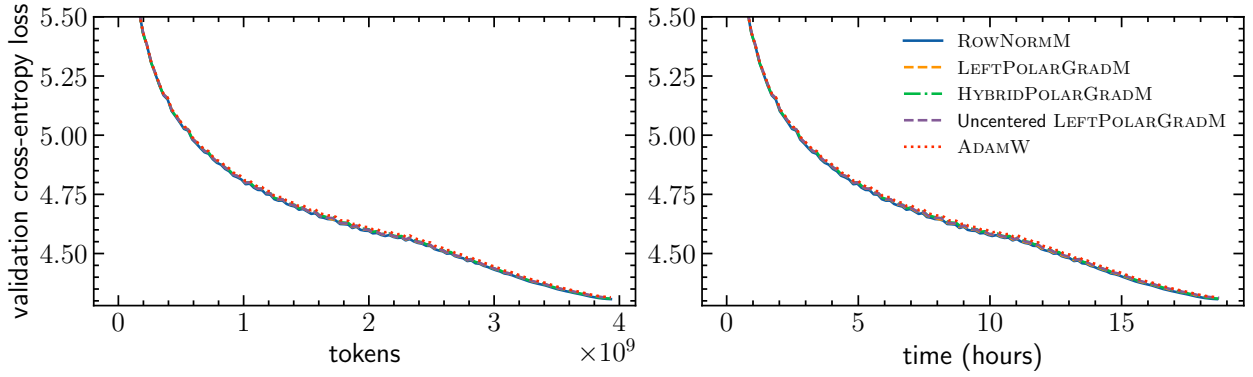


(a) Load entropy and load coefficient of variation. Higher entropy and lower coefficient of variation indicate more balanced expert assignment.

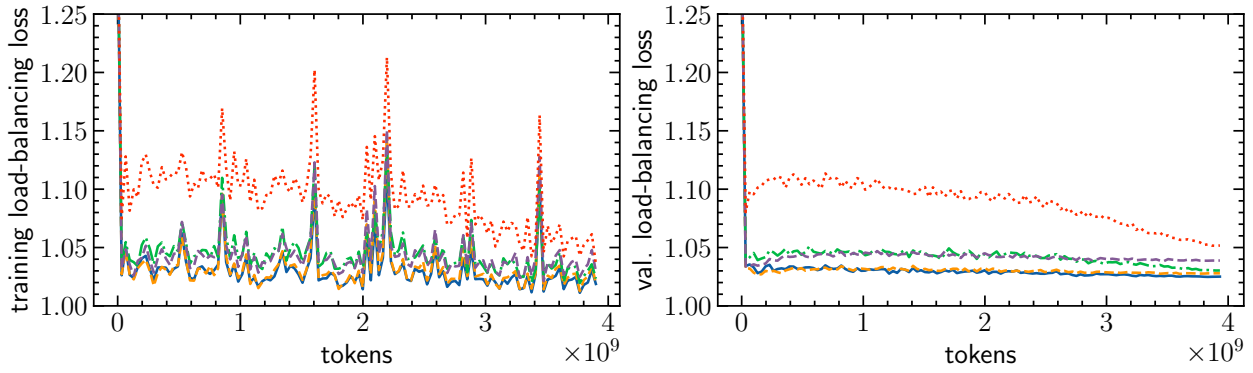


(b) Dead-expert fraction and maximum expert load. Lower values indicate fewer unused experts and less expert dominance.

Figure G.7: Expert-assignment balance summaries for downsized gpt-oss pre-training. These diagnostics directly measure whether routed assignments are evenly distributed across experts.



(a) Validation cross-entropy loss.



(b) Training and validation load-balancing loss.

Figure G.8: Validation loss and measured load-balancing loss for downsized gpt-oss pre-training with auxiliary router losses enabled. We report validation cross-entropy as the primary language modeling metric because the total training objective also includes auxiliary router losses.

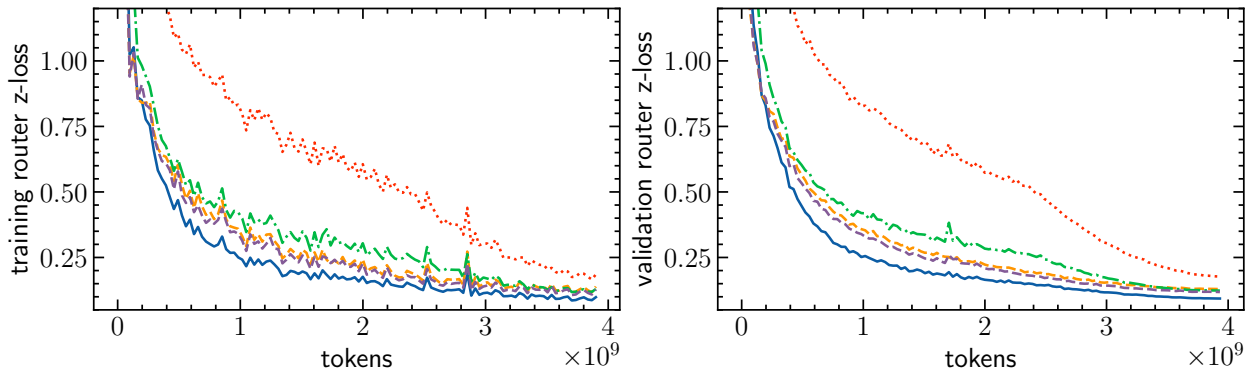
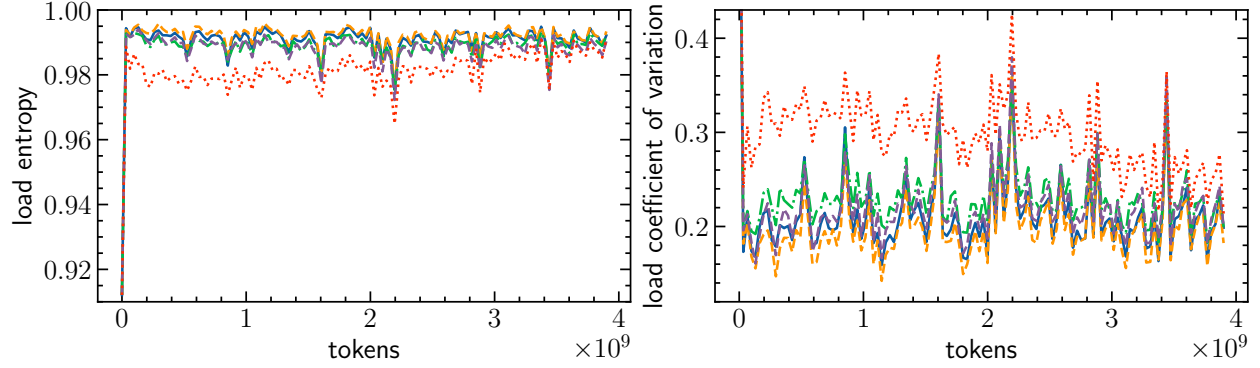
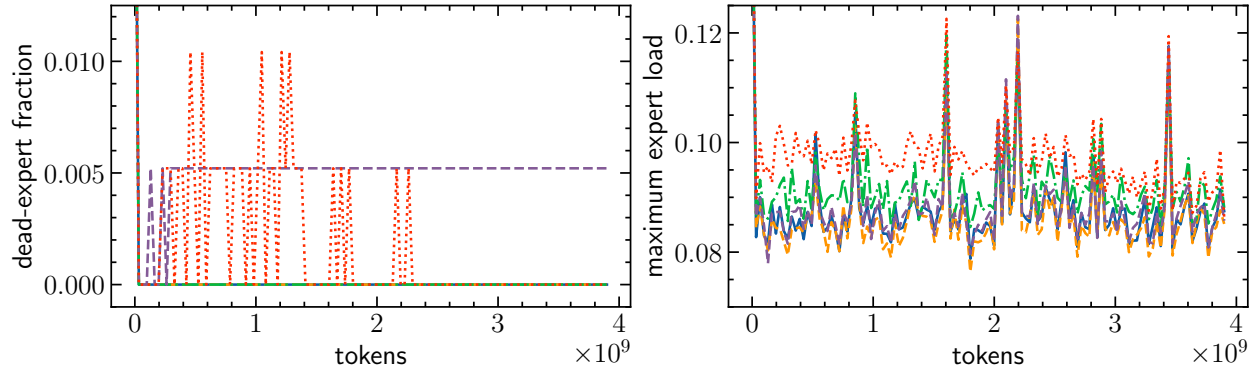


Figure G.9: Training and validation router z-loss for downsized gpt-oss pre-training with auxiliary router losses enabled. Lower router z-loss indicates better control of router-logit scale.



(a) Load entropy and load coefficient of variation. Higher entropy and lower coefficient of variation indicate more balanced expert assignment.



(b) Dead-expert fraction and maximum expert load. Lower values indicate fewer unused experts and less expert dominance.

Figure G.10: Expert-assignment balance summaries for downsized gpt-oss pre-training with auxiliary router losses enabled. These diagnostics directly measure whether routed assignments are evenly distributed across experts.